Topology

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These notes were prepared for a lecture course on topology at the Technical University of Munich in the Summer Semester of 2026. I have relied in part on the sources listed in the bibliography in preparing these notes; any errors are my own. The figures for these notes were made using TikZ and asymptote unless otherwise specified. All material not yet covered in lecture is subject to change without notice.

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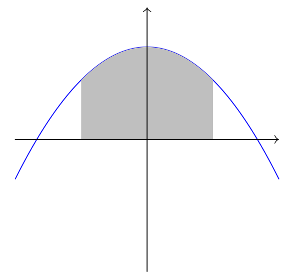
Topologies and continuity

First lecture on 13. April

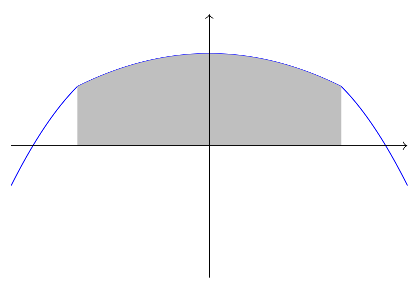
When we first encounter continuity — usually in an analysis class — we define continuous functions in terms of distances. The classic ϵ-δ definition fundamentally requires the structure of, for instance, a metric space to define continuity and associated notions.

The fundamental observation of topology is that continuity does not depend on the metric, but rather only on the open sets defined by the metric.[1] This opens the door to generalizing continuity-related arguments to a wide range of non-metric spaces: posets, polynomial rings over arbitrary fields, and spaces with indistinguishable points are all examples of topological spaces which do not admit a metric.

[1] As one way to illustrate that continuity doesn't depend on the specific metric, but rather on the open and closed sets, let's consider the graph of a continuous function $f : \mathbb{R} \to \mathbb{R}$.



We've highlighted a section of the x-axis, and we now stretch that section out, while leaving the other distances unchanged.



The function remains continuous, even though we've clearly changed the metric we're using on $\mathbb{R}$.

Preliminary to our exploration of topology, let us briefly recall the classic definition of continuity, and see how it only depends on open sets.

Definition 1.0.1. A *metric space* (X, d) is a pair consisting of a set X and a *metric* d on X, i.e., a function

$$d : X \times X \longrightarrow \mathbb{R}_{\geq 0}$$

satisfying the following properties.

1. *Symmetry*: For all $x, y \in X$,

$$d(x, y) = d(y, x).$$

2. *Identity of indiscernables*: For $x, y \in X$, $x = y$ if and only if $d(x, y) = 0$.

3. *Triangle inequality*: For $x, y, z \in X$,

$$d(x, y) + d(y, z) \geq d(x, z).$$

A function $f : (X, d_X) \to (Y, d_Y)$ is called *continuous at* $x \in X$ if, for every $\epsilon \in \mathbb{R}$ with $\epsilon > 0$, there exists a $\delta \in \mathbb{R}$ with $\delta > 0$ such that, for all $z \in X$,

$$d_X(x, z) < \delta \Longrightarrow d_Y(f(x), f(y)) < \epsilon.$$

The function f is called *continuous* if it is continuous at all points $x \in X$.

Definition 1.0.2. If (X, d_X) is a metric space, $x \in X$ and $r \in \mathbb{R}_{>0}$, we define the *open ball of radius r around x* to be

$$B_r(x) := \{y \in X \mid d_X(x, y) < r\} \subset X.$$

We call a subset $U \subset X$ *open* if, for every $x \in U$, there exists a radius $r \in \mathbb{R}_{>0}$ such that $B_r(x) \subset U$.

We can then formalize our earlier observation about continuity.

Proposition 1.0.3. *Let (X, d) and (Y, s) be metric spaces. Then a function $s : X \to Y$ is continuous if and only if, for every open subset $U \subset Y$, the subset $f^{-1}(U) \subset X$ is open.*

Proof. First suppose that s is continuous and let $U \subset Y$ be an open set. Suppose $x \in f^{-1}(U)$. Since U is open, we can find an $\epsilon > 0$ such that the ball $B_\epsilon(f(x)) \subset U$. By continuity, there exists a $\delta > 0$ such that $f(B_\delta(x)) \subset B_\epsilon(f(x))$, and thus $B_\delta(x) \subset f^{-1}(U)$, proving that $f^{-1}(U)$ is open.

Now suppose that every preimage under f of an open set is open. In particular, given $x \in X$, and $\epsilon > 0$, the set $f^{-1}(B_\epsilon(f(x)))$ is open. Therefore, there is a $\delta > 0$ such that $B_\delta(x) \subset f^{-1}(B_\epsilon(f(x)))$, and thus $f(B_\epsilon(x)) \subset B_\epsilon(f(x))$. Hence, f is continuous. □

1.1 The category of topological spaces

Based on Proposition 1.0.3, we define a topological space to be an axiomatization of the key properties of the collection of open sets of a metric space.

Definition 1.1.1. Let X be a set. A *topology on X* is a subset $\tau_X \subset \mathbb{P}(X)$ of the power set such that

1. $X, \varnothing \in \tau_X$.[2]
2. Let $\{U_i\}_{i \in I}$ be a (possibly infinite) collection of sets in τ_X. Then
$$\bigcup_{i \in I} U_i \in \tau_X$$
3. Let $\{U_i\}_{i \in 1}^n$ be a finite collection of sets in τ_X. Then
$$\bigcap_{i=1}^n U_i \in \tau_X.$$

We call the elements $U \in \tau_X$ the *open sets* of the topology on X. We refer to a pair (X, τ_X), where τ_X is a topology on X, as a topological space.[3] We call a subset $C \subset X$ of a topological space *closed* if

$$C^c := X \setminus C$$

is an open set, i.e. is in τ_X.

[2] Applying general set-theoretic conventions, this assertion actually follows from the following two. Since the empty union is empty, and the empty intersection of subsets of X is all of X, condition (2) implies that $X \in \tau_X$, and condition (3) implies that $\varnothing \in \tau_X$

[3] We will often abuse notation and write X for a topological space in cases where the choice of topology is clear from context.

As a first sanity check, we show that this does, indeed, generalize the situation of metric spaces.

Proposition 1.1.2. *Let (X, d) be a metric space. Denote by τ_d the collection of d-open subsets of X. Then τ_d is a topology on X.*[4]

[4] This, together with Proposition 1.0.3 effectively tells us that we can study continuous functions between metric spaces in terms of the associated topological spaces.

Proof. It is immediate that X and $\emptyset$ are elements of τ_d. We now check the remaining properties.

Suppose that $\{U_i\}_{i \in I}$ is a collection of open sets of X, and let $x \in \bigcup_{i \in I} U_i$. Then, in particular, there exists a $j \in I$ such that $x \in U_j$. Since U_j is open, there is a radius r such that $B_r(x) \subset U_j$. However, this implies that $B_r(x) \subset \bigcup_{i \in I} U_i$, so the latter is open.

Now suppose that $\{U_i\}_{i=1}^n$ is a collection of open sets of X. Let $x \in \bigcap U_i$. For each $1 \leq i \leq n$, choose a radius $r_i > 0$ such that $B_{r_i}(x) \subset U_i$. Set $r = \min_i(r_i)$. Then $B_r(x) \subset U_i$ for any $q \leq i \leq n$. Consequently $B_r(x) \subset \bigcap U_i$, and such $\bigcap U_i$ is an open set. $\square$

As alluded to earlier, there are also a wide variety of examples of topological spaces which do not arise as metric spaces.

Example 1.1.3. Suppose that $(P, \leq)$ is a partially ordered set (poset).[5] We can define a topology $\tau_{\leq}$ on P as follows. We define a set $U \subset P$ to be *downwards closed* if, for every $x \in U$ and $y \in P$, if $y \leq x$, then $y \in U$. We claim that the downwards closed sets form a topology τ_P on P. To see this, we check axioms (2) and (3) in the definition of topological spaces

2. Suppose that $\{U_i\}_{i \in I}$ is a collection of downwards-closed sets. Let $x \in \bigcup_{i \in I} U_i$ and $y \in P$ such that $y \leq x$. Then there is some $j \in I$ such that $x \in U_j$. Thus $y \in U_j$, and so $y \in \bigcup_{i \in I} U_i$. Thus, $\bigcup_{i \in I} U_i$ is downwards-closed.

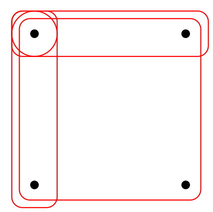
3. Suppose $\{U_i\}_{i \in I}$ is a collection of downwards-closed sets. Let $x \in \bigcap_{i \in I} U_i$ and $y \in P$ such that $y \leq x$. Then for each $i \in I$, $x \in U_i$, and so $y \in U_i$. Thus $y \in \bigcap_{i \in I} U_i$, and so $\bigcap_{i \in I} U_i$ is a downwards-closed set.

Notice that this topology has a curious feature: an *arbitrary* intersection of open sets is still open. This is not true, for instance, in the metric topology on $\mathbb{R}^n$.

[5] Let us consider a specific poset — the power set of $\{1, 2\}$ ordered by inclusion. We can draw the order relation as arrows:

$$\begin{tikzcd} \varnothing \arrow[r] \arrow[d] \arrow[dr] & \{1\} \arrow[d] \\ \{2\} \arrow[r] & \{1,2\} \end{tikzcd}$$

Since this poset is finite, we can also draw all of the corresponding non-empty open sets.



This topology is not trivial in any way — it encodes information about the original order relation. Enough, in fact, that we can recover the poset structure from the topology.

Example 1.1.4. Let k be a field, and consider the polynomial ring $k[x_1, \ldots, x_n]$ in n variables. We define the *Zariski topology* on k^n as follows. Let $I \subset k[x_1, \ldots, x_n]$ be an ideal. We define a corresponding subset of k^n, the *vanishing set* of I, to be

$$V(I) := \{a \in k^n \mid p(a) = 0 \;\; \forall p \in I\}.$$

Let us explore the behavior of the $V(I)$ under intersections and unions.

- Given I, J ideals, we can compute that

$$V(I) \cap V(J) = V(I + J)$$

where $I+J$ is the ideal consisting of elements of the form $p+q$ for $p \in I$ and $q \in J$. More generally, for an arbitrary collection $\{I_s\}_{s\in S}$ of ideals, we have

$$\bigcap_{s\in S} V(I_s) = V\left(\left\langle \bigcup_{s\in S} I_s \right\rangle\right)$$

where $\langle \bigcup_{s\in S} I_s \rangle$ denotes the ideal generated by all of the elements in the I_s.

- For a finite set S and a collection of ideals $\{I_s\}_{s\in S}$, we have[6]

$$\bigcup_{s\in S} V(I) = V\left(\prod_{s\in S} I_s\right).$$

[6] The interested reader may wish to attempt the following.

Exercise. Say what goes wrong here if we consider an infinite set of ideals.

This tells us that the collection of zero-sets of ideals is closed under arbitrary intersection and finite union — the opposite of what we want for a topology. However, this is a simple fix for this issue. We define the *Zariski topology* on k^n to be the topology whose open sets are of the form $k^n \setminus V(I)$ for some ideal $I \subset k[x_1, \ldots, x_n]$.

Definition 1.1.5. There are two key examples of topological spaces which often show up in computations. One is the *empty space* $\varnothing$, whose underlying set is the empty set and whose topology is $\{\varnothing\}$. The other is the *singleton space* — also called the *one-point space* or simply *the point* — denoted by $*$ whose underlying set is a singleton set $\{a\}$ and whose topology is $\mathbb{P}(\{a\}) = \{\varnothing, \{a\}\}$.

Definition 1.1.6. Let τ_1 and τ_2 be two topologies on a set X. If $\mathrm{id}_X : (X, \tau_2) \to (X, \tau_1)$ is continuous, we say that τ_1 is *coarser* than τ_2 or that τ_2 is *finer* than τ_1.

Examples. Some interesting examples of coarseness/fineness are the most extreme. Let X be a set

1. Define a topology τ_{dis} on X by declaring *every* subset of X to be an element of τ_{dis}. We call this the *discrete topology* on X. This is the finest possible topology on X, and it has a very interesting property. Let Y be any topological space, and let $f : X \to Y$ be any map of underlying sets. Then $f : (X, \tau_{dis}) \to (Y, \tau_Y)$ is continuous.
2. Define a topology τ_{ind} on X by $\tau_{ind} := \{\emptyset, X\}$. We call this the *indiscrete topology* on X – the coarsest possible topology on X. For any topological space Y and any map of sets $f : Y \to X$, the map $f : (Y, \tau_Y) \to (X, \tau_{ind})$ is continuous.

These two examples are *dual* to one another, in the same sense that the singleton and the empty space are dual to each other. We will explore this further in the next section.

Exercise 1.1.7. Show that the following statements are equivalent:

1. $\tau_1 \subset \tau_2$
2. τ_1 is coarser than τ_2.

Definition 1.1.8. A map $f : X \to Y$ between (the underlying sets of) topological spaces (X, τ_X) and (Y, τ_Y) is called a *continuous map of topological spaces* if, for every open subset $U \in \tau_Y$, the preimage $f^{-1}(U)$ is an element of τ_X. We say that f is a *homeomorphism* if it is bijective and both f and f^{-1} are continuous.[7] Further, we call f an *open map* if $f(U) \in \tau_Y$ for every $U \in \tau_X$, and a *closed map* if $f(C)$ is closed in Y for every closed set $C \subset X$.

[7] This is simply formalizing what we discovered about metric spaces into a definition. We can now talk about continuity of maps between general topological spaces and know that this subsumes the case of metric spaces.

Example 1.1.9. Define $B^n := \{x \in \mathbb{R}^n \mid |x| < 1\}$ to be the open unit ball in $\mathbb{R}^n$ (both spaces equipped with the topologies induced by the metrics). Define maps

$$\begin{aligned} f : B^n &\longrightarrow \mathbb{R}^n \\ x &\longmapsto \tfrac{x}{1-|x|} \end{aligned}$$

and

$$\begin{aligned} g : \mathbb{R}^n &\longrightarrow B^n \\ x &\longmapsto \tfrac{x}{1+|x|} \end{aligned}$$

Then f is a homeomorphism with inverse g.

Lemma 1.1.10. *Continuous maps have the following properties.*

1. *The composition of two continuous maps is continuous.*

2. *The identity map on any topological space is continuous.*

Proof. Left to the reader. □

We will not yet define categories, but we already have enough information to describe our key object of study: the category of topological spaces. Roughly speaking, a *category* is an axiomatization of a "place to do a certain kind of mathematics". It consists of a collection of "mathematical objects" — which in some simple cases can be thought of as sets equipped with additional structures like topologies or binary operations — and a collection of "morphisms" between any two objects — which in the aforementioned simple cases can be thought of as structure-preserving maps of sets.[8]

[8] While we will work with the category of topological spaces very heuristically here, we will return to delve more formally into category theory in Chapter 3. The reader desiring a more formal definition of category before continuing can consult Definition 3.1.1 and the follow pages for a more detailed exposition and exploration.

Definition 1.1.11. For topological spaces X and Y, we denote the set of continuous maps from X to Y by $\mathsf{Top}(X, Y)$.[9] The *category* Top *of topological spaces and continuous maps* consists of:

1. The collection[10] of all topological spaces. These will be referred to as the *objects* of Top. In a mild abuse of notation, we will often write $X \in \mathsf{Top}$ to mean "X is a topological space"

2. For every pair of spaces $X, Y \in \mathsf{Top}$, the set
$$\mathsf{Top}(X, Y)$$
of continuous maps from X to Y. These continuous maps are also called the *morphisms* of Top.

3. For every triple of spaces $X, Y, Z \in \mathsf{Top}$, the composition map
$$\mathsf{Top}(Y, Z) \times \mathsf{Top}(X, Y) \xrightarrow{-\circ-} \mathsf{Top}(X, Z)$$
which sends a pair of composable continuous maps to their composite.[11]

[9] This is also variously denoted by $C^0(X, Y)$ or $\mathrm{Hom}_{\mathsf{Top}}(X, Y)$.

[10] Not, sadly, a set, owing to Russel's Paradox.

[11] This is well-defined by Lemma 1.1.10.

Notice that the composition map is associative and has units in the sense that, for $f : X \to Y$, $f \circ \mathrm{id}_X = f = \mathrm{id}_Y \circ f$.

Exercise 1.1.12. Show that a continuous map $f : X \to Y$ is a homeomorphism if and only if it is both bijective and open.

1.2 Constructing new topologies

—— Second lecture on 14. April

Our next order of business is to understand how to build new topological spaces out of old ones. As one might expect, there are numerous different ways to approach all of these constructions, but we will focus on two: the *point-set* approach, where we explicitly write a set and characterize its open sets, and the *categorical* approach, where we specify a space by specifying all of the continuous maps out of it (or into it). Thus, each definition — phrased entirely in terms of maps — will be accompanied by a construction, which explains how to specify a topological space which satisfies the given definition. This is in contrast to many texts, in which the construction is given as the definition of a space.

Warning 1.2.1. ⚠ Our definitions here in terms of continuous mappings are more usually called *universal properties*. Our terminology is chosen to emphasize that these properties characterize a topological space *uniquely up to unique homeomorphism*, and so can be considered as definitions of the space in question.

As a tool to ease the point-set approach, we first introduce *bases* for topologies.

Definition 1.2.2. Let X be a set, and $\mathcal{S} \subset \mathbb{P}(X)$ a subset of the power set. The *topology generated by* $\mathcal{S}$ is the coarsest topology containing $\mathcal{S}$. Equivalently, this is the intersection of all topologies on X which contain $\mathcal{S}$. If τ is the topology on X generated by $\mathcal{S}$, we call $\mathcal{S}$ a *subbasis* of τ.

Lemma 1.2.3. *Let X be a set and $\mathcal{S} \subset \mathbb{P}(X)$ a subset of the power set. The topology τ generated by $\mathcal{S}$ consists of the arbitrary unions of finite intersections of elements of $\mathcal{S}$.*

Proof. Since the topology generated by $\mathcal{S}$ clearly must contain the unions of finite intersections of elements of $\mathcal{S}$, it suffices to simply verify that these sets form a topology. This follows immediately from the fact that unions of unions are unions, and intersections distribute over unions. $\square$

If we place some conditions on $\mathcal{S}$, however, the topology generated by $\mathcal{S}$ admits a simpler description.

Definition 1.2.4. We call $\mathcal{S} \subset \mathbb{P}(X)$ a *basis* if it satisfies the condition that, for any $U, V \in \mathcal{S}$ and any $x \in U \cap V$, there exists $W \in \mathcal{S}$ with $x \in W \subset U \cap V$.

Lemma 1.2.5. *If $\mathcal{S} \subset \mathbb{P}(X)$ is a basis, then the topology τ generated by $\mathcal{S}$ consists of the unions of elements of $\mathcal{S}$.*

Proof. By Lemma 1.2.3, it suffices to show that every finite intersection of elements of $\mathcal{S}$ can be written as a union of elements of $\mathcal{S}$. Given $x \in \bigcap_{i=1}^n U_i$, we first choose $V_2 \in \mathcal{S}$ such that $x \in V_1 \subset U_1 \cap U_2$. We then choose $V_3 \in \mathcal{S}$ such that $x \in V_2 \subset V_1 \cap U_3 \subset U_1 \cap U_2 \cap U_3$. Iterating, we find $V_x \in \mathcal{S}$ with $x \in V \subset \bigcap_{1=1}^n U_i$. By construction, $\bigcup_{x \in \bigcap U_i} V_x$ is $\bigcap_{i=1}^n U_i$, completing the proof. $\square$

Lemma 1.2.6. *Let (X, τ_X) and (Y, τ_Y) be topological spaces, let $f : X \to Y$ be a map of underlying sets, and let $\mathcal{B}$ be a basis for the topology τ_X. Then f is continuous if and only if, for every open $U \in \tau_Y$ and every $x \in f^{-1}(U)$, there is a basis element $V_{x,U} \in \mathcal{B}$ such that $x \in V_{x,U} \subset f^{-1}(U)$.*

Proof. For a fixed $U \in \tau_Y$, taking the union over all $x \in f^{-1}(U)$ of the corresponding $V_{x,U}$'s yields $f^{-1}(U)$, showing that $f^{-1}(U)$ is open. On the other hand, if f is continuous, and $U \in \tau_Y$, then $f^{-1}(U) \in \tau_X$. Since $\mathcal{B}$ is a basis, this means that $f^{-1}(U)$ is a union of elements of $\mathcal{B}$, and so every element of $f^{-1}(U)$ is contained in one such. □

Lemma 1.2.7. *Let (X, τ_X) and (Y, τ_Y) be topological spaces, let $f : X \to Y$ be a map of underlying sets, and let $\mathcal{B}$ be a basis for the topology τ_Y. Then f is continuous if and only if $f^{-1}(U)$ is open for every $U \in \mathcal{B}$.*

Proof. One direction is immediate from the definitions. To demonstrate the other, suppose that preimages of basis elements are open, and let $U \in \tau_Y$. Then Choose basis elements $\{U_i\}_{i\in I}$ such that

$$\bigcup_{i\in I} U_i = U.$$

Then

$$f^{-1}(U) = f^{-1}\left(\bigcup_{i\in I} U_i\right) = \bigcup_{i\in I} f^{-1}(U_i)$$

which is a union of open sets, and thus open. □

To deal more easily with categorical definitions, we need *commutative diagrams*. The definition given here is preliminary, and will be revised substantially once we discuss category theory more formally.

Definition 1.2.8. A *diagram* in Top is a directed graph[12] whose vertices are labeled by objects of Top (topological spaces) and whose arrows are labeled by morphisms (continous maps) between the appropriate objects. A diagram is said to *commute* if, for any two directed paths between the same pair of vertices, the composites of the associated morphisms are equal.[13]

Examples 1.2.9.

1. There are several trivial examples of commutative diagrams: any diagram associated to the graph with a single vertex and no arrows trivially commutes, as does any diagram associated to the graph with two vertices and a single arrow between them.

2. The simplest non-trivial example of a commutative diagram consists of three topological spaces X, Y, and Z and three morphisms (continuous maps) f, g and h assigned to a triangle-shaped graph. We draw this commutative diagram as

$$\begin{tikzcd} & Y \arrow[dr, "f"] & \\ X \arrow[ur, "g"] \arrow[rr, "h"'] & & Z \end{tikzcd}$$

The condition that this diagram commutes mean that the composite $f \circ g$ (associated to the upper directed path) and the composite h associated to the lower, 1-edge, path, must be equal.

[12] In principle, we can interpret the term *directed graph* broadly here, allowing parallel edges. In practice, the *commutative* diagrams we draw will almost always lack these. This is because the condition that a diagram commutes means that parallel edges must be assigned the same morphism.

[13] A better, and less long-winded, definition of a commutative diagram is given in Chapter 3 as Definition 3.2.1.

3. Another common shape of commutative diagram is the *commutative square*:

$$\begin{tikzcd} X \arrow[r, "g"] \arrow[d, "h"'] & Y \arrow[d, "f"] \\ Z \arrow[r, "k"'] & W \end{tikzcd}$$

The condition that this diagram commute is that $f \circ g = k \circ h$.

Initially, we will give definitions both using equations of continuous maps and (in the sidebar) the corresponding commutative diagrams. Eventually, as we become more accustomed to categorical terminology and notation, we will switch to using commutative diagrams almost exclusively.

1.2.1 Products and coproducts

One of the themes we will encounter in this class is that categorical definitions come in pairs, related by reversing the directions of all of the morphisms involved. These definitions are said to be *dual* to one another, and the name of one is usually constructed by affixing a "co-" to the front of the name of the other. Our first example of this phenomenon is products and coproducts. We will return to products and coproducts more formally Example 3.6.6 (2) as examples of limits and colimits in a category.

The categorical definition of the product is phrased entirely in terms of the objects and morphisms of Top, but as we will see, uniquely specifies the same space.

Definition 1.2.10 (Universal property of the product topology)**.** Let I be a set, and $\{(X_i, \tau_i)\}_{i \in I}$ a collection of topological spaces indexed by I. A *product* of the X_i is a topological space Y together with a collection of continuous maps $\{\ p_i : Y \longrightarrow X_i\ \}_{i \in I}$, such that, for any other space Z and collection of continuous maps $\{\ q_i : Z \longrightarrow X_i\ \}_{i \in I}$ there is a unique continous map $f : Z \to Y$ such that, for every $i \in I$,[14]

$$p_i \circ f = q_i.$$

Definition 1.2.11 (Universal property of the coproduct topology)**.** Let I be a set, and $\{(X_i, \tau_i)\}_{i \in I}$ a collection of topological spaces indexed by I. A *coproduct* of the X_i is a topological space Y together with a collection of continuous maps $\{\ Y \longleftarrow X_i : \iota_i\ \}_{i \in I}$, such that, for any other space Z and collection of continuous maps $\{\ Z \longleftarrow X_i : h_i\ \}_{i \in I}$ there is a unique continous map $f : Y \to Z$ such that, for every $i \in I$,[15]

$$f \circ \iota_i = h_i.$$

There are three important points about these definitions to notice:

1. The continuous maps p_i are *part of the data of a product.*

2. The definition of a coproduct is obtained from the definition of a product by reversing the directions of all of the arrows, and keeping everything else the same.

[14] Equivalently, this is the condition that all of the diagrams

$$\begin{tikzcd} & Y \arrow[dr, "p_i"] & \\ Z \arrow[ur, "f"] \arrow[rr, "q_i"'] & & X_i \end{tikzcd}$$

commute. The definition of the product is also sometimes written, schematically, as a commutative diagram

$$\begin{tikzcd} & Z \arrow[d, dashed, "\exists!"] \arrow[ddl, "q_i"'] \arrow[ddr, "q_j"] & \\ & Y \arrow[dl, "p_i"] \arrow[dr, "p_j"'] & \\ X_i & \cdots & X_j \end{tikzcd}$$

Here, the dashed arrow must exist uniquely for Y to be a product of the X_i.

[15] Equivalently, this is the condition that all of the diagrams

$$\begin{tikzcd} & Y \arrow[dl, "f"'] & \\ Z & & X_i \arrow[ul, "\iota_i"'] \arrow[ll, "h_i"] \end{tikzcd}$$

commute. The definition of the product is also sometimes written, schematically, as a commutative diagram

$$\begin{tikzcd} & Z & \\ & Y \arrow[u, dashed, "\exists!"] & \\ X_i \arrow[ur, "\iota_i"'] \arrow[uur, "h_i"] & \cdots & X_j \arrow[ul, "\iota_j"] \arrow[uul, "h_j"'] \end{tikzcd}$$

Here, the dashed arrow must exist uniquely for Y to be a product of the X_i.

3. We have defined **a** product, rather than **the** product, and a priori, our definition may not specify a unique space (or any space at all). Our next order of business will be to show that products exist, and are unique up to unique homeomorphism. Since homeomorphic spaces are effectively the same, this means we may safely speak of **the** categorical product.

The introduction of the product topology introduces some ambiguity to the notation $\prod_{i=1}^n X_i$ when the topologies on the X_i are induced by metrics. It is, however, not hard to prove that if $\{(X_i, d_i)\}_{i=1}^n$ are metric spaces, and τ_i are the corresponding topologies, then the product metric

$$d(x, y) = \left(\sum_{i=1}^n d_i(x_i, y_i)^2\right)^{\frac{1}{2}}$$

induces the product topology.

In particular, when we write $\mathbb{R}^n$, we can equivalently view the topology as being induced by the standard metric, or as being the product topology induced by viewing $\mathbb{R}^n$ as the product of n copies of $\mathbb{R}$.

Construction 1.2.12. Let I be a set, and $\{(X_i, \tau_i)\}_{i\in I}$ a collection of topological spaces indexed by I.

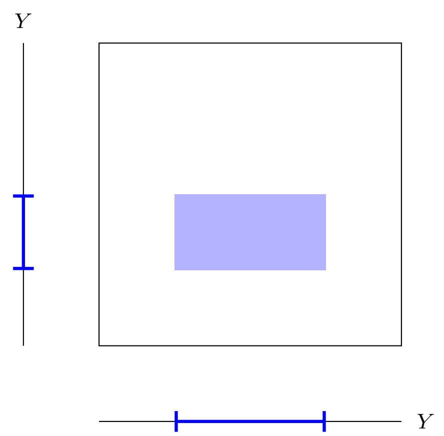
1. The *product* of the X_i, denoted by $\prod_{i\in I} X_i$, is the topological space whose underlying set is the Cartesian product of the sets X_i and whose topology is generated by the basis

$$\left\{\prod_{i\in I} U_i \;\middle|\; U_i \in \tau_i \text{ and } U_i \neq X_i \text{ for finitely many } i \in I\right\}$$

2. The *coproduct* of the X_i, denote by $\coprod_{i\in I} X_i$, is the topological space whose underlying set is the disjoint union, and whose topology consists of those subsets $U \subset \coprod X_i$ such that $U \cap X_i$ is open for all $i \in I$.

We leave it to the reader to verify that the set in part (1) is, in fact, a basis.

The conventional picture of basis elements in the product topology on $X \times Y$ depicts X and Y as intervals and U and V as open intervals.



Proposition 1.2.13. *Let I be a set and $\{X_i\}_{i\in I}$ a collection of spaces indexed by I. Let $\prod_{i\in I} X_i$ be the product space as given in Construction 1.2.12, and denote by*

$$p_j : \prod_{i\in I} X_i \longrightarrow X_j$$

denote the projection map of underlying sets.

1. *The maps p_i are continuous.*
2. *The space $\prod_{i\in I} X_i$ together with the morphisms $\{p_i\}_{i\in I}$ satisfy Categorical Definition 1.2.10, i.e., form a product in Top.*

Proof. To prove part (1), we use Lemma 1.2.6. For an open subset $U \subset X_i$,

$$p_i^{-1}(U) = \prod_{j\in I} V_j$$

where $V_i = U$ and $V_j = X_j$ for $j \neq i$. As such, p_i is continuous.

To prove part (2), suppose we are given another space Z with continuous maps $q_i : Z \to X_i$. There is a unique map of sets

$$\begin{aligned} f : Z &\longrightarrow \prod_{i\in I} X_i \\ z &\longmapsto (q_i(z))_{i\in I} \end{aligned}$$

so it will suffice for us to check that it is continuous. Using Lemma 1.2.7, it suffices for us to check this on basis elements. Consider a basis element $U = \prod_{i\in I} U_i$, and let $i_1, \dots, i_k$ be the indices so that $U_{i_j} \neq X_{i_j}$. Then

$$f^{-1}(U) = \bigcap_{j=1}^{k} q_{i_j}^{-1}(U_{i_j})$$

which is a finite intersection of open sets, and thus open. □

Proposition 1.2.14. *Let $\{X_i\}_{i\in I}$ be an index collection of spaces, and let Y, $\{p_i : Y \to X_i\}_{i\in I}$ and Z, $\{q_i : Z \to X_i\}_{i\in I}$ be two products of the X_i in Top. Then there is a unique homeomorphism $f : Z \to Y$ such that $p_i \circ f = q_i$ for all $i \in I$.*

Roughly, diagrammatically, we can draw this argument as

$$\begin{tikzcd} Y \arrow[r, "\exists! f"] \arrow[dr, "p_i"'] \arrow[rr, bend left, "\exists! \mathrm{id}_Y"] & Z \arrow[r, "\exists! g"] \arrow[d, "q_i"] & Y \arrow[dl, "p_i"] \\ & X_i & \end{tikzcd}$$

Proof. Since Y is a product of the X_i's in Top, there is a unique continuous map $f : Z \to Y$ satisfying the desired commutativity condition. Thus it suffices to see that f is a homeomorphism. Similarly, since Z is a product of the X_i's in Top there are unique maps $g : Y \to Z$ such that $q_i \circ g = p_i$ for all $i \in I$.

This means that $f \circ g$ is a continuous map satisfying the condition

$$p_i \circ (f \circ g) = q_i \circ g = p_i.$$

However, the identity id_Y on Y also satisfies this condition. Since Y is a product, there is a unique continuous map satisfying this condition, and so $f \circ g = \mathrm{id}_Y$. An identical argument shows that $g \circ f = \mathrm{id}_Z$, and so f is a homeomorphism. □

Exercise 1.2.15. State and prove the analogues of Propositions 1.2.13 and 1.2.14 for the coproduct of topological spaces.

1.2.2 Subspaces and quotients

The next pair of constructions we consider are the so-called *subspace* and *quotient* topologies. Again, we will give categorical definitions and point-set definitions for each. Similarly, we will also write these definitions in such a way as to make it clear that they are *dual*: each definition is obtained from the other by reversing the directions of all of the arrows.[16]

To ease the categorical side of our definitions, we will introduce another category.

Definition 1.2.16. For sets X and Y, we denote the set of functions from X to Y by $\mathsf{Set}(X, Y)$.[17] The *category* Set *of sets and functions* consists of:

1. The collection[18] of all sets. These will be referred to as the *objects* of Set. In a mild abuse of notation, we will often write $X \in \mathsf{Set}$ to mean "X is a set"

2. For every pair of sets $X, Y \in \mathsf{Set}$, the set

$$\mathsf{Set}(X, Y)$$

of functions from X to Y. These functions are also called the *morphisms* of Set.

[16] Unlike with products and coproducts, we will not return directly to these subspaces and quotients when we discuss category theory. However, it turns out that they are examples of *equalizers* and *coequalizers*, see Example 3.6.6 (5).

[17] This is also sometimes denoted by $\mathrm{Hom}_{\mathsf{Set}}(X, Y)$.

[18] As was the case for Top, this is not a set.

3. For every triple of spaces $X, Y, Z \in \mathsf{Set}$, the composition map

$$\mathsf{Set}(Y,Z) \times \mathsf{Set}(X,Y) \xrightarrow{-\circ-} \mathsf{Set}(X,Z)$$

which sends a pair of composable functions to their composite.

Notice that the composition map is associative and has units in the sense that, for $f : X \to Y$, $f \circ \mathrm{id}_X = f = \mathrm{id}_Y \circ f$.

We can also now define our first *functor*.[19]

[19] Note that, as with categories, we have not given a formal definition of functors yet.

Definition 1.2.17. The *forgetful functor from* Top *to* Set, written $U : \mathsf{Top} \to \mathsf{Set}$ consists of the following assignments.

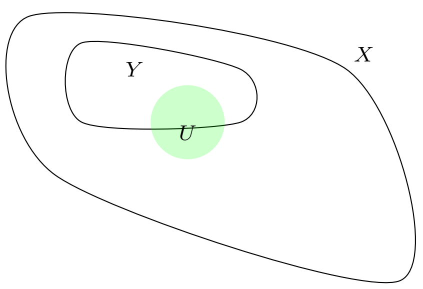
1. U sends a topological space $X \in \mathsf{Top}$ to the underlying set $U(X) \in \mathsf{Set}$. That is, U forgets the topology on X.
2. U sends a continuous map $f : X \to Y$ in Top to the underlying function between sets $U(f) : U(X) \to U(Y)$ in Set.

Notice that U preserves composition and identities: $U(\mathrm{id}_X) = \mathrm{id}_{U(X)}$ and $U(f \circ g) = U(f) \circ U(g)$.

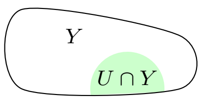
There are other useful functors between these two categories, but we will defer discussion of these to a later section.

—— Third lecture on 20. April

A schematic depiction of the open sets in the supspace topology is as follows. In the first drawing we have a space X, a subset Y, and an open set U of X.



In the second, we have the corresponding open subset $Y \cap U$ of Y in the subspace topology



Definition 1.2.18 (Universal property of the subspace topology)**.** Let $X \in \mathsf{Top}$ and let $\iota : Y \to U(X)$ be an injective morphism of Set. We call $(Y, \tau) \in \mathsf{Top}$ a *subspace topology on* Y if ι is continuous with respect to τ and, for every continuous map $f : Z \to X$ such that $U(f) : U(Z) \to U(X)$ factors through ι, the induced map $Z \to (Y, \tau)$ is continuous.[20]

[20] Notice that this is secretly saying that there exists a unique continuous map satisfying a condition: there is a unique continuous map $g : Z \to (Y, \tau)$ such that $\iota \circ U(g) = f$.

Construction 1.2.19. Let X be a topological space, and let $\iota : Y \to U(X)$ be an injective map in Set — effectively a subset of the underlying set of X. The *subspace topology* on Y is the topology $\tau_{Y \subset X}$ defined by requiring that V is open in Y if and only if there is an open set U of X such that $\iota^{-1}(U) = V$.[21]

[21] Note that when Y is truly a subset, this reads $Y \cap U = V$.

Proposition 1.2.20. *Let X be a topological space, and $\iota : Y \to U(X)$ an injective map of sets.*

1. *The subspace topology $(Y, \tau_{Y \subset X})$ of Construction 1.2.19 satisfies Definition 1.2.18.*
2. *If two topologies on Y satisfy Definition 1.2.18, then they are equal.*

Proof. It is immediate from the definitions that $\iota : (Y, \tau_{Y \subset X}) \to X$ is continuous. Suppose given a continuous map $g : Z \to X$ such that $U(g)$ factors through ι as $\iota \circ h = g$. We claim that $h : Z \to (Y, \tau_{Y \subset X})$ is continuous. To see this, let $U \in \tau_{Y \subset X}$. Then there is a $V \in \tau_X$ with $\iota^{-1}(V) = U$. Since $g = \iota \circ h$, we have that $g^{-1}(V) = h^{-1}(\iota^{-1}(V)) = h^{-1}(U)$, which is open by the continuity of g. Thus, h is continuous.

To prove the second claim, suppose that τ_1 and τ_2 are topologies on Y satisfying Categorical Definition 1.2.18. Since $\iota : Y \to U(X)$ clearly factors through Y (as id_Y), it follows that $\mathrm{id}_Y : (Y, \tau_1) \to (Y, \tau_2)$ and $\mathrm{id}_Y : (Y, \tau_2) \to (Y, \tau_1)$ are both continuous. Thus $\tau_1 \subset \tau_2$ and $\tau_2 \subset \tau_1$, so the topologies are equal. □

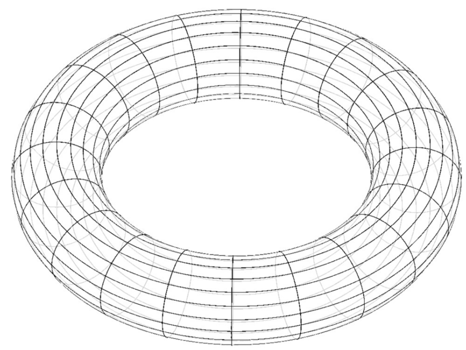
As with the product topology, there is an potential ambiguity when talking about subspaces of metric spaces. However, it is once again easy to show that the restriction of a metric to a subset induces the same topology as the subspace topology.

Example 1.2.21. For $n \geq 0$, the *n-sphere* S^n is the subset

$$S^n := \{x \in \mathbb{R}^{n+1} \mid |x| = 1\} \subset \mathbb{R}^{n+1}$$

equipped with the subspace topology inherited from the standard topology on $\mathbb{R}^n$. Equivalently, we will often view the *circle* S^1 as a subspace of $\mathbb{C}$ (with the metric topology).

When drawn, the torus is a surface which looks a bit like a doughnut:



Example 1.2.22. The *torus* $T^2 = S^1 \times S^1$ is another fundamental example of a topological space. We can view the torus either as the product of two copies of the circle (with the product topology) or as a subspace of $\mathbb{R}^2$. It is a useful exercise to show that these definitions coincide.

Definition 1.2.23. A continuous map $f : X \to Y$ between topological spaces is called an *embedding* if f induces a homeomorphism between X and the image of X, where the latter is equipped with the subspace topology.

In effect, an embedding $f : X \to Y$ identifies X with a subspace of Y.

Proposition 1.2.24. *Let $f : X \to Y$ be a continuous injective map of topological spaces and suppose that f is either closed or open. Then f is an embedding.*

Proof. We give the proof for an open map, the proof for a closed map is similar. Suppose that f is open. Then for any open set $U \subset X$, $f(U) = f(U) \cap f(X)$, and so the continuous map onto the image $f(X)$

$$\widetilde{f} : X \longrightarrow f(X)$$

is open as well. Since $\widetilde{f}$ is continuous, open, and bijective by assumption, it follows that $\widetilde{f}$ is a homeomorphism, completing the proof. □

The dual definition to the subspace topology is the *quotient* topology. Notice that this definition really does arise by simply reversing the arrows in Definition 1.2.18.

Definition 1.2.25 (Universal property of the quotient topology)**.** Let $X \in \mathsf{Top}$, and let $\pi : U(X) \to Y$ be a surjective[22] morphism in Set. We call (Y, τ) a *quotient topology* if π is continuous with respect to τ, and, for every morphism $f : X \to Z$ in Top such that $U(f)$ factors through π, the induced map $(Y, \tau) \to Z$ is continuous.[23]

Construction 1.2.26. Let X be a topological space, let Y be a set, and let $\pi : X \to Y$ be a surjective map of sets. The *quotient topology on Y* is defined by declaring $U \subset Y$ to be open if and only if $f^{-1}(U)$ is open.

[22] In Set, surjectivity is formally dual to injectivity in a way we will not discuss in this class. The interested student can read about *epimorphisms* and *monomorphisms* for further details.

[23] We will often call a continuous map $\pi : X \to Y$ which satisfies this universal property (i.e., such that the topology on Y is the quotient topology) a *quotient map*.

Proposition 1.2.27. *Let $X \in \mathsf{Top}$, and let $\pi : U(X) \to Y$ be a surjective map of sets.*

1. *The quotient topology of Construction 1.2.26 satisfies Definition 1.2.25.*
2. *If τ_1 and τ_2 are two topologies satisfying Definition 1.2.25, then $\tau_1 = \tau_2$.*[24]

Proof. Exercise. □

[24] Previously, we said that our categorical definitions specify a topological up to unique homeomorphism. Here, however, we see that we obtain *equal* topologies on the same space. This is because we wrote Definitions 1.2.18 and 1.2.25 in such a way that the underlying set and the underlying map of sets is fixed.

Notation 1.2.28.

1. The most common source of quotient maps comes from equivalence relations. If X is a topological space, and $\sim$ is an equivalence relation on X, we will typically write $X_{/\sim}$ for the quotient set equipped with the quotient topology inherited via the quotient map $X \to X_{/\sim}$.
2. As a special case of the previous example, if G is a group which acts on a topological space, we obtain an equivalence relation by setting $x \sim g \cdot x$ for $g \in G$. In this case, we write the quotient space as $X_{/G}$.

It will be useful for us later to quickly recognize quotient maps.

Lemma 1.2.29. *Let $\pi : X \to Y$ be a continuous, surjective map of topological spaces. If π is either an open map or a closed map, then π is a quotient map.*

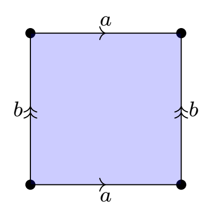
Proof. First suppose π is an open map, and let $U \subset Y$ such that $\pi^{-1}(U)$ is open. Then $\pi(\pi^{-1}(U)) = U$, and since π is an open map, U is open. Thus, the topology on Y is the quotient topology.

Now suppose π is a closed map, and let $U \subset Y$ such that $\pi^{-1}(U)$ is open. Then $\pi^{-1}(U)^c$ is closed. Since π is surjective, $\pi(\pi^{-1}(U)^c) = U^c$, and since π is a closed map, this implies U^c is closed. Thus U is open, and the topology on Y is the quotient topology. □

Lemma 1.2.30. *A product of continuous open surjections is open.*

Proof. Let $\{X_i\}_{i\in I}$ and $\{Y_i\}_{i\in I}$ be collections of spaces indexed by I, and let $f_i : X_i \to Y_i$ be continuous open surjections. Write $f : \prod_{i\in I} X_i \to \prod_{i\in I} Y_i$ for the product map which sends $(x_i)_{i\in I} \mapsto (f(x_i))_{i\in I}$. If $\prod_{i\in I} U_i$ is a basic open in $\prod_{i\in I} X_i$, then
$$f\left(\prod_{i\in I} U_i\right) = \prod_{i\in I} f(U_i)$$
is a basic open (the requirement that at most finitely many factors are not equal to the whole space is satisfied by surjectivity). Taking unions implies that f maps open sets to open sets. □

We will often draw pictures meant to convey quotients in terms of equivalence relations. For example, the quotient of $[0,1] \times [0,1]$ from Example 1.2.31 might be drawn as



The labels tell us which intervals should be identified homeomorphically, and the arrows tell us whether to reverse the direction when we glue or not. The corresponding equivalence relation is generated by $(s,0) \sim (s,1)$ and $(0,t) \sim (1,t)$.

Example 1.2.31 (The torus as a quotient in two ways)**.** Consider the surjective, continuous map
$$\begin{aligned} p: \quad \mathbb{R}^2 &\longrightarrow S^1 \times S^1 \\ (x,y) &\longmapsto (e^{2\pi i x}, e^{2\pi i y}). \end{aligned}$$

It is sufficient for us to show that the exponential map is continuous, surjective, and open. We know the former facts from analysis. We turn to seeing that the exponential map is open. Let $x \in S^1$ be a point in the circle, and let $t_0 \in \mathbb{R}$ be such that $e^{2\pi i t} = x$. Then for $|y - x|$ sufficiently small, we have that the function $y \mapsto \frac{1}{2\pi}(t_0 + \arccos(\mathrm{Re}(y)))$ is a continuous section to $t \mapsto e^{2\pi i t}$. A brief calculation shows that

$$|e^{2\pi_i t} - e^{2\pi i t_0}| = \sqrt{2}|\sin(2\pi|x-y|)|$$

Thus, for sufficiently small $r \geq 0$, $B_r(t)$ is mapped homeomorphically to $B_{\sqrt{2}|\sin(2\pi r)|}(x)$. Since the open balls of radius below any fixed value greater than 0 form a basis for the topology on $\mathbb{R}$, it follows that p is an open map. As such, the torus is a quotient space of $\mathbb{R}^2$ with quotient map p.

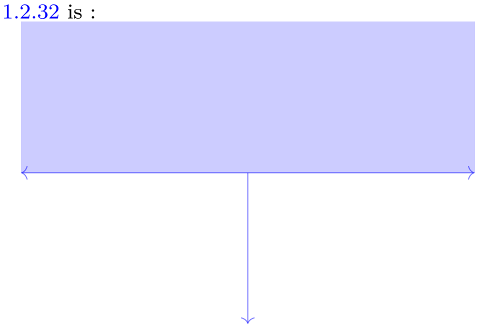
Consider the composite

$$[0,1] \times [0,1] \hookrightarrow \mathbb{R}^2 \xrightarrow{p} S^1 \times S^1$$

of p with the inclusion, which we will temporarily term $\overline{p}$. A similar argument to the above can be used to show that this is a closed map, so that $\overline{p}$ is a quotient map.

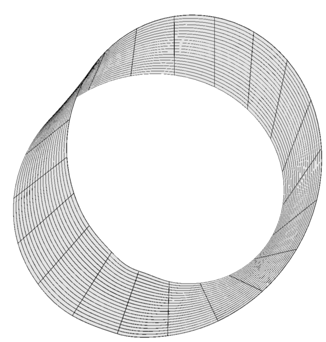
⚠ The map p is not closed, and the map $\overline{p}$ is not open.

A schematic depiction of the space from Example 1.2.32 is :



Example 1.2.32. ⚠ A quotient map can be neither open nor closed. Consider the subspace $\mathbb{R} \times \{0\} \cup [0,\infty) \times \mathbb{R} \subset \mathbb{R}^2$. Then the projection onto the first coordinate is a quotient map that is clearly neither open nor closed.

The Möbius band is a rare example of a *non-orientable* surface which is easy to visualize:



Indeed, one can easily construct a Möbius band from paper or fabric.

Examples 1.2.33.

1. Consider $\mathbb{R}$ equipped with the metric topology, and define an equivalence relation on $\mathbb{R}$ by $x \sim y$ if and only if there exists $n \in \mathbb{Z}$ with $x = y + n$. Then the quotient space $\mathbb{R}_{/\sim}$ is homeomorphic to S^1. This is effectively the same as the quotient map explored in Example 1.2.31.

2. Let $X = [0,1] \times [-1,1] \subset \mathbb{R}^2$, equipped with the subspace topology. Define an equivalence relation on X by setting $(0,x) \sim (1,-x)$. The quotient space $(X_{/\sim}, \tau_\sim)$ is called the *Möbius band*.

Example 1.2.34. Consider $\mathbb{R}^{n+1} \setminus \{0\}$ with the subspace topology coming from $\mathbb{R}^{n+1}$. Define an equivalence relation on $\mathbb{R}^{n+1} \setminus \{0\}$ by $x \sim \lambda x$ for every $\lambda \in \mathbb{R} \setminus \{0\}$. We define the *$n$-dimensional real projective space* $\mathbb{R}P^n$ to be the quotient space $(\mathbb{R}^{n+1} \setminus \{0\})_{/\sim}$.

Note that we can also consider $\mathbb{R}P^n$ as the quotient of the unit sphere S^n by the equivalence relation $x = -x$.

—— Fourth lecture on 21. April

1.2.3 Pullbacks and pushouts

Our final set of constructions building new spaces from old (at least for the time being) are pullbacks and pushouts. These are again dual constructions — each categorical definition obtained from the other by reversing all of the arrows. In this case, however, we have a new an interesting feature: each of our new constructions can be obtained by performing two of our previous constructions in sequence.[25]

Our definition of pullback will be the first in which we give the definition using commutative diagrams, rather than equations of continuous maps.

Categorical Definition 1.2.35 (Universal property of the pullback)**.** Let $X, Y, Z \in$ Top, and let $f : X \to Z$ and $g : Y \to Z$ be morphisms of Top. A *pullback* of the diagram

$$\begin{tikzcd} & Y \arrow[d, "g"] \\ X \arrow[r, "f"'] & Z \end{tikzcd}$$

is an object $P \in \mathsf{Top}$ equipped with morphisms $p_1 : P \to X$ and $p_2 : P \to Y$ such that the diagram

$$\begin{tikzcd} P \arrow[r, "p_2"] \arrow[d, "p_1"'] & Y \arrow[d, "g"] \\ X \arrow[r, "f"'] & Z \end{tikzcd}$$

commutes, and satisfying the property that, for any $W \in \mathsf{Top}$ equipped with morphisms $q_1 : W \to X$ and $q_2 : W \to Y$ such that the diagram

$$\begin{tikzcd} W \arrow[r, "q_2"] \arrow[d, "q_1"'] & Y \arrow[d, "g"] \\ X \arrow[r, "f"'] & Z \end{tikzcd}$$

commutes, there is a unique morphism $u : W \to P$ such that the diagram

$$\begin{tikzcd} W \arrow[dr, "u"] \arrow[drr, bend left, "q_2"] \arrow[ddr, bend right, "q_1"'] & & \\ & P \arrow[r, "p_2"] \arrow[d, "p_1"'] & Y \arrow[d, "g"] \\ & X \arrow[r, "f"'] & Z \end{tikzcd}$$

commutes.

[25] As before, we will return to pullbacks and pushouts when we discuss categories. See Example 3.6.6 3.

We can compactly summarize the universal property with the single commutative diagram

$$\begin{tikzcd} W \arrow[dr, dashed, "\exists!"] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\ & P \arrow[r, "p_2"] \arrow[d, "p_1"'] & Y \arrow[d, "g"] \\ & X \arrow[r, "f"'] & Z \end{tikzcd}$$

We immediately give the dual definition — *pushouts*.

Categorical Definition 1.2.36 (Universal property of the pushout)**.** Let $X, Y, Z \in$ Top and let $i : Z \to X$ and $j : Z \to Y$ be morphisms of Top. A *pushout* of X and Y over Z is an object $P \in \mathsf{Top}$ equipped with morphism $k_1 : X \to P$ and $k_2 : Y \to P$ satisfying the universal property expressed by the commutative diagram

$$\begin{tikzcd} Z \arrow[r, "i"] \arrow[d, "j"'] & X \arrow[d, "k_1"] \arrow[ddr, bend left] & \\ Y \arrow[r, "k_2"'] \arrow[drr, bend right] & P \arrow[dr, dashed, "\exists!"] & \\ & & W \end{tikzcd}$$

Construction 1.2.37. Let X, Y, and Z be topological spaces, and let $f : X \to Z$ and $g : Y \to Z$ be continuous maps. The *pullback*[26] $X \times_Z Y$ of X and Y over Z[27] is the subspace of the product

$$X \times_Z Y := \{(x, y) \in X \times Y \mid f(x) = g(y)\}.$$

Construction 1.2.38. Let X, Y, and Z be topological spaces, and let $i : Z \to X$ and $j : Z \to Y$ be continuous maps. The *pushout* $X \amalg_Z Y$ of X and Y over Z[28] is the quotient of the coproduct

$$(X \amalg Y)_{/\sim}$$

by the equivalence relation generated by $i(z) \sim j(z)$ for $z \in Z$.

Proposition 1.2.39. *Let X, Y, and Z be topological spaces, and let $f : X \to Z$ and $g : Y \to Z$ be continuous maps.*

1. *For any two pullbacks P, $p_1 : P \to X$, $p_2 : P \to Y$, and Q, $q_1 : Q \to X$ and $q_2 : Q \to Y$, there is a unique homeomorphism from P to Q which commutes with the p_i and q_i.*

2. *Construction 1.2.37 satisfies Definition 1.2.35.*

Proof. Part (1) is the usual routine. By Categorical Definition 1.2.35, there are unique morphisms $h : P \to Q$ and $k : Q \to P$ which commute with the p_i's and q_i's. The composite $k \circ h$ commutes with the p_i's, and so by the uniqueness guaranteed by Categorical Definition 1.2.35 must equal id_P. Similarly, $h \circ k = \mathrm{id}_Q$, and so h and k are mutually inverse homeomorphisms.

Part (2) follows by considering the corresponding statements for products and subspaces. Let $W \in \mathsf{Top}$, and let $q_1 : W \to X$ and $q_1 : W \to Y$ be two maps. By Categorical Definition 1.2.10, there is a unique map $u : W \to X \times Y$ that commutes with the p_i's and q_i's. Moreover, the underlying map of sets of u factors through the subset $X \times_Z Y$ if and only if $f \circ q_1 = g \circ q_2$, and so by Categorical Definition 1.2.18, there is a unique continuous map $\overline{u} : W \to X \times_Z Y$ making the desired diagram commute. □

Exercise 1.2.40. Formulate and prove the dual statement (i.e., the analogue for pushouts) of Proposition 1.2.39.

The pushout has a particularly nice visual interpretation as a *gluing*: The idea is that we first take $X \amalg Y$ — a space which has two unrelated pieces, and then glue them together according to the maps provided.

Examples 1.2.41.

1. Consider the diagram

$$\begin{tikzcd} \{0,1\} \arrow[r, hook] \arrow[d] & {[0,1]} \\ * & \end{tikzcd}$$

[26] Also sometimes called the *fiber product* or *fibered product*.

[27] ⚠ This is misleading terminology and notation, though extremely common, since the pullback depends on f and g, rather than only depending on the spaces X, Y, and Z.

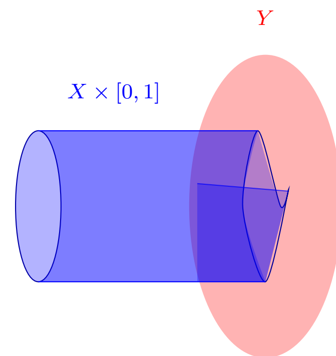
[28] ⚠ The previous warning about pullbacks applies here as well. The notation is misleading, and the pushout fundamentally depends on the maps i and j.

where $*$ represents the one-point space, and $\{0,1\}$ is equipped, equivalently, with either the discrete topology, or the subspace topology inherited from $[0,1]$. Then the pushout is (homeomorphic to) $[0,1]$ quotiented by the relation $0 \sim 1$, and thus is homeomorphic to S^1.

2. Let $f : X \to Y$ be a continuous map of topological spaces. The *mapping cylinder* M_f of f is the pushout of the diagram

$$\begin{tikzcd} X \times \{0\} \arrow[r, hook] \arrow[d, "f"'] & X \times [0,1] \\ Y & \end{tikzcd}$$

We can picture the mapping cylinder as follows



Remark 1.2.42. Effectively, the universal property of pushouts (Definition 1.2.36) is the reason that we can define functions piecewise. If, for example, we note that for any $b \in \mathbb{R}$

$$\mathbb{R} \cong (-\infty, b] \amalg_{\{b\}} [b, \infty)$$

then Definition 1.2.36 tells us that we can define a continuous function f on all of $\mathbb{R}$ uniquely by specifying continuous functions on $(-\infty, b]$ and $[b, \infty)$, and checking that they agree on b.

—— fifth lecture on 27. April

1.2.4 Directed (co)limits (direct and inverse limits)

Our final dual pair of constructions are fundamentally infinite in nature. These two constructions, and particularly directed colimits, are the key tool by which infinite, iterative constructions of topological spaces are achieved, as we will see in the next section.[29]

[29] See Example 3.6.6 (4) for the categorical approach to this construction.

Our notation for directed and inverse systems is meant to suggest something that we will see later: that these *functors* from $\mathbb{N}$ to Top. Compare with Definition 1.2.17.

Definition 1.2.43. A *directed system* of topological spaces consists of

1. An object $F_i \in \mathsf{Top}$ for each $i \in \mathbb{N}$.
2. Whenever $i \leq j$ a morphism $f_{i,j} : F_i \to F_j$ in Top.

subject to the conditions

1. The morphisms $f_{i,i}$ are identity morphisms.
2. For $i \leq j \leq k$, we have
$$f_{i,k} = f_{j,k} \circ f_{i,j}.$$

Note that these conditions imply that the directed system is completely and uniquely determined by the spaces F_i and the morphisms $f_{i,i+1} : F_i \to F_{i+1}$. We will abbreviate the latter by f_i, and write $F := (\{F_i\}, \{f_i\})$ for the directed system.

An *inverse system* of topological spaces is the dual notion, given by reversing the directions of all of the $f_{i,j}$'s.

Definition 1.2.44. A *cone under* a directed system F is an object $L \in \mathsf{Top}$ equipped with morphisms $\ell_i : F_i \to L$ for all $i \in \mathbb{N}$, and such that, for all $i \leq j$ in $\mathbb{N}$, $\ell_j \circ f_{i,j} = \ell_i$. Equivalently, this can be visualized as a commutative diagram

$$\begin{tikzcd}
F_1 \arrow[r, "f_1"] \arrow[drrr, "\ell_1"'] & F_2 \arrow[r, "f_2"] \arrow[drr, "\ell_2"] & F_3 \arrow[r] \arrow[dr, "\ell_3"] & \cdots \\
 & & & L
\end{tikzcd}$$

A *colimit*[30] of the directed system F is a cone $(L, \{\ell_i\})$ such that, for every other cone $(M, \{m_i\})$ under F, there is a unique morphism $g : L \to M$ such that, for every $i \in \mathbb{N}$, $g \circ \ell_i = m_i$.

[30] In older references, this is sometimes called a *direct limit*, but this terminology should be avoided if at all possible.

A *cone over* an inverse system F is the dual notion to a cone under a directed system, and a *limit* of an inverse system is the dual notion of a colimit of a directed system.

We can now give an explicit characterization of direct and inverse limits.

Construction 1.2.45. Given a directed system F, we define a space $\operatorname{colim}_{\mathbb{N}} F$ to be the quotient space

$$\operatorname*{colim}_{\mathbb{N}} F := \left(\coprod_{i \in \mathbb{N}} F_i \right)_{\sim}$$

by the equivalence relation generated by $x \sim f_{i,j}(x)$ for $i, j \in \mathbb{N}$ and $x \in F_i$. There is a canonical cone $(\operatorname{colim}_{\mathbb{N}} F, \{\iota_i\})$ under F whose components are given by the composites

$$F_i \longrightarrow \textstyle\coprod_{i \in \mathbb{N}} F_i \longrightarrow \operatorname{colim}_{\mathbb{N}} F$$

of the quotient map with the summand inclusions.

Proposition 1.2.46. *Given a directed system F, the cone $(\operatorname{colim}_{\mathbb{N}} F, \{\iota_i\})$ is a colimit of F.*

Proof. Given another cone $(L, \{\ell_i\})$ under F, we need to construct a unique morphisms $g : \operatorname{colim}_{\mathbb{N}} \to L$ which commutes with the ℓ_i's and ι_i's. We first note that, by Definition 1.2.11 (the universal property of the coproduct), there is a unique morphism

$$\widetilde{g} : \textstyle\coprod_{i \in \mathbb{N}} F_i \longrightarrow L$$

which commutes with the product. It therefore suffices for us to show that this descends to a unique morphism on the quotient. However, by Definition 1.2.25 (the universal property of the quotient), it suffices to show that $\widetilde{g}$ descends uniquely on the level of sets, i.e., that $\widetilde{g}$ respects the equivalence relation. However, by construction, we have that, for $x \in F_i$ and $i < j$,

$$\widetilde{g}(\iota_j(f_{i,j}(x))) = \ell_j(f_{i,j}(x)) = \ell_i(x) = \widetilde{g}(\iota_i(x))$$

and so $\widetilde{g}$ respects the equivalence relation, completing the proof. $\square$

Construction 1.2.47. Given an inverse system F we define a space $\lim_{\mathbb{N}} F$ to be the subspace

$$\lim_{\mathbb{N}} F := \left\{ (x_i) \in \prod_{\mathbb{N}} F_i \;\middle|\; f_{i,j}(x_j) = x_i \right\} \subset \prod_{i \in \mathbb{N}} F_i.$$

There is a canonical cone $(\lim_{\mathbb{N}} F, \{\pi_i\})$ over F whose components are given by the composites

$$\lim_{\mathbb{N}} F \longrightarrow \prod_{i\in\mathbb{N}} F_i \longrightarrow F_j$$

of the subspace inclusion with the product projections.

Proposition 1.2.48. *Given an inverse system F, the cone $(\lim_{\mathbb{N}} F, \{\pi_i\})$ is a limit of F.*

Proof. Dual to the proof of Proposition 1.2.46. □

Exercise 1.2.49. Show that colimits of directed systems and limits of inverse systems are unique up to unique homeomorphism which commutes with the defining cones.

Remark 1.2.50. In the case where we are given a directed system F consisting of subspace inclusions

$$F_1 \subset F_2 \subset F_3 \subset \cdots$$

then the underlying set of $\operatorname{colim}_{\mathbb{N}} F$ can be identified with $\bigcup_{i\in\mathbb{N}} F_i$, and a set U is open in the topology of $\operatorname{colim}_{\mathbb{N}} F$ if and only if $U \cap F_i$ is open for every i. It is common to write $\bigcup_{i\in\mathbb{N}} F_i$ for the colimit in this instance.

Warning 1.2.51. ⚠ Given a fixed space X and a directed system of subspace inclusions

$$F_1 \subset F_2 \subset F_3 \subset \cdots \subset X,$$

the topology on $\operatorname{colim}_{\mathbb{N}} F$ may not be the same as the subspace topology on $\bigcup_{i\in\mathbb{N}} F_i \subset X$.

Example 1.2.52 (The Cantor Set)**.** We define an inverse system of subspaces $K_i \subset [0, 1] \subset \mathbb{R}$ recursively.

- We set $K_0 = [0, 1]$.
- We set
$$K_{i+1} = \frac{1}{3} K_i \cup \left(\frac{2}{3} + \frac{1}{3} K_i \right)$$

This construction of K_i amounts to taking the (closed) upper and lower thirds of every closed interval in K_i. The inclusions form an inverse system of topological spaces, and we define the Cantor set C to be the limit of the K_i. This can be identified with the subspace of $\mathbb{R}$ whose underlying set is the intersection of the K_i. One can show that the Cantor set has the cardinality of $\mathbb{R}$.

1.2.5 CW complexes

We now leverage the constructions we have provided to give a description of a particularly useful class of spaces: *CW complexes*. These, loosely speaking are the spaces that can be build by gluing together n-dimensional disks.

Notation 1.2.53. We denote by D^n the closed unit disk in $\mathbb{R}^n$:

$$D^n := \{x \in \mathbb{R}^n \quad | \quad |x| \leq 1\}$$

There is a canonical inclusion $S^{n-1} \hookrightarrow D^n$ which views S^{n-1} as the 'boundary'[31] of D^n.

Notice that $D^0 = *$ is the one-point space. We fix the convention that $S^{-1} = \varnothing$, as the "boundary" of D^0.[32]

[31] We will shortly give a topological definition of the boundary, but it is not necessary for our purposes here

[32] Note that this convention accords with our definition of S^n. Since there are no points of norm 1 in $\mathbb{R}^0$, $S^{-1} = \varnothing$.

Definition 1.2.54. A *CW complex* K consists of the following data

- For every $n \in \mathbb{N}$, a set $\mathrm{Cell}_n(K)$, called the *set of n-cells* of K.
- A sequence of topological spaces spaces K_n for $n \geq -1$ with $K_{-1} = \varnothing$.
- For every $n \in \mathbb{N}$ and every $i \in \mathrm{Cell}_n(K)$, a continuous map

$$\iota_i : S^{n-1} \longrightarrow K_{n-1}$$

  called the *attaching map* of the cell i.
- For each $n \in \mathbb{N}$, pushout diagrams

$$\begin{tikzcd} \coprod_{i \in \mathrm{Cell}_n(K)} S^{n-1} \arrow[r, "\amalg \iota_i"] \arrow[d, hook] & K_{n-1} \arrow[d, "j_n"] \\ \coprod_{i \in \mathrm{Cell}_n(K)} D^n \arrow[r] & K_n \end{tikzcd}$$

We call the space

$$|K| = \operatorname*{colim}_{\mathbb{N}} K_n = \bigcup_{i \in \mathbb{N}} K_n$$

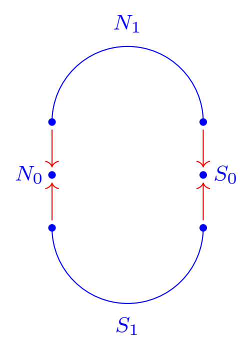
the *realization* of the complex K. We call a CW complex *n-dimensional* if $\mathrm{Cell}_m(K) = \varnothing$ for $m > n$.

In a sense, a CW complex is a "recipe" for building a space. We start, *tabula rasa*, with an empty space K_{-1}. We then add a (possibly infinite) collection of points (0-disks), connect these points with a (possibly infinite) collection of line segments (1-disks), and so on. If there are an infinite number of steps, we combine them all into one space by taking a "union" (directed colimit).

Example 1.2.55. There is a standard (finite-dimensional) CW complex structure on the n-sphere S^n given as follows. In each dimension i less than or equal to n, there are two cells, which I will call N_i and S_i. In the case where $n = 0$, this gives the 0-sphere, which is simply two points, without any further data. We then inductively suppose that the cells up to dimension n give us the n-sphere, and define the attaching maps for N_{n+1} and S_{n+1} to be the identity maps $\mathrm{id} : S^n \to S^n$.

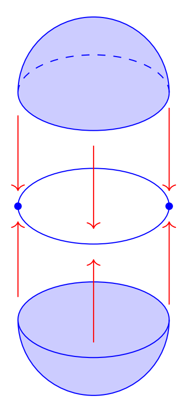
We can illustrate the CW-structure of the n-sphere from Example 1.2.55 as follows. We first add the cells N_0 and S_0, to get the 0-sphere.

We then attach the two 1-cells (intervals) as follows



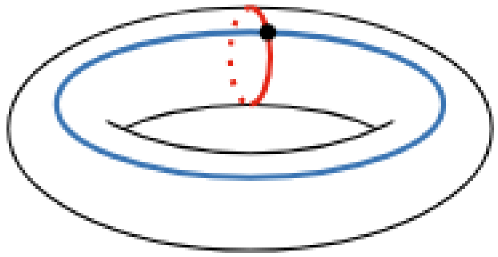
to get the circle S^1.

Next, we attach a pair of 2-cells (2-dimensional disks)



to obtain the sphere S^2.

Example 1.2.56. There is a standard CW structure on the torus $S^1 \times S^1$ consisting of one 0-cell, two 1-cells, and one 2-cell. We can picture this as



(Image created by Ulrich Bauer)

Sixth lecture on 28. April

1.3 First properties of topological spaces

Now that we have built up a small bestiary of topological spaces, we begin to explore the properties these spaces can have and the construction which can take place internal to a given space.

1.3.1 Interior, closure, and boundary

Already intuitively familiar from analysis are the interrelated notions of interior, closure, and boundary. Our task now is simply to understand how best to formulate them in terms of open and closed sets.

Definition 1.3.1. Let X be a topological space, and $A \subset X$ a subset.

1. The *interior of A in X*, denoted by $\mathring{A}$, is the largest open set of X contained in A, or, equivalently, the union of all open sets of X contained in A.

2. The *closure of A in X*, denoted by $\overline{A}$, is the smallest closed set containing A, or, equivalently, the intersection of all closed sets of X containing A.

3. The *boundary of A in X*, denoted by ∂A, is the set-theoretic difference $\partial A = \overline{A} \setminus \mathring{A}$.

Lemma 1.3.2. *Let X be a topological space and $A \subset X$. Then*

1. $\partial A = \overline{A} \cap \overline{X \setminus A}$.

2. *$x \in \overline{A}$ if and only if, for every open set U containing X, $U \cap A \neq \varnothing$.*

3. *$x \in \mathring{A}$ if and only if there is an open set U with $x \in U \subset A$.*

Proof. Exercise. □

Definition 1.3.3. A subset $A \subset X$ of a topological space X is called

1. *dense* if $\overline{A} = X$, and

For any topological space, there is a way of constructing a preorder from that space which has a close connection to the construction of topologies from posets.

Example. Let X be a topological space. The *specialization preorder on* X is a preorder on the underlying set of X defined by declaring $x \leq y$ if $x \in \overline{y}$.

This construction, in fact, defines an *adjoint functor* (a topic we will not explore here, but about which there is copious literature) to a functor from preorders to topological spaces defined analogously to Example 1.1.3. In fact, we can consider preorders to be a special case of topological spaces.

2. *nowhere dense* if $\overline{\mathring{A}} = \varnothing$.

Remark 1.3.4. Let (X, d) be a metric space, and $A \subset X$ be a subset. Then the metric definitions of closure, interior, dense, and nowhere dense agree with the topological definitions.

1.3.2 Connectedness and path-connectedness

Above, we constructed the *disjoint union* of topological spaces, $X \amalg Y$, which can be viewed as consisting of two separate 'parts': X and Y. However, given a topological space (X, τ_X), we do not yet have any way of testing whether it has been built in this way. Such a criterion is provided by notions of *connectedness*.

Definition 1.3.5. Let X be a topological space. If, for every pair of non-empty open sets $U, V \subset X$ such that $U \cup V = X$ the intersection $U \cap V \neq \emptyset$, we say that X is connected. A *connected component* of X is a maximal connected subspace $Y \subset X$.

Lemma 1.3.6. *A topological space X is connected if and only if it has a single connected component: X.*

Proof. Left as an exercise for the reader. □

Remark 1.3.7. Notice that if X is not connected, then there are non-empty open sets $U, V \subset X$ such that $U \cup V = X$ and $U \cap V = \varnothing$. As such, U and V are both open and closed. Indeed, this is an equivalent characterization of connectedness.

Lemma 1.3.8. *A topological space X is not connected if and only if it is homeomorphic to a disjoint union $X_0 \amalg X_1$ of spaces.*

Proof. If $X \cong X_0 \amalg X_1$, then it is clear that the open subsets X_0 and X_1 display X as disconnected.

If, on the other hand, X is not connected, let $Y \subsetneq X$ be a proper subset which is both open and closed. Suppose $U \subset X$ is a subset such that $U \cap Y$ and $U \cap (X \setminus Y)$ are open in the subspace topologies. Then, since Y and $(X \setminus Y)$ are open, $U \cap Y$ and $U \cap (X \setminus Y)$ are open in X, and so their union, U must be open as well. Thus U is open if and only if $U \cap Y$ and $U \cap (X \setminus Y)$ are open in the subspace topologies, and so $X \cong Y \amalg (X \setminus Y)$ as desired. □

Examples 1.3.9.

1. A subspace of $\mathbb{R}$ is connected if and only if it is an open, closed, or half-open interval.

2. Any open or closed ball in $\mathbb{R}^n$ is connected.

In a sense made precise by the following proposition, connectedness measures 'discreteness of maps out of X'.

Proposition 1.3.10. *Let X be a topological space. The following are equivalent*

1. *X is connected.*

2. *Every continuous map $f : X \to Y$ to a discrete space is constant.*

Proof. We first show 2. $\Rightarrow$ 1. Suppose that X is not connected. Then there are two non-empty open sets $U, V \subset X$ with $U \cup V = X$ and $U \cap V = \emptyset$. Define a map to $f : X \to \{0, 1\}$ by sending every element of U to 0, and every element of V to 1. It is immediate from the definition that f is continuous with respect to the discrete topology on $\{0, 1\}$ and non-constant.

We now show 1. $\Rightarrow$ 2. Suppose that there is a continuous, non-constant map $f : X \to Y$, where Y is equipped with the discrete topology. In particular, choose two distinct elements y_0 and y_1 in Y such that both are in the image of f. Choose any map of sets $p : Y \to \{0, 1\}$ such that $p(y_0) = 0$ and $p(y_1) = 1$. Since this is continuous with respect to the discrete topologies, we get a non-constant continuous map $p \circ f : X \to \{0, 1\}$. Since this is continuous, the sets $U := (p \circ f)^{-1}(0)$ and $V := (p \circ f)^{-1}(1)$ are open. Since $p \circ f$ is non-constant, both U and V are non-empty. By definition $U \cup V = X$ and $U \cap V \neq X$. $\square$

Definition 1.3.11. Let (X, τ_X) be a topological space, and let $A \subset X$. We define the *closure* of A to be the subset $\overline{A} \subset X$ which is the intersection of all closed subsets of X which contain A.

Lemma 1.3.12. *Let $f : X \to Y$ be a continuous map, and let X be connected. Then $f(X) \subset Y$ is connected.*

Proof. Let $U, V \subset Y$ be open subsets with $f(X) \subset U \cup V$. Then $f^{-1}(U)$ and $f^{-1}(V)$ are open subsets of X whose union equals X. As such, there is an $x \in f^{-1}(U) \cap f^{-1}(V)$ since X is connected, and thus, $f(x) \in U \cap V$, so $f(X)$ is connected. $\square$

Proposition 1.3.13. *Let X be a topological space, and A a connected subset. If $B \subset X$ such that $A \subset B \subset \overline{A}$, then B is connected.*

Proof. Suppose that there were two open sets $U, V \subset X$ such that $U \cup V = B$ and $U \cap V \cap B = \emptyset$. Since A is connected, we must then have that $A \subset U$ or $A \subset V$. WLOG, assume $A \subset U$. But then, for $b \in B$, we have that $b \in \overline{A}$ and V is an open subset containing b. Therefore, $V \cap A \neq \emptyset$, and thus, $V \cap U \cap B \neq \emptyset$, which is a contradiction. $\square$

Definition 1.3.14. A topological space X is called *totally disconnected* if every connected component is a singleton subset.

Example 1.3.15. The Cantor set of Example 1.2.52 is totally disconnected.

So if connectedness measures the discreteness of maps *out* of X, can we also measure the discreteness of maps *into* X?

Definition 1.3.16. A *path* in a topological space X is a continuous map $\alpha : [0, 1] \to X$, where $[0, 1]$ is equipped with the subspace topology inherited from $\mathbb{R}$. We say that α is a *path from x to y* if $\alpha(0) = x$ and $\alpha(1) = y$.

We define an equivalence relation on X by $x \sim y$ if and only if there exists a path in X from x to y. A *path component of* x is an equivalence class $[x] \in X_{/\sim}$, viewed as a subspace $[x] \subset X$. We denote by $\pi_0(X)$ the set of path components of X. We say that X is *path connected* if $\pi_0(X)$ is a singleton.

Proposition 1.3.17. *The relation $\sim$ of Definition 1.3.16 is indeed an equivalence relation on X.*

Proof. Reflexivity follows immediately from the existence of constant paths. Given $x \in X$, there is a path

$$c_x : [0,1] \longrightarrow X$$

from x to x which sends every point of $[0,1]$ to x.[33]

[33] This is clearly continuous, since the preimage of any set U are either $[0,1]$ itself or $\varnothing$, depending on whether U contains x.

Given a path α from x to y and a path β from y to z, we can define a path $\beta * \alpha$ as a piecewise function

$$\beta * \alpha(t) = \begin{cases} \alpha\,(2) & t \in \left[0, \frac{1}{2}\right] \\ \beta(2(t-1)) & t \in \left[\frac{1}{2}, 1\right] \end{cases}$$

which goes from x to z, showing transitivity.[34]

[34] For continuity, see Remark 1.2.42.

Finally, for symmetry, notice that the map

$$\begin{aligned} \tau : [0,1] &\longrightarrow [0,1] \\ t &\longmapsto 1-t \end{aligned}$$

is continuous, and if α is a path from x to y, then $\alpha \circ \tau$ is a path from y to x. □

Proposition 1.3.18. *Let X be a path-connected topological space. Then X is connected.*

Proof. Suppose X is not connected. Then there exists a continous map $f : X \to \{0,1\}$ (where $\{0,1\}$ is equipped with the discrete topology) such that f is non-constant. Let $x \in f^{-1}(0)$ and $y \in f^{-1}(1)$. A path $\alpha : [0,1] \to X$ from x to y would then yield a continuous, non-constant map $f \circ \alpha : [0,1] \to \{0,1\}$. Since $[0,1]$ is connected, this cannot occur, and thus, X is not path connected. □

Warning 1.3.19. ⚠ The converse of Proposition 1.3.18 is *not* true. There are connected spaces which are not path-connected. We need to make additional assumptions about our space X if we want connectedness and path-connectedness to be equivalent.

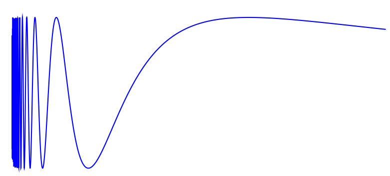
A standard counterexample in topology is the *topologist's sine curve*. Let $X \subset \mathbb{R}^2$ be the collection of all points $(x, \sin(\frac{1}{x}))$ for $x > 0$, together with the segment of the y-axis from -1 to 1. This inherits a topology from $\mathbb{R}^2$.

Lemma. *The topologist's sine curve is connected.*

Proof. We simply need note that the subspace $A := \{(x, \sin(\frac{1}{x})) \mid x > 0\}$ is path-connected, and thus connected. Moreover $A \subset X \subset \overline{A}$. Therefore, by Proposition 1.3.13, X is connected. □

Lemma. *The topologist's sine curve is not path connected.*

Proof. Suppose we have a path $p : [0,1] \to X$ going from $(1, \sin(1))$ to $(0,0)$. Consider the component functions $p_x, p_y : [0,1] \to \mathbb{R}$. Since p_x is continuous, its image is connected, and therefore is the interval $[0,1]$. But then, p is the map $t \mapsto (t, \sin(\frac{1}{t}))$. But for every $\delta > 0$, there is a $0 < t < \delta$ such that $\sin(\frac{1}{t}) = 1$, i.e $|p(t) - (0,0)| > 1$. Therefore, p cannot be continuous. □



Definition 1.3.20. We call a topological space X *locally path-connected* if, for every $x \in X$ and every open U containing x, there is an open V with $x \in V \subset U$ such that V is path connected.

Proposition 1.3.21. *If X is a connected, locally path-connected topological space, then X is path-connected.*

Proof. Let $x \in X$, and denote by $[x] \subset X$ the path component of x in X. We claim that $[x]$ is both open and closed, so since $[x]$ is non-empty and X is connected, $[x]$ must equal X.

To see that $[x]$ is closed, let $y \in \overline{[x]}$, and use the locally path-connected property of X to choose an open path-connected subset U containing y. Then $Y \cap [x]$ is non-empty, and so by the transitivity of the path equivalence relation $y \in [x]$. Thus, $\overline{[x]} = [x]$, and so $[x]$ is closed.

To see that $[x]$ is open, let $y \in [x]$, and choose an open path-connected subset U containing Y. Then by the transitivity of the path equivalence relation $U \subset [x]$, and so every point of $[x]$ has an open neighborhood contained in $[x]$. Thus $[x]$ is open. □

Proposition 1.3.22. *The assignment* $\mathsf{Top} \to \mathsf{Set}$ *which sends every space X to its set of path components is* functorial. *That is,*

1. *For every morphism $f : X \to Y$ in* Top*, there is a morphism $\pi_0(f) : \pi_0(X) \to \pi_0(Y)$ in* Set*.*
2. *For the identity morphism $\mathrm{id}_X : X \to X$ in* Top*, $\pi_0(\mathrm{id}_X)$ is the identity morphism on $\pi_0(X)$.*
3. *For composable morphisms*

$$X \xrightarrow{g} Y \xrightarrow{f} Z$$

in Top*,*

$$\pi_0(f \circ g) = \pi_0(f) \circ \pi_0(g).$$

Proof. Since the composition of a path with a continuous map preserves the equivalence relation $\sim$ of Definition 1.3.16, there is a unique map $\pi_0(f)$ induced by each continuous f which sends $[x] \mapsto [f(x)]$, showing part (1). To see part (2), note that

$$\pi_0(\mathrm{id}_X)([x]) = [\mathrm{id}_X(x)] = [x].$$

Finally, to see part (3), notice that

$$\pi_0(f \circ g)([x]) = [f(g(x))] = \pi_0(f)([g(x)]) = \pi_0(f)(\pi_0(g)([x])).$$ □

2

Classical topology

Seventh lecture on 4. May

We now move from first constructions to the exploration of key theorems in point-set topology. Most of the material we will cover here is very classical in flavor, and, as such, the category-theoretic considerations introduced in the first chapter will fall briefly by the wayside. They will, however, return when we begin our exploration of homotopy theory.

2.1 Separation axioms

Our first topic in this section will be conditions on topological spaces which help to ensure that they better match our intuition. To get a sense of why we might want this, let us first consider a counter-example.

Example 2.1.1. Consider any set X equipped with the *indiscrete topology*: the topology in which on $\varnothing$ and X are open. In this topology, the open sets tell us nothing about the points of X — every point of X is contained in exactly the same open sets.

Definition 2.1.2. Given a topological space X, we call a subset $U \subset X$ containing a point $x \in X$ and such that there is an open V in X with $x \in V \subset U$ a *neighborhood of x*. We call U an *open neighborhood* or a *closed neighborhood* of x when U is open or closed, respectively. We write N_x for the set of neighborhoods of x in X.

Two points $x, y \in X$ are called *topologically distinquishable* if $N_x \neq N_y$.

Our first separation axiom will be the mildest

Schematically, we can draw the T_0 property as

Definition 2.1.3. A topological space X is called T_0 or a *Kolmogorov space* if, for every two points $x_1, x_2 \in X$, there exists $i \in \{1, 2\}$ and an open neighborhood U of x_i which does not contain the other point.

Lemma 2.1.4. *For a topological space X, the following are equivalent.*

1. *The space X is T_0.*

2. *Any two points of X are topologically distinguishable.*

Proof. It is immediate that in a T_0 space, every two points are topologically distinguishable. Suppose on the other hand that every two points of X are topologically distinguishable. For $x, y \in X$, there is then (without loss of generality) a neighborhood C of X which is not a neighborhood of y. In particular, there is an open set V with $x \in V \subset C$. Since C is not a neighborhood of y, $y \notin V$, and so X is T_0. $\square$

Lemma 2.1.5. *Let $(P, \leq)$ be a poset, and let $\tau_{\leq}$ be the poset topology on P.*[1] *Then $(P, \tau_{\leq})$ is T_0.*

[1] See Example 1.1.3.

Proof. Suppose that $x, y \in P$ such that every open set containing one contains the other. Since the sets

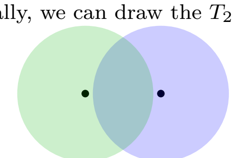
$$\{z \in P \mid z \leq x\} \quad \text{and} \quad \{z \in \mid z \leq y\}$$

are open, it follows that $x \leq y$ and $y \leq x$, and so $x = y$. $\square$

Definition 2.1.6. A topological space is called T_1, or *Fréchet*[2] if, for every $x, y \in X$, there exists open neighborhoods U_x of x and U_y of y such that $x \notin U_y$ and $y \notin U_x$.

[2] Note that this has nothing to do with the functional-analytic notion of a Fréchet space.

Schematically, we can draw the T_2 property as



Lemma 2.1.7. *For a topological space X, the following are equivalent.*

1. *The space X is T_1.*
2. *Every singleton $\{x\} \subset X$ is closed.*
3. *Every subset $A \subset X$ is the intersection of all open sets containing A.*

Proof. To see that (1)⇒(2), let $x \in X$, and for every other point $y \in X \setminus \{x\}$, choose an open neighborhood V_y of y such that $x \notin V_y$. Then

$$X \setminus \{x\} = \cup_{y \in X \setminus \{x\}} V_y$$

is open, and so $\{x\}$ is closed.

To see that (2)⇒(3), let $A \subset X$. Then for $x \in X \setminus A$, $X \setminus \{x\}$ is open. Since

$$A = \bigcap_{x \in X \setminus A} X \setminus \{x\}$$

we have in particular that A is the intersection of the open sets containing A.

Finally, to see that (3)⇒(1), let $x \neq y$. Then since

$$\{x\} = \bigcap_{x \in U \overset{\circ}{\subset} X} U$$

there must be at least one open neighborhood V of x which does not contain y. $\square$

Example 2.1.8. There are poset topologies which are not T_1. Indeed, suppose $(P, \leq)$ is a poset in which the order is not trivial. Then there are distinct elements $x, y \in P$ such that $x < y$, and so every open set containing y also must contain x.

Example 2.1.9. Let X be a set. The *cofinite topology* on X is the topology given by declaring a set U to be open if and only if $X \setminus U$ is finite. The cofinite topology on an infinite set is always T_1.

Definition 2.1.10. A topological space X is called T_2 or *a Hausdorff space* if, for every $x \neq y$ in X, there are open neighborhoods U of x and V of y such that $U \cap V = \varnothing$.

Schematically, we can draw the Hausdorff property as

Lemma 2.1.11. *For a topological space X, the following are equivalent.*

1. *The space X is T_2.*
2. *Every point is the intersection of all its closed neighborhoods.*

Proof. To show that (1)⇒(2), it suffices to show that for every x and every $y \neq x$, there is a closed neighborhood of x not containing y. Let U and V be the open neighborhoods of x and y guaranteed by the Hausdorff property. Then $x \in U \subset X \setminus V$, so $X \setminus V$ is a closed neighborhood of x not containing y.

To show that (2)⇒(1), let $x, y \in X$ with $x \neq y$ and choose a closed neighborhood C of x not containing y. Then there is an open U with $x \in U \subset C$, and so U and $X \setminus C$ are the desired open neighborhoods in the Hausdorff property. □

Remark 2.1.12. One key takeaway from the preceding lemma is that points are closed in Hausdorff spaces. In particular, every Hausdorff space is T_1.

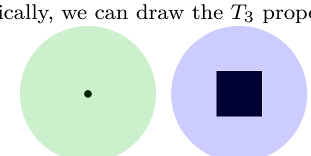
Example 2.1.13. Every metric space (X, d) is a Hausdorff space. Every CW complex is a Hausdorff space.

Warning 2.1.14. ⚠ Our next few separation axioms create problems, because the terminology is not standardized. I am opting to use the same terminology as the previous version of this course, but bear in mind that in many sources (including Wikipedia), the meaning of the terms, e.g., T_3 and *regular* are reversed.

Definition 2.1.15. A topological space X is called T_3 if, for every closed set $C \subset X$ and every point $x \in X \setminus C$, there exists open sets $U, V \subset X$ with $x \in U$, $C \in V$, and $U \cap V \neq \varnothing$.

The space X is called *regular* if it is T_3 and T_1 (or, equivalently, if it is T_3 and Hausdorff).

Schematically, we can draw the T_3 property as



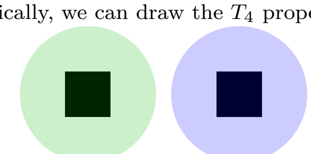
Remark 2.1.16. Note that T_3 does not imply Hausdorff, precisely because it may not guarantee that singletons are closed.

The failure of a T_3 space to be Hausdorff is always caused by points not being closed sets. For example, if we consider the space X with underlying set $\{a, b, c, d\}$ and non-trivial opens given by $\{a, b\}$ and $\{c, d\}$, then it is easily checkable that X is T_3. However, the only non-trivial closed sets are two-point sets, and so X cannot be Hausdorff.

Definition 2.1.17. A topological space X is called T_4 if, for any closed sets $C, K \subset X$ such that $C \cap K = \varnothing$, then there are open sets U and V, which respectively contain C and K and such that $U \cap V = \varnothing$.

The space X is called *normal* if it is T_4 and T_1 (or, equivalently, T_4 and Hausdorff, or, equivalently, T_4 and regular).

Schematically, we can draw the T_4 property as



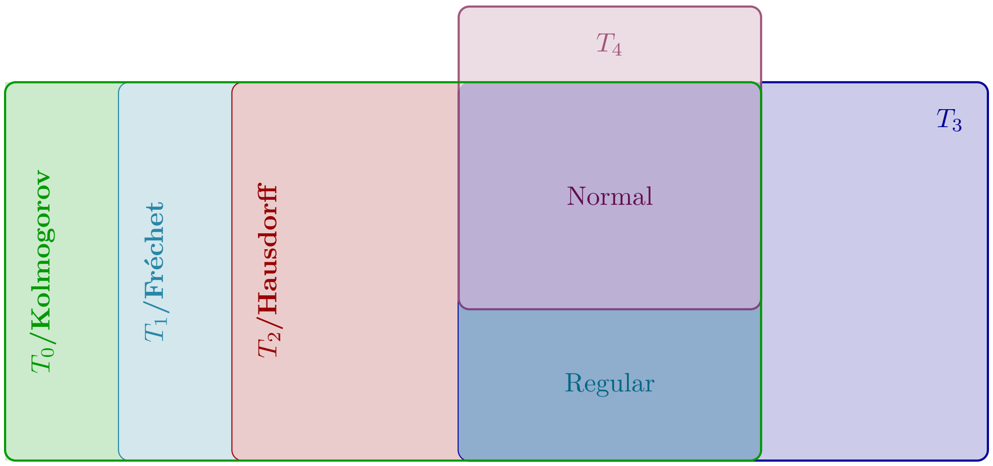
Example 2.1.18. Every CW-complex is normal. We will see later that every metric space is also normal.

The standard example of a topological space which is T_4 but not T_3 is the *Sierpinski space* S, whose underlying set is $\{0,1\}$ and whose open sets are $\varnothing$, $\{1\}$ and S. Since any two non-trivial closed sets are not disjoint, S vacuously satisfies the T_4 axiom. However, one cannot separate the point 1 and the closed set $\{0\}$ by opens, and so S is not T_3. As before, this distinction evaporates when one requires that points are closed.

Lemma 2.1.19. *A topological space X is T_4 if and only if, for every pair of disjoint closed sets A and B in X, there is an open set U containing A such that $\overline{U}$ is disjoint from B.*

Proof. This is effectively just a reformulation. If X satisfies the property specified in the lemma, we can set $V = X \setminus \overline{U}$ to obtain the T_4 property. On the other hand, if U and V are the sets guaranteed by the T_4 property, $X \setminus V$ is closed, and so $\overline{U} \subset X \setminus V \subset X \setminus B$. $\square$

Before we get to our first major theorem of the course, let us briefly provide a graphical summary of the implications among the various separations axioms.



2.1.1 The Tietze Extension Theorem

We now come to the core of this section: the theorems of Urysohn and Tietze.

Definition 2.1.20. Let X be a topological space, and let $A, B \subset X$ be disjoint closed subsets of X. An *Urysohn function* is a continuous function

$$f : X \longrightarrow [0,1]$$

such that $f(A) = 0$ and $f(B) = 1$

Our first technical result which will pave the way to proving Urysohn's Metrization Theorem is the following proposition.[3]

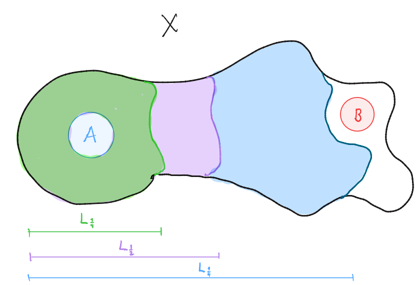
Proposition 2.1.21 (Urysohn's Lemma)**.** *A topological space is T_4 if and only if for any closed, disjoint subsets $A, B \subset X$, there is an Urysohn function for A and B.*

[3] The grounds for calling this proposition "Urysohn's Lemma" are somewhat obscure — likely because it is now seen as a simple helpmeet in the proof of the Tietze Extension Theorem and Urysohn's Metrization Theorem. However, the proof is somewhat technical, and in Urysohn's original paper proving it (*Über die Mächtigkeit der zusammenhängenden Mengen*, 1925) the statement appears as a "Satz" (Theorem, or Proposition) rather than as a "Hilfssatz" (Lemma).

Proof. If, for any such $A, B \subset X$ there exists an Urysohn function $f : X \to [0,1]$, then $f^{-1}([0, \frac{1}{2}))$ and $f^{-1}((\frac{1}{2}, 1])$ are the disjoint open sets containing A and B required for X to be T_4.

If, on the other hand, X is T_4, the proof is more involved. The strategy will be to define a countable sequence of open subsets of X which contain A and expand out to "almost fill up" $X \setminus B$. These subspaces will be indexed by binary fractions in the interval $[0,1]$. From such a sequence, we then construct a function by using the density of binary fractions in the unit interval.

We can picture the first few steps of this construction as follows.



Constructing the spaces: We first construct a space $L_{\frac{1}{2}}$. Using Lemma 2.1.19, there is an open set $L_{\frac{1}{2}}$ such that

$$A \subset L_{\frac{1}{2}} \subset \overline{L_{\frac{1}{2}}} \subset X \setminus B.$$

We then apply Lemma 2.1.19 twice, (to the pairs $(A, X \setminus L_{\frac{1}{2}})$ and $(\overline{L_{\frac{1}{2}}}, B)$) to obtain a sequence

$$A \subset L_{\frac{1}{4}} \subset \overline{L_{\frac{1}{4}}} \subset L_{\frac{1}{2}} \subset \overline{L_{\frac{1}{2}}} \subset L_{\frac{3}{4}} \subset \overline{L_{\frac{3}{4}}} \subset X \setminus B$$

Iterating, we get open subspaces

$$\{L_{\frac{m}{2^n}} \mid m, n \in \mathbb{N} \text{ and } 0 < m < 2^n\}$$

such that for binary fractions $p < q$, we have $\overline{L_p} \subset L_q$.

Defining the function: Write $D \subset (0,1)$ for the set of binary fractions. We define

$$f(x) := \inf(\{p \in D \mid x \in L_p\} \cup \{1\}).$$

We notice that, for $b \in B$, $f(b) = 1$, since b cannot be contained in any of the L_p, and for $a \in A$, $f(a) = 0$ since a is contained in every L_p.

Proving continuity: As the final step, we show that f is continuous. Let $U \subset [0,1]$ be an open set, and let $x \in f^{-1}(U)$. We have three cases:

1. If $f(x) = 0$, then choose $p \in D$ with $[0, p) \subset U$. Then L_p is an open subset containing x which maps into U.
2. If $f(x) = 1$, then choose $p \in D$ with $(p, 1] \subset U$. Then $X \setminus \overline{L_p}$ is an open subset containing x which maps into U.
3. If $0 < f(x) < 1$, choose $p, q \in D$ with $p < f(x) < q$ and such that $(p, q) \subset U$. Then $L_q \setminus \overline{L_p}$ is an open subset containing x which maps into U.

thus f is continuous, completing the proof. □

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Definition 2.1.22. Let X be a topological space, and let $\{f_n : X \to \mathbb{R}\}_{n \in \mathbb{N}}$ be a sequence of continuous real-valued functions on X. We say that the f_n's *converge uniformly* to a function $f : X \to \mathbb{R}$ if, for every $\epsilon > 0$ there is an $m_\epsilon \in \mathbb{N}$ such that, for all $x \in X$ and all $n \geq m$,

$$|f_n(x) - f(x)| < \epsilon.$$

Lemma 2.1.23. *Let X be a topological space, and suppose that $\{f_n : X \to \mathbb{R}\}_{n\in\mathbb{N}}$ is a sequence of functions which converge uniformly to $f : X \to \mathbb{R}$. If the f_n are continuous, then f is continuous.*

Proof. We check continuity on the basis of open ϵ-balls. Consider $y \in \mathbb{R}$ and an ϵ-ball $B_\epsilon(y)$. Let $x \in f^{-1}(B_\epsilon(y))$ and let $r = |f(x) - y| < \epsilon$. For any fixed $\gamma > 0$ we can choose $n \in \mathbb{N}$ and an open subset $U \subset X$ containing x such that (1) for $z \in X$ we have

$$|f_n(z) - f(z)| < \gamma$$

and (2) for $z \in U$ we have $|f_n(z) - f_n(x)| < \gamma$. Thus, for $z \in U$

$$\begin{aligned} |f(z) - y| &= |f(z) - f_n(z) + f_n(z) - f_n(x) + f_n(x) - f(x) + f(x) - y| \\ &\leq |f(z) - f_n(z)| + |f_n(z) - f_n(x)| + |f_n(x) - f(x)| + |f(x) - y| \\ &< 3\gamma + r. \end{aligned}$$

taking γ to be less than $\frac{1}{3}(\epsilon - r)$ thus shows that U is an open set containing x and contained in $f^{-1}(B_\epsilon(y))$, completing the proof. □

Theorem 2.1.24 (Tietze's Extension Theorem)**.** *Let X be a T_4 topological space, let $A \subset X$ be closed, and let $f : A \to [-1, 1]$ be a continuous function. Then there is a continuous extension of f to X, i.e., a continuous function*

$$F : X \longrightarrow [-1, 1]$$

such that $F|_A = f$.

Proof. We will use Urysohn's lemma to construct a sequence of functions which converge to the desired F. The central notion is an *approximate ϵ-extension* of f: A continuous function g such that, for all $a \in A$,

$$|f(a) - g(a)| \leq \epsilon.$$

The key step of the proof will be to construct a better approximation from a worse one.

Improving extensions: Let $g : X \to [-1, 1]$ be an approximate ϵ-extension of f. Consider the closed subsets

$$C = (f - g)^{-1}([-\epsilon, -\frac{1}{3}\epsilon]), \quad D = (f - g)^{-1}([\frac{1}{3}\epsilon, \epsilon])$$

These are precisely the sets such that the error in the approximation is greater than $\frac{\epsilon}{3}$. By Proposition 2.1.21 (The Urysohn Lemma), there is a function

$$\gamma : X \longrightarrow [-\tfrac{\epsilon}{3}, \tfrac{\epsilon}{3}]$$

which takes value $-\frac{\epsilon}{3}$ on C and $\frac{\epsilon}{3}$ on D. Define $h = g + \gamma$. Then for, e.g., $a \in A \cap C$, we have

$$\begin{aligned} f(x) - h(x) &= f(x) - g(x) - \gamma(x) \\ &> -\epsilon + \frac{\epsilon}{3} = -\frac{2\epsilon}{3} \end{aligned}$$

and

$$\begin{aligned} f(x) - h(x) &= f(x) - g(x) - \gamma(x) \\ &< -\frac{\epsilon}{3} + \frac{\epsilon}{3} = 0 \end{aligned}$$

Similar arguments on $A \cap D$ and $A \setminus (C \cup D)$ show that h is an approximate $\frac{2\epsilon}{3}$-extension of f.

The remainder of the proof is easy. We note that the constant zero function $g = 0$ is an approximate 1-extension of f. We then use our step above to improve it to an approximate $\frac{2}{3}$-extension g_1 of f. This is improved to an approximate $(\frac{2}{3})^2$-extension of f, and so on. This yields a sequence $\{g_n\}$ of approximate $(\frac{2}{3})^n$-extensions which converge uniformly to some function F which agrees with f on A. By Lemma 2.1.23, this is continuous, completing the proof. □

2.1.2 Metrization

We now can give a partial answer to the question of when a topology is induced by a metric.

Definition 2.1.25. A topological space X is called *metrizable* if there exists a metric d on X which induces the topology on X.

Lemma 2.1.26. *Every metric space (X, d) is normal.*

Proof. It is easy to see that any metric space is Hausdorff. We thus prove only the T_4 axiom. Let $A, B \subset X$ be closed disjoint subsets. We define

$$\begin{aligned} d_A : X &\longrightarrow [0, \infty) \\ x &\longmapsto \inf\{d(a, x) \mid a \in A\} \end{aligned}$$

and note that for $x, y \in X$[4]

$$|d_A(x) - d_A(y)| \leq d(x, y).$$

[4] This follows from applying the infimum to the triangle inequality:

$$\inf\{d(a,x) \mid \in A\} \leq \inf\{d(a,y) + d(y,x) \mid a \in A\}$$

and

$$\inf\{d(a,y) \mid \in A\} \leq \inf\{d(a,x) + d(x,y) \mid a \in A\}.$$

If $d_A(x) = 0$, we can chose a sequence $\{a_n\}_{n \in \mathbb{N}}$ which converges to x. Since A is closed, this implies $x \in A$. As such, since A and B are disjoint, it follows that

$$d_A(x) + d_B(x) > 0$$

for all $x \in X$.

We then define a continuous function $f : X \to [0, 1]$ by

$$f(x) = \frac{d_A(x)}{d_A(x) + d_B(x)}$$

which is clearly an Urysohn function for A and B. Thus by Proposition 2.1.21, X is T_4. □

Urysohn's Metrization theorem provides a partial converse, giving us a broad class of examples in which topological spaces admit metrics.

Definition 2.1.27. A topological space X is called *second countable*[5] if there is a countable basis $\mathcal{B}$ for the topology on X.

[5] While there is a notion of first countability, we will not discuss it in this course.

Example 2.1.28. For any $n \in \mathbb{N}$, the set of balls around rational points with rational radii provides a countable basis for $\mathbb{R}^n$. Similarly, S^n, $\mathbb{R}P^n$, T^2, etc. are second countable. Subspaces of second countable spaces are second countable, as are countable products of second countable spaces.

Theorem 2.1.29 (Urysohn's Metrization Theorem)**.** *If X is a second countable, normal topological space, then X is metrizable.*

Proof. We will prove the theorem as a sequence of claims.

Claim 1: There is a countable collection of continuous functions

$$\{ \ f_\alpha : X \to [0,1] \ \}_{\alpha \in A}$$

which *separates points from closed sets*. That is, for every $x \in X$ and $C \subset X$ closed such that $x \notin C$, there is an $\alpha \in A$ such that $f_\alpha(x) \notin \overline{f_\alpha(C)}$.

Proof of Claim 1. Let A be the set of pairs (U, V) of basis elements such that $\overline{U} \subset V$ and for $(U, V) \in A$, let $f_{U,V}$ be an Urysohn function for $\overline{U}$ and $X \setminus V$. We claim that this (countable) set of continuous maps separates points from closed sets. To see this, let $C \subset X$ be closed and let $x \notin C$. Choose a basis element V with $x \in V$ and $V \cap C = \varnothing$. Then $X \setminus V$ is closed, and so we can find a further basis element U with $x \in U$ and $\overline{U} \subset V$. Thus, $f_{(U,V)}$ separates x and C by construction. ■

Claim 2: Let $\{f_\alpha\}_{\alpha\in A}$ be the functions from Claim 1. The product $F = \prod_{\alpha \in A} f_\alpha$ defines an embedding $F : X \to [0,1]^A$.

Proof of Claim 2: Note that by construction, F separates points from closed subsets. Since X is T_1, singletons are closed, and so F separates distinct points, i.e., is injective.

We then show that F is a closed map onto its image. Let $C \subset X$ be closed, and suppose $x \in X$ such that $f(x) \in \overline{F(C)}$. This implies $x \in C$, since F separates points from closed sets. Thus, $F(X) \cap \overline{F(C)} = F(C)$, and so $F(C)$ is closed in $F(X)$. ■

We now can complete the proof. Since $[0,1]^A$ is a countable product of metric spaces, it is a metric space. Thus, X is (homeomorphic to) a subspace of a metric space, and so is itself metrizable. □

2.2 Compactness

We now briefly digress into one of the most important (and, at times, mysterious) properties in topology and analysis: compactness. The definition is familiar from analysis courses, but it takes on new importance in topology.

Definition 2.2.1. Let X be a topological space. An *(open) cover* of X is a set $\mathcal{U}$ of open subsets of X such that

$$X = \bigcup_{U \in \mathcal{U}} U.$$

A *subcover* of $\mathcal{U}$ is a subset $\mathcal{V} \subset \mathcal{U}$ such that $\mathcal{V}$ is still a cover of X.

The space X is said to be *compact* if every open cover of X admits a finite subcover.

Examples 2.2.2.

1. Any finite space X is compact.

2. A subset of $\mathbb{R}^n$ is compact if and only it is closed and bounded. This is the Heine-Borel Theorem, which we will prove in the next section.

3. The spheres S^n are all compact, as we will see shortly.

Remark 2.2.3. If we consider a subspace $A \subset X$, we can equivalently define compactness by considering covers of A *by open subsets of* X, using the definition of the subspace topology.

Example 2.2.4 (non-example)**.** Let $p, q \in \mathbb{Q}$ with $p < q$. Then the subset $[p, q] \subset \mathbb{Q}$ is not compact. To see this, let $r \in \mathbb{R} \setminus \mathbb{Q}$ with $p < r < q$. Let $m = \min\{q - r, r - p\}$, and consider the cover $\{U_\epsilon\}_{0<\epsilon<m}$ of $[p, q]$ by the opens

$$U_\epsilon = \{x \in \mathbb{Q} \mid |x - r| > \epsilon\}.$$

If there were a finite subcover $\{U_{\epsilon_1}, \dots, U_{\epsilon_n}\}$, then there would be an i such that $\epsilon_i = \min\{\epsilon_j \mid 1 \le j \le n\}$. This would imply that $[p, q] \subset U_{\epsilon_i}$. However, by the density of the rationals in $\mathbb{R}$, there is a rational between r and $r + \epsilon_i$, a contradiction.

Lemma 2.2.5. *Let $f : X \to Y$ be continuous, and let X be compact. Then $f(X)$ is compact.*

Proof. Given a cover $\mathcal{U}$ of $f(X)$ (by open subsets of X), the preimages of the sets in $\mathcal{U}$ are open in X by continuity. As such,

$$\mathcal{V} = \{f^{-1}(U) \mid U \in \mathcal{U}\}$$

is an open cover of X. Let $\{V_1, \dots, V_n\}$ be a finite subcover. Then $\{f(V_1), \dots, f(V_n)\} \subset \mathcal{U}$, and covers $f(X)$. Thus $f(X)$ is compact. □

Ninth lecture on 11. May

Lemma 2.2.6. *Let X be compact and $A \subset X$ be closed. Then A is compact.*

Proof. Given a cover $\mathcal{U}$ of A by open subsets of X, then $\mathcal{U} \cup \{X \setminus A\}$ is an open cover of X. The compactness of X gives us a finite subcover $\mathcal{V}$, and thus $\mathcal{V} \setminus \{X \setminus A\}$ is a finite subcover of $\mathcal{U}$. □

Lemma 2.2.7. *If X is a Hausdorff space and $A \subset X$ is compact, then A is closed.*

Proof. For any fixed point $x \in X \setminus A$ and every point $a \in A$, we can use the Hausdorff property to find disjoint open subsets with $a \in U_{x,a}$ and $x \in V_{x,a}$. Then $\{U_{x,a} \mid a \in A\}$ is an open cover of A, so admits a finite subcover $U_{x,a_1}, \dots, U_{x,a_k}$ by compactness. Then

$$V_x := \bigcap_{i=1}^{k} V_{x,a_i}$$

is an open neighborhood of x which is disjoint from A. Taking the union over all $x \in X \setminus A$, we find that $X \setminus A$ is open, as desired. □

Corollary 2.2.8. *Let $f : X \to Y$ be a continuous map. If X is compact and Y is a Hausdorff space, then f is a closed map.*

Proof. Let $A \subset X$ be closed. Then since X is compact, A is compact by Lemma 2.2.6, and so $f(A)$ is compact by Lemma 2.2.5. Thus by Lemma 2.2.7, $f(A)$ is closed. □

Remark 2.2.9. In practice, this makes it particularly easy to display compact Hausdorff spaces as quotients of compact spaces. For instance, the surjective map

$$[0, 1] \longrightarrow S^1$$

is a surjective map from a compact space to a Hausdorff space, and so is closed by Corollary 2.2.8. As such, by Lemma 1.2.29, it follows that it is a quotient map.

Lemma 2.2.10. *For a topological space X, any finite union of compact subspaces of X is itself compact.*

Proof. Omitted. □

Finally, we prove an interesting proposition about the relationship of compactness to separation axioms.

Proposition 2.2.11. *If X is a compact Hausdorff space, then X is normal.*

Proof. Let X be a compact Hausdorff space. And let $A, B \subset X$ be disjoint closed subsets of X. By Lemma 2.2.6, it follows that both A and B are compact. Fix $a \in A$. Then by the Hausdorff property, for every $b \in B$, we can find disjoint open sets with $a \in U_{a,b}$ and $b \in V_{a,b}$. Since B is compact, we can choose a finite set $\{b_1, \dots, b_k\} \subset B$ such that $B \subset \bigcup_{i=1}^{k} V_{a,b_i}$. Setting $V_a = \bigcup_{i=1}^{k} V_{a,b_i}$ and $U_a = \bigcap_{i=1}^{k} U_{a,b_i}$,we have found two disjoint open sets with $a \in U_a$ and $B \subset V_a$.[6]

Now choose such a U_a and V_a for each $a \in A$. Since the U_a's cover A and A is compact, there is a finite set $\{a_1, \dots, a_\ell\}$ such that $A \subset \bigcup_{i=1}^{\ell} U_{a_i}$. Thus

$$U = \bigcup_{i=1}^{\ell} U_{a_i} \quad \text{and} \quad V = \bigcap_{i=1}^{\ell} V_{a_i}$$

are disjoint open sets containing A and B, respectively. This shows that X is T_4, and since X is Hausdorff, this implies X is normal. □

[6] Notice that this argument is basically showing X is regular.

2.3 The Subbase Theorem, Tychonoff, and Heine-Borel

We now come to the last major theorem of the chapter, and two of its corollaries.[7]

[7] There are many proofs of the two corollaries we give here, and they are often proven without reference to Alexander's subbase theorem. We here opt for a unified treatment, but the interested reader can find a variety of other proofs in standard texts on the subject.

Theorem 2.3.1 (Alexander's subbase theorem)**.** *Let X be a topological space, and let $\mathcal{B}$ be a subbasis of X. Then X is compact if and only if X fulfills the following condition*

> For any cover $\mathcal{U}$ of X by elements of $\mathcal{B}$, there is a finite subcover of $\mathcal{U}$.

Proof. Since every element of $\mathcal{B}$ is open, it is clear that compact spaces satisfy the given condition.

On the other hand, suppose X satisfies the given condition, and let $\mathbf{N}$ be the set of open covers of X not admitting a subcover. Suppose, by way of contradiction, that $\mathbf{N}$ is non-empty.

By Zorn's Lemma[8], there is a maximal cover $\mathcal{U}$ (under subset inclusion, rather than refinement of covers). Then $\mathcal{B} \cap \mathcal{U}$ is not a cover of X, for if it were, it would admit a finite subcover, contradicting our definition of $\mathcal{U}$. Thus, we can choose a point x in the complement of the points covered by $\mathcal{B} \cap \mathcal{U}$, and an open $V \in \mathcal{U}$ containing x. Since $\mathcal{B}$ is a subbase, we can choose $B_1, \ldots, B_n \in \mathcal{B}$ whose intersection contains x and lies in V.

[8] To apply Zorn's Lemma, we must show that every chain has an upper bound. However, a finite subcover of a union of covers must, in particular be a subset of one of the original covers, and so the union of a chain of covers in $\mathbb{N}$ must also not admit a finite subcover.

Since the B_i are not in $\mathcal{B} \cap \mathcal{U}$[9] each of the covers $\mathcal{U}_i = \{B_i\} \cup \mathcal{U}$ strictly contains $\mathcal{U}$. By the maximality of $\mathcal{U}$, it follows that each of these covers admits a finite subcover $\mathcal{F}_i$. It follows that $\mathcal{F}_i \setminus \{B_i\}$ must cover $X \setminus B_i$. However, this means that

[9] Since otherwise x would be covered by $\mathcal{B} \cap \mathcal{U}$.

$$\mathcal{F} := \bigcup_{i=1}^{n} (\mathcal{F}_i \setminus \{B_i\})$$

covers

$$\bigcup_{i=1}^{n}(X \setminus B_i) = X \setminus \left(\bigcap_{i=1}^{n} B_i \right)$$

and so, since the latter intersection is contained in V, $\mathcal{F} \cup \{V\}$ covers X. This is a finite subcover of $\mathcal{U}$ — a contradiction. □

Corollary 2.3.2 (Tychonoff's Theorem)**.** *Let $\{X_i\}_{i \in I}$ be an collection of topological spaces indexed by a set I. Then $\prod_{i \in I} X_i$ is compact if and only if each X_i is.*

Proof. On the one hand, the continuous image of a compact subset is compact by Lemma 2.2.5. Since the projections $p_j : \prod_{i \in I} X_i \to X_j$ are continuous and surjective, it follows that if the product is compact, each factor is compact.

On the other hand, suppose that each X_i is compact. Let $p_j : \prod_i X_i \to X_j$ denote the projections. By Theorem 2.3.1, it suffices to consider covers coming from the subbase

$$\{p_i^{-1}(U_i) \mid i \in I \text{ and } U_i \in \tau_i\}.$$

Let $\mathcal{U}$ be such a cover. If there is a $j \in I$ such that

$$\mathcal{U}_j := \{p_j(V) \mid V \in \mathcal{U}\} \setminus \{X_j\}$$

is an open cover of X_j, then we are done by the compactness of X_j. If we assume this is not the case, then consider the set

$$\prod_{i\in I}\left(X_i \setminus \bigcup_{U\in\mathcal{U}_i} U\right).$$

By the Axiom of Choice, this is non-empty, and so contains a point $x = (x_i)_{i\in I}$. However, for every $V \in \mathcal{U}$ with $p_j(V) \neq X_j$, $x_j \notin p_j(V)$, and so $x \notin V$. This contradicts the assumption that $\mathcal{U}$ was a cover, completing the proof. □

Corollary 2.3.3 (The Heine-Borel Theorem)**.** *Let $C \subset \mathbb{R}^n$. Then C is compact if and C is closed and bounded.*

Proof. On the one hand, if C is bounded, then it is contained within a product $\prod_{i=1}^n [a_i, b_i]$, which is closed by Corollary 2.3.2. If C is a additionally closed, then as closed subset of a compact space, C is compact by lem 2.2.6.

On the other hand, if C is compact, then since $\mathbb{R}^n$ is a Hausdorff space, C is closed by Lemma 2.2.7. Since $\{B_r(0)\}_{r>0}$ is an open cover, there is a finite subcover of C. Thus, in particular, there is a maximal radius R to this subcover, and so $C \subset B_R(0)$, meaning C is bounded. □

2.4 Locally compact Hausdorff spaces

A general principle in topology is that local versions of properties often suffice to guarantee good behavior. The local version of a property P is often of the form "X is *locally* P if for every point x of X and neighborhood U of x, there is a subneighborhood $V \subset U$ of x which satisfies P." In the case of compactness, this becomes the following definition.

Definition 2.4.1. A topological space X is called *locally compact* if, for every open $U \subset X$, every point $x \in U$ admits an open neighnorhood $x \in V$ such that $\overline{V}$ is compact and $\overline{V} \subset U$.

A quick survey of the literature would reveal, however, that there are dozens of other conditions referred to as local compactness, and they are not, in general, equivalent. This is because local compactness is of greatest use when the space in question is, additionally, Hausdorff, and under that hypothesis, all of the common definitions coincide.

Proposition 2.4.2. *Let X be a Hausdorff space. The following are equivalent.*

1. *X is locally compact.*
2. *For every point $x \in X$, there is an open set U with $x \in U$ and $\overline{U}$ compact.*
3. *For every compact set $K \subset X$, there is an open set U with $\overline{U}$ compact.*
4. *For every compact set $K \subset X$ and every open subset $U \subset X$ with $K \subset U$, there is an open subset V with $K \subset V$, $\overline{V} \subset U$, and $\overline{V}$ compact.*

Proof. It is clear that (4)$\Rightarrow$ (3) $\Rightarrow$ (2).

Now suppose that (2) holds. Let $U \subset X$ be open and let $x \in U$. Using condition (2), we find an open set L with $x \in L$ and $\overline{L}$ compact. It then follows from Proposition 2.2.11 that $\overline{L}$ is normal. The set $U \cap L$ is an open neighborhood of x in $\overline{L}$, and so since $\overline{L}$ is normal, there is an open neighborhood V of X in L whose closure is contained in $U \cap L$. However, since $V \subset U \cap L$ is open in the open set $U \cap L$, it follows that V is open in X, and since $\overline{V}$ is a closed subset of the compact space $\overline{L}$, it follows that $\overline{V}$ is compact. Thus, we have $x \in V \subset \overline{V} \subset U$, with V open and $\overline{V}$ compact, so (1) holds.

Finally, we show that (1) implies (4). Suppose that (1) holds. Let $K \subset X$ be compact and let U be open, with $K \subset U$. For each $x \in K$, choose an open V_x with $x \in V_x$, $\overline{V}_x$ compact, and $\overline{V}_x \subset U$. Then using compactness, there is a finite set $\{V_{x_1}, \dots, V_{x_n}\}$ which covers K. Setting $V = \bigcup_{i=1}^n V_{x_i}$, we see that

$$\overline{V} = \bigcup_{i=1}^{n} \overline{V_{x_i}}$$

is compact and contained in U, precisely as desired. □

We will henceforth only consider locally compact Hausdorff spaces, so the the above conditions coincide.

Examples 2.4.3. $\mathbb{R}$ and $\mathbb{C}$ are locally compact, however, $\mathbb{Q}$ is not. To see this, note that every open set (and thus every neighborhood) of a point $x \in \mathbb{Q}$ must contain a closed interval $[p, q]$. The same basic argument as Example 2.2.4 then shows that no neighborhood can be compact.

2.4.1 One-point compactification

Our first use for locally compact Hausdorff spaces is a general construction to canonically embed a locally compact Hausdorff space into a compact space by adding only a single point.

Definition 2.4.4. Let X be a non-compact topological space. A *compactification* of X is an embedding $\iota : X \to \overline{X}$ such that $\overline{X}$ is compact and the image of X under ι is dense in $\overline{X}$.[10]

[10] As we have not used these definitions often, a reminder: density is Definition 1.3.3 and embeddings are defined in Definition 1.2.23.

Construction 2.4.5 (The one-point compactification)**.** Let X be a locally compact Hausdorff space. We construct a new space X^* called the *one-point compactification*[11] of X as follows. The underlying set of X^* is $X \amalg \{\infty\}$. The topology on X^* is defined by declaring the following sets to be open:

- The open sets of X.
- The sets $U \subset X^*$ whose compliments are compact subsets of $X \subset X^*$.

[11] Or, less often, the *Alexandrov compactification* of X.

Proposition 2.4.6. *Let X be a non-compact, locally compact Hausdorff space. The inclusion $X \subset X^*$ into the one-point compactification is a compactification of X. Moreover, X^* is Hausdorff.*

Proof. It is immediate from the definitions that the inclusion is an embedding. It thus suffices for us to show that X^* is compact and that X is dense in X.

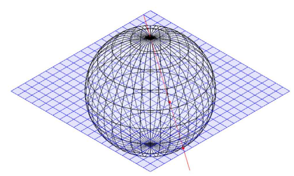
To see compactness, let $\{U_i\}_{i\in I}$ be an open cover of X^*. Let j be an index such that $\infty \in U_j$. Then by the construction of the one-point compactification, $K := U_j^c$ is compact. Since the U_i are a cover of X^*, the remaining U_i's form a cover of K. Since K is compact, we can find a finite subcover $\{U_{i_1}, \dots, U_{i_\ell}\}$ of K, and adding U_j to this yields a finite subcover of X^*.

To see density, let $U \subset X^*$ be an open set containing ∞. Then since X is not compact and $U^c \subset X$ is, we have that $U \cap X \neq \varnothing$. Thus, every open set containing ∞ intersects X and so X is dense in X^*.

Finally, to see that X^* is Hausdorff, note that the required separation property holds by definition for $x, y \in X \subset X$. It thus suffices to show that for $x \in X$ and ∞, the desired separation property holds. However, since X is locally compact Hausdorff, x has a compact neighborhood K. Letting V be an open set with $x \in V \subset K$, we then have that V and K^c are disjoint open sets in X^* containing x and ∞, respectively. □

Example 2.4.7. The one-point compactification of $\mathbb{R}^n$ is homeomorphic to the sphere S^n via the stereographic projection.[12]

[12] The stereographic projection is a continuous map $S^n \setminus \{(0, 0, \dots, 0, 1)\} \to \mathbb{R}^n$ given by projecting a point x in the sphere to the unique intersection between the line through $(0, 0, \dots, 0, 1)$ and x and the hyperplane $\mathbb{R}^n \subset \mathbb{R}^{n+1}$ defined by setting the $n + 1^{\text{st}}$ coordinate equal to 0. Schematically, the stereographic projection maps the lower red point in the following image (which lies on the sphere) to the upper one (which lies on the plane)



2.4.2 The compact-open topology

The compact-open topology is a topology on the set of continuous maps between two topological spaces, which is of particular use in the study of homotopy theory. It has the useful property that, under some mild conditions, maps out of a product of two spaces correspond to maps from one of the two spaces into the compact-open topology of maps out of the other.

Definition 2.4.8. Let X and Y be topological spaces. The *compact-open topology* is a topology on $\mathsf{Top}(X, Y)$ given as follows. For $K \subset X$ compact and $U \subset Y$ open, define

$$B_{K,U} := \{f \in \mathsf{Top}(X, Y) \mid f(K) \subset U\}.$$

The compact-open topology is then the topology generated by the sub-basis[13]

$$\{B_{K,U} \mid K \subset X \text{ compact, and } U \subset Y \text{ open}\}.$$

[13] Parsing this in the case when the target is a metric space, this can be viewed as the topology of uniform convergence on compact intervals.

The topological space defined by the compact-open topology is called the *mapping space* from X to Y. We will denote this space by $\mathrm{Map}(X, Y)$.[14]

[14] The mapping space is also often denoted by Y^X, in recognition of the fact that, for nice enough spaces, it is an *exponential object* in the category Top. We will not discuss exponential objects further here, but the interested reader can find myriad treatments both online and in standard category theory texts.

Example 2.4.9. A key example of mapping spaces is the construction of *path spaces*. Given any space X, the space $\mathrm{Map}([0, 1], X)$ can be interpreted as the space of paths in X. Each point of $\mathrm{Map}([0, 1], X)$ is a path in X, and the topology allows us to think about the "proximity" of paths. The path space is often considered together with the map

$$\begin{aligned} \mathrm{Map}([0,1], X) &\longrightarrow X \times X \\ \alpha &\longmapsto (\alpha(0), \alpha(1)) \end{aligned}$$

or the map

$$\begin{aligned} \mathrm{Map}([0,1],X) &\longrightarrow X \times X \\ \alpha &\longmapsto \alpha(1). \end{aligned}$$

By Corollary 2.4.12 below, both of these maps are continuous; the latter is commonly known as the *path space fibration*.

Lemma 2.4.10. *Let X, Y, and Z be topological spaces, with Y locally compact Hausdorff. Then the composition map*

$$\begin{aligned} \mathrm{comp} : \mathrm{Map}(Y,Z) \times \mathrm{Map}(X,Y) &\longrightarrow \mathrm{Map}(X,Z) \\ (f,g) &\longmapsto f \circ g \end{aligned}$$

is continuous.

Proof. We check continuity on a subbasis. Let $B_{K,U} \subset \mathrm{Map}(X,Z)$ be a sub-basic open and let $(f,g) \in \mathrm{comp}^{-1}(B_{K,U})$. As the continuous image of a compact set, $g(K)$ is compact. Since Y is locally compact Hausdorff, we can find an open V with $\overline{V}$ compact and

$$g(K) \subset V \subset \overline{V} \subset f^{-1}(U).$$

Then $B_{K,g^{-1}(V)} \times B_{\overline{V},U} \subset \mathrm{Map}(Y,Z) \times \mathrm{Map}(X,Y)$ is an open containing (f,g) and such that $B_{K,g^{-1}(V)} \times B_{\overline{V},U} \subset \mathrm{comp}^{-1}(B_{K,U})$, proving continuity. □

Lemma 2.4.11. *Let X be a topological space. Then the map*

$$\begin{aligned} \mathrm{Map}(*,X) &\longrightarrow X \\ f &\longmapsto f(*) \end{aligned}$$

is a homeomorphism.

Proof. Let $U \subset X$ be open. Then $B_{*,U}$ is a subbasic open of $\mathrm{Map}(*,X)$ which maps to U. This immediately shows that the topology generated by the $B_{*,U}$'s is precisely the topology of X, as desired. □

Corollary 2.4.12. *Let X and Y be spaces, where X is locally compact Hausdorff. Then the evaluation map*

$$\begin{aligned} \mathrm{ev} : X \times \mathrm{Map}(X,Y) &\longrightarrow Y \\ (x,f) &\longmapsto f(x) \end{aligned}$$

is continuous.

Proof. The evaluation map is simply the composite

$$X \times \mathrm{Map}(X,Y) \longrightarrow \mathrm{Map}(*,X) \times \mathrm{Map}(X,Y) \xrightarrow{\mathrm{comp}} \mathrm{Map}(*,Y) \longrightarrow Y$$

and so is continuous by the previous two lemmata. □

The most important property of the compact-open topology is the *tensor-hom adjunction*[15]

[15] Also called the *currying isomorphism*, if you happen to be a computer scientist.

Proposition 2.4.13. *Let X, Y, and Z be topological spaces, and suppose that Y is a locally compact Hausdorff space. The map*

$$\begin{aligned} \mathsf{Top}(X \times Y, Z) &\longrightarrow \mathsf{Top}(X, \mathrm{Map}(Y, Z)) \\ f &\longmapsto (x \mapsto (y \mapsto f(x,y))). \end{aligned}$$

is a well-defined bijection. Moreover, when X is also locally compact Hausdorff, then this map induces a homeomorphism of spaces[16]

$$\mathrm{Map}(X \times Y, Z) \longrightarrow \mathrm{Map}(X, \mathrm{Map}(Y, Z)).$$

[16] These homeomorphisms, and the bijections underlying them, are *natural* — a term we will define in the next chapter. While we will not need to use this fact, and thus do not discuss it further, it is worth keeping in mind, as it presages an important topic in category theory, algebra, and topology: adjunctions.

Proof. Given $f : X \times Y \to Z$, define $f_x : Y \to Z$ to be the map $y \mapsto f(x,y)$. If f is continuous, then f_x is the composite

$$* \times Y \longrightarrow X \times Y \xrightarrow{f} Z$$

and thus is continuous. Denote by $c_f : X \to \mathrm{Map}(Y, Z)$ the map which sends x to f_x. Then if f is continuous, let $B_{K,U} \subset \mathrm{Map}(Y, Z)$ be a subbasic open, and let $x \in c_f^{-1}(B_{K,U})$, i.e., $\{x\} \times K \subset f^{-1}(U)$. Since $f^{-1}(U)$ is open, we can choose, for each $k \in K$, opens $x \in V_k \subset X$ and $k \in W_k \subset Y$ such that $V_k \times W_k \subset f^{-1}(U)$. Thus the W_k cover K, and so we can find a finite subcover. Taking the intersection of the corresponding V_k's, we obtain an open V such that $\{x\} \times K \subset V \times K \subset f^{-1}(U)$.[17] As such, $x \in V \subset c_f^{-1}(B_{K,U})$, so that c_f is continuous.

[17] This construction is often given separately as the *Tube Lemma*.

On the other hand, let $g : X \to \mathrm{Map}(Y, Z)$ be continuous, and denote by $\widehat{g} : X \times Y \to Z$ the map $(x, y) \mapsto g(x)(y)$. Let $U \subset Z$ be open and let $(x, y) \in \widehat{g}^{-1}(U)$. Since $g(x)$ is continuous and Y is locally compact Hausdorff, we can choose $y \in V \subset \overline{V} \subset g(x)^{-1}(U)$ such that V is open and $\overline{V}$ is compact. Then $W = g^{-1}(B_{\overline{V},U})$ is open and contains x. Thus $W \times V$ is an open containing (x, y) and contained in $\widehat{g}^{-1}(U)$, showing continuity.

The first statement follows immediately, since it is clear that the constructions $f \mapsto c_f$ and $g \mapsto \widehat{g}$ are mutually inverse. To see that second statement is simply checking the continuity of these two constructions, which we leave as an exercise. □

3

Interlude: some category theory

In this chapter, we digress into the basics of category theory and make rigorous some of the concepts we have begun using in the previous chapters.

3.1 *Categories, functors, and natural transformations*

In some sense, a category is a *setting in which to do mathematics*. It tells us what kind of mathematical objects we are considering, and how we relate them. More formally

Definition 3.1.1. A *category* $\mathcal{C}$ consists of

- A set[1] $\mathrm{Ob}(\mathcal{C})$, called the *objects* of $\mathcal{C}$.
- For each pair of objects $x, y \in \mathrm{Ob}(\mathcal{C})$, a set[2] $\mathcal{C}(x, y)$ called the *morphisms* from x to y. We will write $f : x \to y$ for a morphism $f \in \mathcal{C}(x, y)$.
- For every triple of objects $x, y, z \in \mathrm{Ob}(\mathcal{C})$, a map of sets
$$\circ : \mathcal{C}(y, z) \times \mathcal{C}(x, y) \longrightarrow \mathcal{C}(x, z)$$
$$(f, g) \longmapsto f \circ g$$
called *composition*.
- For every $x \in \mathrm{Ob}(\mathcal{C})$, a distinguished element $\mathrm{id}_x \in \mathcal{C}(x, x)$ called *the identity on x*.

These data are then required to satisfy the following conditions.

- (Associativity) For every triple of morphisms $x \xrightarrow{h} y$, $y \xrightarrow{g} z$, and $z \xrightarrow{f} w$, the composites
$$f \circ (g \circ h) = (f \circ g) \circ h$$
are equal.
- (Unitality) For $f : x \to y$ and $g : y \to x$, we have
$$f \circ \mathrm{id}_x = f \quad \text{and} \quad \mathrm{id}_x \circ g = g.$$

[1] What we mean here should be made a little more precise, as the definition here could lead to set-theoretic paradoxes. I will, for the most part, sweep such technicalities under the rug, and ask the reader to trust that they can be resolved. The more naturally suspicious reader may satisfy their curiousity in, for example, [6].

[2] *Vide supra.*

The associativity and unitality conditions are often presented diagrammatically. They then say that, for all $a, b, c, d \in \mathrm{Ob}(\mathcal{C})$, the diagram

$$\begin{tikzcd}
\mathcal{C}(c,d) \times \mathcal{C}(b,c) \times \mathcal{C}(a,b) \arrow[r, "\mathrm{id}\times\circ"] \arrow[d, "\circ\times\mathrm{id}"] & \mathcal{C}c, d \times \mathcal{C}(a,c) \arrow[d, "\circ"] \\
\mathcal{C}(b,d) \times \mathcal{C}(a,b) \arrow[r, "\circ"'] & \mathcal{C}(a,d)
\end{tikzcd}$$

the diagram

$$\begin{tikzcd}
* \times \mathcal{C}(a,b) \arrow[r, "\{\mathrm{id}_b\}\times\mathrm{id}"] \arrow[dr, "\cong"'] & \mathcal{C}(b,b) \times \mathcal{C}(a,b) \arrow[d, "\circ"] \\
& \mathcal{C}(a,b)
\end{tikzcd}$$

and the diagram

$$\begin{tikzcd}
\mathcal{C}(a,b) \times * \arrow[r, "\mathrm{id}\times\{\mathrm{id}_a\}"] \arrow[dr, "\cong"'] & \mathcal{C}(a,b) \times \mathcal{C}(a,a) \arrow[d, "\circ"] \\
& \mathcal{C}(a,b)
\end{tikzcd}$$

all commute.

This seems like a lot to unpack, but really it amounts to an axiomatization of the things which we do with functions in mathematics. We can compose functions, there are identity functions, and these behave in precisely the way we described above. So how does a category represent a 'place to do mathematics'?

Examples 3.1.2.

1. There is a category Set whose objects are sets[3], and such that, for every two sets X and Y, we have

$$\mathsf{Set}(X,Y) := \{\text{functions } f : X \to Y\}.$$

The composition is the usual composition of functions, and the identies are the identity maps.

[3] Here's where the set-theoretic technicalities come into play: This definition, on its face, would require us to have a 'set of all sets' which ends up leading us down the dark path to Russell's paradox. The usual way of resolving this is to define 'little sets' and 'big sets' such that the set of all little sets is a big set, and then defining Set to be the category of all little sets. More technically, the frameworks of *inaccessible cardinals* and *Grothendieck universes* are typical ways to address this issue, as is the Von Neumann-Bernays-Godel axiomatization of sets and classes. A good summary of approaches to the foundations of category theory can be found in [3].

2. There are categories Grp and Ab whose objects are groups and abelian groups, respectively, and whose morphisms are group homomorphisms.

3. There is a category Top whose objects are topological spaces, and whose morphisms are continuous maps of topological spaces.

4. There is a category Top_* whose objects are *pointed topological spaces* — pairs (X, x), where X is a topological space and $x \in X$ is a basepoint. Morphisms in Top_* from $(X, x) \to (Y, y)$ are continuous maps $f : X \to Y$ such that $f(x) = y$.

5. Let $(P, \geq)$ be a partially ordered set. We can define a category $\mathcal{P}$ with $\mathrm{Ob}(\mathcal{P}) := P$, and a unique morphism $p \to p'$ if and only if $p \leq p'$.

We will often have cause to sketch categories arising from posets. Let us consider, for instance, the category associated to the power set $\mathbb{P}(\{0,1\})$, ordered by inclusion. We will draw an arrow for every non-identity morphism:

$$\begin{tikzcd} \{0\} \arrow[r] & \{0,1\} \\ \emptyset \arrow[u] \arrow[ur] \arrow[r] & \{1\} \arrow[u] \end{tikzcd}$$

In the picture, we assume that the category is a poset, e.g. that the arrow from $\emptyset \to \{0,1\}$ is the composite of the arrows $\emptyset \to \{1\} \to \{0,1\}$ and the composite of the arrows $\emptyset \to \{0\} \to \{0,1\}$. For ease of drawing, we will sometimes omit composite arrows:

$$\begin{tikzcd} \{0\} \arrow[r] & \{0,1\} \\ \emptyset \arrow[u] \arrow[r] & \{1\} \arrow[u] \end{tikzcd}$$

6. Let G be a group. Then there is a category BG with a single object $*$, and $BG(*, *) = G$. The composition is given by $g \circ f = g \cdot f$, and the identity is the neutral element of G.

7. We define *the simplex category* Δ to be the category whose objects are the totally ordered sets $[n] := \{0, 1, \ldots, n\}$ for $n \geq 0$, and whose morphisms $[n] \to [m]$ are maps of sets $f : [n] \to [m]$ such that, if $a \leq b$, $f(a) \leq f(b)$.

8. Let $\mathcal{C}$ be a category. We can define the *opposite category* $\mathcal{C}^{\mathrm{op}}$ to have $\mathrm{Ob}(\mathcal{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathcal{C})$, and, for every $x, y \in \mathrm{Ob}(\mathcal{C}^{\mathrm{op}})$

$$\mathcal{C}^{\mathrm{op}}(x,y) = \mathcal{C}(y,x).$$

The composition and identities are the same maps as in $\mathcal{C}$. One thinks of the opposite category as 'reversing the arrows' of $\mathcal{C}$, i.e. reversing the direction of morphisms.

9. There is a category **Metric** whose objects are metric spaces, and whose morphisms are continuous maps.

10. Given two categories $\mathcal{C}$ and $\mathcal{D}$, we can form a new category $\mathcal{C} \times \mathcal{D}$, called the *product category.* The objects are $\mathrm{Ob}(\mathcal{C} \times \mathcal{D}) = \mathrm{Ob}(\mathcal{C}) \times \mathrm{Ob}(\mathcal{D})$, and the morphisms are given by

$$\mathcal{C} \times \mathcal{D}((x_1, x_2), (y_1, y_2)) := \mathcal{C}(x_1, y_1) \times \mathcal{D}(x_2, y_2).$$

The identities are $\mathrm{id}_{(x,y)} = (\mathrm{id}_x, \mathrm{id}_y)$.

Definition 3.1.3. Let $\mathcal{C}$ be a category, and let $f : x \to y$ be a morphism in $\mathcal{C}$. We call $g : y \to x$ a *right inverse*[4] of f if $f \circ g = \mathrm{id}_y$. We say that g is a *left inverse* of f if $g \circ f = \mathrm{id}_x$. If g is both a left and right inverse to f, then we call g the *inverse* of f, and say that f is an isomorphism in $\mathcal{C}$.

[4] This is also sometimes called a *section* of f.

Example 3.1.4. In a category $\mathcal{C}$, every identity morphism id_x is an isomorphism. In the category BG associated to a group G, every morphism is an isomorphism.

Exercise 3.1.5. Justify calling g *the* inverse of f by showing that any two inverses of f are equal. Show that every identity is an isomorphism.[5]

[5] This is analogous to showing that inverses are unique in a group G, and showing that the neutral element $e \in G$ has an inverse.

Exercise 3.1.6. Show that the isomorphisms in Set, Top and Grp are, respectively, the bijections, homeomorphisms, and group isomorphisms. Show that the only isomorphisms in the simplex category Δ are the identities.

Lemma 3.1.7. *Let $\mathcal{C}$ be a category. The isomorphisms of $\mathcal{C}$ satisfy the* 2-out-of-3 *property: given morphisms $g : x \to y$ and $f : y \to z$ in $\mathcal{C}$, if any two of the morphisms f, g, and $f \circ g$ are isomorphisms, then the third must be as well.*[6]

[6] In particular, the composition of isomorphisms is an isomorphism.

Proof. Since the inverse of an isomorphism is itself an isomorphism, it suffices for us to write the proof in the case where f and g are isomorphisms. Let $f^{-1} : z \to y$ and $g^{-1} : y \to x$ be their respective inverses. Then we can simply compute

$$(f \circ g)(g^{-1} \circ f^{-1}) = f \circ (g \circ g^{-1}) \circ f^{-1} = f \circ \mathrm{id}_y \circ f^{-1} = f \circ f^{-1} = \mathrm{id}_z$$

and similarly

$$(g^{-1} \circ f^{-1}) \circ (f \circ g) = \mathrm{id}_x.$$

Thus $g^{-1} \circ f^{-1}$ is inverse to $f \circ g$ and the lemma is proved. □

Lemma 3.1.8. *Functors preserve isomorphism. That is, given a functor $F : \mathcal{C} \to \mathcal{D}$, and an isomorphism $f : x \to y$ in $\mathcal{C}$, then $F(f) : F(x) \to F(y)$ is also an isomorphism.*

Proof. Follows by unwinding the definitions. □

From categories we can also extract more familiar algebraic structures: monoids and groups.

Definition 3.1.9. Let $\mathcal{C}$ be a category and $x \in \mathrm{Ob}(\mathcal{C})$ an object. An *endomorphism* of x in $\mathcal{C}$ is a morphism $f : x \to x$ in $\mathcal{C}$. An *automorphism* of $x \in \mathcal{C}$ is an isomorphism $f : x \to x$ in $\mathcal{C}$. We write

$$\mathrm{End}_{\mathcal{C}}(x) := \mathcal{C}(x, x)$$

for the set of endomorphisms of x, and $\mathrm{Aut}_{\mathcal{C}}(x) \subset \mathrm{End}_{\mathcal{C}}(x)$ for the subset consisting of automorphisms. Notice that the composition of morphisms of $\mathcal{C}$ provides a binary operation

$$\mathrm{End}_{\mathcal{C}}(x) \times \mathrm{End}_{\mathcal{C}}(x) \longrightarrow \mathrm{End}_{\mathcal{C}}(x)$$

and, by Lemma 3.1.7, this induces a binary operation

$$\mathrm{Aut}_{\mathcal{C}}(x) \times \mathrm{Aut}_{\mathcal{C}}(x) \longrightarrow \mathrm{Aut}_{\mathcal{C}}(x).$$

Lemma 3.1.10. *Let $\mathcal{C}$ be a category and $x \in \mathrm{Ob}(\mathcal{C})$ an object. The composition of morphisms makes $\mathrm{End}_{\mathcal{C}}(x)$ into a monoid with neutral element id_x and makes $\mathrm{Aut}_{\mathcal{C}}(x)$ into a group with neutral element id_x.*

Proof. Associativity and unitality follow directly from associativity and unitality for the composition of morphisms in a category. In the case of $\mathrm{Aut}_{\mathcal{C}}(x)$, the inverse morphism defines an inverse element in the usual group-theoretic sense. □

So how do we relate different categories to one another? We need to define a 'map of categories' which tells us not only how objects relate, but also how morphisms relate.

Definition 3.1.11. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A *functor* $F : \mathcal{C} \to \mathcal{D}$ consists of a map of sets $F : \mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D})$, and, for each $x, y \in \mathrm{Ob}(\mathcal{C})$ a map of sets $F : \mathcal{C}(x, y) \to \mathcal{D}(F(x), F(y))$ such that:

- For every $x \in \mathrm{Ob}(\mathcal{C})$, $F(\mathrm{id}_x) = \mathrm{id}_{F(x)}$.
- For every pair of composable morphisms $x \xrightarrow{g} y \xrightarrow{f} z$ in $\mathcal{C}$, we have that $F(f \circ g) = F(f) \circ F(g)$.

One can compose functors in the obvious way — simply compose the defining maps of sets. We will leave it as an exercise to the reader to check that this is actually a functor:

Exercise. Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{B} \to \mathcal{C}$ be functors. Give a definition of the composite functor $F \circ G : \mathcal{B} \to \mathcal{D}$. Show that your definition is, in fact, a functor.

Examples 3.1.12.

1. For every category $\mathcal{C}$, there is an 'identity' functor $\mathrm{id}_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$ which sends $x \mapsto x$ and $f \mapsto f$.
2. There is a functor $U : \mathsf{Top} \to \mathsf{Set}$ which sends each topological space to its underlying set, and each continuous map to the underlying map of sets. (See 1.2.17)
3. Our results in Propositions 1.0.3 and 1.1.2 can be used to show that there is a functor $F : \textbf{Metric} \to \mathsf{Top}$, which sends each metric space to it's underlying topological space.

Lemma 3.1.13. *For a group G, a functor $F : BG \to \mathsf{Set}$ consists of a set X and a group action $G \curvearrowright X$.*

Proof. Let as an exercise. □

Construction 3.1.14. Let $\mathcal{C}$ be a category, and $S \subset \mathrm{Ob}(\mathcal{C})$ be a set of objects. We define the *full subcategory of $\mathcal{C}$ on S* to be the category $\mathcal{D}$ with $\mathrm{Ob}(\mathcal{D}) = S$ and, for each $x, y \in S$

$$\mathcal{D}(x, y) = \mathcal{C}(x, y).$$

Definition 3.1.15. There is a category Cat whose objects are categories, and whose morphisms are functors.[7]

[7] This runs into the same set-theoretic difficulties alluded to earlier. While we won't go into these here, it is important to note that Cat, though it is a category, is not an object of Cat. That is, the category of categories does not contain itself.

Lemma 3.1.16. *Let $\mathcal{C}$ and $\mathcal{D}$ be categories and let $x \in \mathrm{Ob}(\mathcal{C})$. A functor $F : \mathcal{C} \to \mathcal{D}$ induces a homomorphism*

$$\mathrm{End}_{\mathcal{C}}(x) \longrightarrow \mathrm{End}_{\mathcal{D}}(F(x))$$

of monoids and a homomorphism

$$\mathrm{Aut}_{\mathcal{C}}(x) \longrightarrow \mathrm{Aut}_{\mathcal{C}}(x)$$

of groups.

Proof. In both cases the binary operation is composition. Since F preserves composition, the lemma follows. $\square$

Exercise 3.1.17. The totally ordered sets $[n]$ from the definition of the simplex category are, in particular, posets, and thus give rise to categories $\overline{n}$. Show that the simplex category Δ can be identified with the full subcategory of Cat on the objects $\overline{n}$, $n \geq 0$.

3.2 A note on commuting diagrams

Before we continue, let us briefly revise Definition 1.2.8, and talk about commutative diagrams.

Definition 3.2.1. A *commutative diagram* in a category $\mathcal{C}$ is a functor $F : P \to \mathcal{C}$ out of a poset category P.

This definition is very important, so let us unpack some basic facts about it.

Firstly, if the partial order $\leq$ on P is generated under transitivity and reflexivity (i.e., by adding composites and identities) by some set $\{x_i \leq y_i\}_{i \in I}$ of relations, then the functor F is uniquely determined by its value on the objects of P and its value on the (unique) morphisms $x_i \to y_i$ in P. For example, if we let P be the poset of subsets of $\{0, 1\}$ ordered by inclusion, then we can simply remember the values of F on the morphisms $\varnothing \to \{0\}$, $\varnothing \to \{1\}$,$\{0\} \to \{0, 1\}$, and $\{1\} \to \{0, 1\}$. We very often draw these data, as we have already been doing with commutative diagrams, as follows.

$$\begin{tikzcd} F(\varnothing) \arrow[r] \arrow[d] & F(\{0\}) \arrow[d] \\ F(\{1\}) \arrow[r] & F(\{0,1\}) \end{tikzcd}$$

Secondly, the condition that P is a poset means that, in our picture above, any two composite morphisms which can be equal will be equal. This is, indeed, a common definition of commutative diagrams. If, for instance, I draw a square

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "k"'] & Y \arrow[d, "g"] \\ Z \arrow[r, "h"'] & W \end{tikzcd}$$

and say that it *commutes*, this means that the data pictured define a functor out of the (unique up to isomorphism) poset generated by the arrows, in this case, the power set of $\{0,1\}$.

3.3 A note on duality

One key construction above was the opposite category $\mathcal{C}^{\mathrm{op}}$ of a category $\mathcal{C}$. This is like $\mathcal{C}$ in all ways, except that we view the morphisms as going in the opposite direction. As we have already seen, definitions in category theory come in pairs, with each definition in the pair given by reversing the direction of all the arrows in the definition. The opposite category gives us a nice way to formalize it.

Definition 3.3.1. If a *foo in* $\mathcal{C}$ is a categorical definition[8] then a *cofoo in* $\mathcal{C}$ is defined to be a foo in $\mathcal{C}^{\mathrm{op}}$.

[8] By this, we loosely mean a definition made in terms of objects, morphisms, composition, and identities.

3.4 Naturality and equivalence

Functors play the same role in relating categories to one another as homomorphisms play in relating groups to one another. However, there is a bit of an issue. The isomorphisms in Cat (called, with the blithe directness typical of mathematicians, *isomorphisms of categories*) are a little useless.

Example 3.4.1. Let's define a category $\mathsf{Set}^{\mathrm{un}}$ to have precisely one object of each cardinality, and morphisms the maps of sets. It shouldn't really matter whether we work in Set or $\mathsf{Set}^{\mathrm{un}}$, but the two categories aren't isomorphic. To see this, we need only note that the isomorphism class of singletons in $\mathsf{Set}^{\mathrm{un}}$ consists of a single object, and the isomorphism class of singletons in Set is infinite.

How do we find a way around this problem? Well, we can effectively do what we did when talking about homotopy: try to define a relation between functors, and then define a weaker notion of equivalence.

Definition 3.4.2. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and $F, G : \mathcal{C} \to \mathcal{D}$ be functors. A *natural transformation from F to G* $\alpha : F \Rightarrow G$ consists of a morphism $\alpha_x : F(x) \to G(x)$ in $\mathcal{D}$ for every $x \in \mathrm{Ob}(\mathcal{C})$ such that, for every morphism $f : x \to y$ in $\mathcal{C}$ the following diagram commutes:[9]

$$\begin{tikzcd} F(x) \arrow[r, "\alpha_x"] \arrow[d, "F(f)"'] & G(x) \arrow[d, "G(f)"] \\ F(y) \arrow[r, "\alpha_y"] & G(y). \end{tikzcd}$$

[9] When we say a diagram commutes, we mean that it follows the convention described above for drawing posets: if two composites can be equal, they are equal. In this case, this would mean in equations that $G(f) \circ \alpha_x = \alpha_y \circ F(f)$.

We call a natural transformation a *natural isomorphism* if each of its components α_x is an isomorphism in $\mathcal{D}$.

Examples 3.4.3.

1. There is an 'identity' natural isomorphism $\mathrm{Id}_F : F \to F$ for every $F : \mathcal{C} \to \mathcal{D}$. The components of Id_F are $(\mathrm{Id}_F)_x := \mathrm{id}_{F(x)}$

2. Consider the two functors $\mathrm{id}, C : \mathsf{Grp} \to \mathsf{Grp}$, where $C(G) = G/[G,G]$ is the abelianization. There is a natural transformation $\alpha : \mathrm{id} \Rightarrow C$ with component α_G given by the quotient map $G \to G/[G,G]$.

Construction 3.4.4. Let $\beta : F \Rightarrow G$ and $\alpha : G \Rightarrow H$ be natural transformations among functors $F, G, H : \mathcal{C} \to \mathcal{D}$. We define the composite $\alpha \circ \beta$ via the formula, for all $x \in \mathcal{C}$, $(\alpha \circ \beta)_x = \alpha_x \circ \beta_x$. To check that this is natural, note that for each morphism $f : x \to y$ in $\mathcal{C}$ we have a pair of squares:

$$\begin{tikzcd}
F(x) \arrow[r, "\beta_x"] \arrow[d, "F(f)"'] & G(x) \arrow[r, "\alpha_x"] \arrow[d, "G(f)"] & H(x) \arrow[d, "H(f)"] \\
F(y) \arrow[r, "\beta_y"'] & G(y) \arrow[r, "\alpha_y"'] & H(y).
\end{tikzcd}$$

where the left and right hand squares both commute. It is easy to verify from the diagram that this means the external square commutes as well[10]

[10] This is a kind of *pasting law*, and we will often make use of such arguments without comment.

Exercise 3.4.5. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. Show that there is a category $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ with objects given by functors from $\mathcal{C}$ to $\mathcal{D}$, and morphisms given by natural transformations.

This notion of natural transformation is our categorical analogue of homotopy. This can even be made more precise, as the following exercise shows:

Exercise 3.4.6. Let $\bar{1}$ be the category from Exercise 3.1.17 — that is, the category with two objects 0 and 1, and a unique non-identity morphism from 0 to 1. Check that, for any category $\mathcal{C}$ There are two functors

$$\begin{aligned} \iota_i : \mathcal{C} &\to \bar{1} \times \mathcal{C} \\ x &\mapsto (i, x) \end{aligned}$$

where $i = 0$ or $i = 1$. Given functors $F, G : \mathcal{C} \to \mathcal{D}$, define

$$\mathrm{Nat}(F, G) := \left\{ H \in \mathrm{Fun}(\bar{1} \times \mathcal{C}, \mathcal{D}) \mid H \circ \iota_0 = F \text{ and } H \circ \iota_1 = G \right\}$$

and show that $\mathrm{Nat}(F, G)$ is in bijection with the set of natural transformations from F to G.

Definition 3.4.7. We say that a functor $F : \mathcal{C} \to \mathcal{D}$ is an *equivalence of categories* if there is a functor $G : \mathcal{D} \to \mathcal{C}$, and natural isomorphisms $\alpha : G \circ F \cong \mathrm{Id}_{\mathcal{C}}$ and $\beta : F \circ G \cong \mathrm{Id}_{\mathcal{D}}$. We then say that the categories $\mathcal{C}$ and $\mathcal{D}$ are *equivalent*, and call G a *weak inverse* to F.

Definition 3.4.8. Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. We call F

1. *full* if, for every $x, y \in \mathcal{C}$, the map $F : \mathcal{C}(x, y) \to \mathcal{D}(F(x), F(y))$ is surjective.

2. *faithful* if , for every $x, y \in \mathcal{C}$, the map $F : \mathcal{C}(x, y) \to \mathcal{D}(F(x), F(y))$ is injective.

3. *fully faithful* if it is both full and faithful.

4. *essentially surjective* if, for every $y \in \mathcal{D}$, there exists an $x \in \mathcal{C}$ such that $F(x)$ is isomorphic to y.

The term *faithful* in category theory has a precise analogue in representation theory. A representation ρ of a group G on a vector space V is said to be *faithful* if every group element $g \in G$ is represented by a *distinct* linear map $\rho(g) : V \to V$. Rephrasing this in categorical terms, a representation of G on a vector space V is the same thing as a functor

$$\rho : BG \longrightarrow \mathsf{Vect}_k$$

which sends the unique object $*$ of BG to the vector space V. A representation is faithful if and only if the associated functor is faithful.

Examples 3.4.9.

1. Consider the functor $F : \mathsf{Top} \to \mathsf{Set}$ which sends each topological space (X, τ_X) to the underlying set X, and each continuous map to the underlying map of sets. This functor is faithful, since a continuous map $X \to Y$ is uniquely determined by the underlying map of sets, and it is essentially surjective, since for any set X, we can define a topology on X. However, F is *not* full, since, given two spaces (X, τ_X) and (Y, τ_Y), not every map of sets $X \to Y$ defines a continuous function.

2. Consider a homomorphism of groups $\phi : G \to H$. The induced functor $BG \to BH$ is always essentially surjective, is full if and only if ϕ is surjective, and is faithful if and only if ϕ is injective.

3. Consider the 'inclusion' $\iota : \Delta \to \mathsf{Cat}$, which sends each poset $[n]$ to the associated poset category, and each monotone map to the associated functor. The functor ι is fully faithful, however, it is not essentially surjective, since not every category is isomorphic to a category with finitely many objects.

4. Consider a category Bun whose objects are triples (M, V, p) where M is a smooth manifold, and $p : V \to M$ is a vector bundle on M. The morphisms of Bun are commutative diagrams

$$\begin{tikzcd} V \arrow[r, "\phi"] \arrow[d, "p"'] & W \arrow[d, "q"] \\ M \arrow[r, "f"'] & N \end{tikzcd}$$

where ϕ is a morphism of vector bundles and f is a smooth map. We can define a functor $F : \text{Bun} \to \mathsf{Top}$ which sends each diagram as above to the underlying continuous map between topological spaces

$$f : M \to N.$$

The functor F is not full, not faithful, and not essentially surjective. It is not full because not every continuous map between smooth manifolds is smooth; it is not faithful because there are multiple bundle maps which lie over the same smooth map; and it is not essentially surjective because not every topological space admits the structure of a smooth manifold.

Proposition 3.4.10. *Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. Then the following are equivalent:*

1. *F is fully faithful and essentially surjective.*

2. F is an equivalence of categories.

Proof. We first show 2. $\Rightarrow$ 1. Let G be a weak inverse to F, and α and β the natural isomorphisms displaying the weak invertibility of F. Given y in $\mathcal{D}$, the y-component of β is an isomorphism $\beta_y : F(G(y)) \xrightarrow{\cong} y$, showing that F is essentially surjective.

Let $f : x \to y$ be a morphism in $\mathcal{C}$. Then, by the naturality of α, the diagram

$$\begin{tikzcd} G(F(x)) \arrow[r, "\alpha_x"] \arrow[d, "G(F(f))"'] & x \arrow[d, "f"] \\ G(F(y)) \arrow[r, "\alpha_y"'] & y \end{tikzcd}$$

commutes. Since α_x is invertible, this means that

$$f = \alpha_y \circ G(F(f)) \circ \alpha_x^{-1},$$

i.e., f is uniquely determined by $G(F(f))$. Thus $G \circ F : \mathcal{C}(x, y) \to \mathcal{C}(G(F(x)), G(F(y)))$ is injective, and hence, $F : \mathcal{C}(x, y) \to \mathcal{D}(F(x), F(y))$ is injective, and F is faithful. Note that we can make the same argument with β to find that G is faithful.

Now let $h : F(x) \to F(y)$ in $\mathcal{D}$ be a morphism, and define $f := \alpha_y \circ G(h) \circ \alpha_x^{-1}$. Again we find a commutative square:

$$\begin{tikzcd} G(F(x)) \arrow[r, "\alpha_x"] \arrow[d, "G(h)"'] & x \arrow[d, "f"] \\ G(F(y)) \arrow[r, "\alpha_y"'] & y \end{tikzcd}$$

Which implies that $G(F(f)) = G(h)$. However, since G is faithful, this implies $F(f) = h$, and thus, $F : \mathcal{C}(x, y) \to \mathcal{D}(F(x), F(y))$ is surjective and F is full.

We now show 1. $\Rightarrow$ 2. Suppose F is essentially surjective and fully faithful. For every $y \in \mathrm{Ob}(\mathcal{D})$, we choose[11] an object which we call $G(y) \in \mathrm{Ob}(\mathcal{C})$ together with an isomorphism $\beta_y : F(G(y)) \xrightarrow{\cong} y$ (by essential surjectivity). We then define, for every $h : x \to y$ in $\mathcal{D}$, the morphism $G(h) := F^{-1}(\beta_y^{-1} \circ h \circ \beta_x) \in \mathcal{C}(G(x), G(y))$.[12]

Since

$$G(\mathrm{id}_x) = F^{-1}(\beta_x^{-1} \circ \mathrm{id}_x \circ \beta_x) = F^{-1}(\beta_x^{-1} \circ \beta_x) = F^{-1}(\mathrm{id}_x),$$

we see that $G(\mathrm{id}_x) = \mathrm{id}_{G(x)}$. Moreover, since

$$G(f \circ g) = F^{-1}(\beta_z^{-1} \circ f \circ g \circ \beta_x) = F^{-1}(\beta_z^{-1} \circ f \circ \beta_y \circ \beta_y^{-1} \circ g \circ \beta_x)$$

we have $G(f \circ g) = G(f) \circ G(g)$. Thus, G defines a functor.[13] Tracing the definitions, it is immediate that the squares

$$\begin{tikzcd} F(G(x)) \arrow[r, "\beta_x"] \arrow[d, "F(G(f))"'] & x \arrow[d, "f"] \\ F(G(y)) \arrow[r, "\beta_y"'] & y \end{tikzcd}$$

[11] Note that this requires the axiom of choice. As a rule, we will assume the axiom of choice in our categorical arguments.

[12] We can do this precisely because F is fully faithful, so the map of sets

$$F : \mathcal{C}(G(x), G(y)) \to \mathcal{D}(F(G(x)), F(G(y)))$$

is a bijection.

[13] We have used the fully faithfulness of F for both computations here, as well as the functoriality of F.

commute for any $f : x \to y$ in $\mathcal{D}$, so that $\beta : F \circ G \Rightarrow \mathrm{Id}_{\mathcal{D}}$ is a natural isomorphism.

We now seek to obtain the natural isomorphism $\alpha : G \circ F \Rightarrow \mathrm{Id}_{\mathcal{C}}$. Note that $\beta_{(F(x))} : F(G(F(x))) \to F(x)$ is an isomorphism. By the fully faithfulness of F, there is a unique isomorphism[14] $\alpha_x := F^{-1}(\beta_{F(x)}) : G(F(x)) \to x$. Since β is a natural transformation, for any $f : x \to y$ in $\mathcal{C}$, we get a commutative square

[14] We leave it as an exercise to the interested reader to check that this is actually an isomorphism using the fully-faithfulness of F.

$$\begin{tikzcd} F(G(F(x))) \arrow[r, "\beta_{F(x)}"] \arrow[d, "F(G(F(f)))"'] & F(x) \arrow[d, "F(f)"] \\ F(G(F(y))) \arrow[r, "\beta_{F(y)}"'] & F(y) \end{tikzcd}$$

i.e. $F(f \circ \alpha_x) = F(\alpha_y \circ G(F(f)))$. But, since F is fully faithful, this implies

$$f \circ \alpha_x = \alpha_y \circ G(F(f))$$

and thus the square

$$\begin{tikzcd} G(F(x)) \arrow[r, "\alpha_x"] \arrow[d, "G(F(f))"'] & x \arrow[d, "f"] \\ G(F(y)) \arrow[r, "\alpha_y"'] & y \end{tikzcd}$$

commutes. This means that α is natural, and thus provides the desired natural isomorphism $\alpha : G \circ F \Rightarrow \mathrm{Id}_{\mathcal{C}}$. □

Exercise 3.4.11. Show that the two categories from Example 3.4.1 are equivalent.

3.5 Universality

One of the most important uses of category theory is to give 'universal' constructions, which determine objects in a category by conditions on their morphisms. Before getting into the general definition of a universal property, let's look at a familiar example.

Example 3.5.1. Let k be a field, and define Vect_k be the category whose objects are k-vector spaces, and whose morphisms are k-linear maps. Given $V, W \in \mathrm{Ob}(\mathsf{Vect}_k)$, a *tensor product* of V and W in Vect_k is an object $V \otimes W$ together with a bilinear map $l : V \times W \to V \otimes W$ satisfying the following

Universal Property: Any bilinear map $f : V \times W \to U$ of k-vector spaces factors uniquely though l. That is, given a bilinear map $f : V \times W \to U$, there is a unique k-linear map $\tilde{f} : V \otimes W \to U$ such that the diagram

$$\begin{tikzcd} V \times W \arrow[r, "f"] \arrow[d, "l"'] & U \\ V \otimes W \arrow[ur, "\tilde{f}"'] & \end{tikzcd}$$

commutes. This establishes a bijection (for any $U \in \mathrm{Ob}(\mathsf{Vect}_k)$) between bilinear maps $V \times W \to U$ and linear maps $V \otimes W \to U$.

There is an explicit construction of $V \otimes W$ as a quotient of $V \times W$ by relations imposing bilinearity. The result of this construction is often called *the* tensor product of V and W. From the perspective of universal properties, this is an abuse of terminology. It would be better to call it *a* tensor product of V and W. We can work equally well with any other tensor product of V and W.

This universal property determines $V \otimes W$ and l up to unique isomorphism as follows. Let $V \boxtimes W$ be another tensor product of V and W, with defining bilinear map $r : V \times W \to V \boxtimes W$. Then, by universal property, r factors uniquely through f as $\tilde{r} : V \otimes W \to V \boxtimes W$, and similarly, l factors uniquely through r as $\tilde{l} : V \boxtimes W \to V \otimes W$. However, this implies that the diagram

$$\begin{tikzcd} V \times W \arrow[r, "l"] \arrow[d, "l"'] & V \otimes W \\ V \otimes W \arrow[ur, "\tilde{r} \circ \tilde{l}"'] & \end{tikzcd}$$

commutes. However, again by universal property, the morphism making this diagram commute is *unique*. Since $\mathrm{id}_{V \otimes W}$ also makes this diagram commute, this means that $\tilde{r} \circ \tilde{l} = \mathrm{id}_{V \otimes W}$. Similarly, by universal property $\tilde{l} \circ \tilde{r} = \mathrm{id}_{V \boxtimes W}$, so $\tilde{r}$ and $\tilde{l}$ are isomorphisms. Since $\tilde{r}$ and $\tilde{l}$ are uniquely determined, we say the universal property determines the tensor product *up to unique isomorphism*.

So how do we formulate universal properties categorically? The answer lies in the *Yoneda lemma*.[15] This simple lemma might reasonably be called the fundamental theorem of category theory, and encapsulates an important philosophy of category theory:

[15] Nobuo Yoneda, a mathematician and computer scientist from Japan, formulated and proved the Yoneda lemma, but never published it. He communicated it to Saunders MacLane, one of the originators of category theory. The name 'the Yoneda Lemma' was given to the lemma by MacLane.

SLOGAN: By their morphisms shall ye know them.

In practice, this means that, if we want to study mathematical objects or structures, we should study the structure-preserving maps between them. It doesn't make much sense to study groups by considering only arbitrary maps of underlying sets. Instead we consider group homomorphisms. Similarly, as we have already seen, the study of topological spaces is, in some sense, the study of abstract continuous maps. The Yoneda lemma makes this precise by showing that, given a category $\mathcal{C}$, an object c in $\mathcal{C}$ is, in a sense, uniquely determined up to unique isomorphism by the collection of all morphisms coming out of c.

Definition 3.5.2. Let $\mathcal{C}$ be a category[16], and $c \in \mathrm{Ob}(\mathcal{C})$. We define a functor

$$\begin{aligned} h^c : \mathcal{C} &\longrightarrow \mathsf{Set} \\ d &\longmapsto \mathcal{C}(c, d) \end{aligned}$$

[16] Once again, set-theoretic technicalities come into play. Technically, what we need for this definition is a *locally small category*, that is, a category such that all of the Hom-sets are 'small sets.'

called the *representable functor* associated to c. For a morphism $f : x \to y$ in $\mathcal{C}$, we define $h^c(f)$ to send $g : c \to x$ to $f \circ g : c \to y$.

Similarly, we obtain a representable functor

$$\begin{aligned} h_c : \mathcal{C}^{\mathrm{op}} &\longrightarrow \mathsf{Set} \\ d &\longmapsto \mathcal{C}(d, c) \end{aligned}$$

This is sometimes called the *contravariant*[17] *representable functor* associated to c.

[17] This is an old terminological convention. A functor $F : \mathcal{C} \to \mathcal{D}$ is sometimes called a *covariant* functor from $\mathcal{C}$ to $\mathcal{D}$, and a functor $F : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}$ is called a *contravariant* functor from $\mathcal{C}$ to $\mathcal{D}$. Covariant functors preserve the direct of morphisms, whereas contravariant functors reverse the direction of morphisms.

Exercise 3.5.3. Show that

$$\begin{aligned} \mathcal{Y} : \mathcal{C} &\longrightarrow \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathsf{Set}) \\ c &\longmapsto h_c \end{aligned}$$

defines a functor. This functor is called the *Yoneda embedding.*

Notation 3.5.4. Let $F, G : \mathcal{C} \to \mathcal{D}$ be functors. We denote the set of natural transformations from F to G by $\mathrm{Nat}(F, G)$. Note that $\mathrm{Nat}(-, G) : \mathrm{Fun}(\mathcal{C}, \mathcal{D})^{\mathrm{op}} \to \mathsf{Set}$ is the representable functor h_G on the category $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$.

Proposition 3.5.5 (The Yoneda Lemma)**.** *Let $F : \mathcal{C}^{\mathrm{op}} \to \mathsf{Set}$ be a functor and $c \in \mathrm{Ob}(\mathcal{C})$. There is an isomorphism*

$$\mathrm{Nat}(h_c, F) \overset{\cong}{\Rightarrow} F(c)$$

Moreover, this isomorphism is natural in both c and F.

Before beginning the proof, we first comment on what we mean by "natural in c and F". We view $\mathrm{Nat}(h_{(-)}, -)$ as a functor $\mathcal{C}^{\mathrm{op}} \times \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathsf{Set}) \to \mathsf{Set}$, which sends (x, F) to $\mathrm{Nat}(h_x, F)$, and sends a pair of morphisms $f : x \to y$ in $\mathcal{C}$ and $\alpha : F \Rightarrow G$ to the morphism

$$\begin{aligned} \mathrm{Nat}(h_y, F) &\longrightarrow \mathrm{Nat}(h_x, G) \\ \beta &\longmapsto (\alpha \circ \beta \circ \mathcal{Y}(f)) \end{aligned}$$

Similarly, we define a functor $(c, F) \mapsto F(c)$ with the obvious functoriality.

Proof. We first prove that there is an isomorphism for any c and F. We define a map

$$\begin{aligned} \chi_{(c,F)} : \mathrm{Nat}(h_c, F) &\longrightarrow F(c) \\ \alpha &\longmapsto \alpha_c(\mathrm{id}_c). \end{aligned}$$

We will show that this is a bijection (i.e. an isomorphism in Set.)

First, let $\alpha : h_c \Rightarrow F$ be an arbitrary natural transformation, and let $f : d \to c$ be a morphism in $\mathcal{C}$ (i.e., a morphism $c \to d$ in $\mathcal{C}^{\mathrm{op}}$). Then naturality yields a commutative square

$$\begin{tikzcd} \mathcal{C}(c,c) \arrow[r, "h_c(f)"] \arrow[d, "\alpha_c"'] & \mathcal{C}(d,c) \arrow[d, "\alpha_d"] \\ F(c) \arrow[r, "F(f)"'] & F(d) \end{tikzcd}$$

Since $h_c(f)(\mathrm{id}_c) = f$, this, in particular implies that $\alpha_d(f) = F(f)(\alpha_c(\mathrm{id}_c))$. Therefore, α is uniquely determined by $\alpha_c(\mathrm{id}_c)$, and thus, $\chi_{(c,F)}$ is injective.

Now, given $x \in F(c)$ we define a natural transformation β with by setting $\beta_d(f) := F(f)(c)$. We now check that β is natural. Let $g : e \to d$ be a morphism in $\mathcal{C}$, and let $f \in \mathcal{C}(d, c)$. We then have that

$$\alpha_e(h_c(g)(f)) = F(h_c(g)(f))(\mathrm{id}_c) = F(g \circ f)(\mathrm{id}_c) = F(g)(F(f)(\mathrm{id}_c))$$

and by definition

$$F(g)(\alpha_d(f)) = F(g)(F(f)(\mathrm{id}_c)).$$

So the diagram

$$\begin{tikzcd} \mathcal{C}(d,c) \arrow[r, "h_c(g)"] \arrow[d, "\alpha_c"'] & \mathcal{C}(e,c) \arrow[d, "\alpha_d"] \\ F(d) \arrow[r, "F(g)"'] & F(e) \end{tikzcd}$$

commutes. Hence, β is a natural transformation with $\chi_{(c,F)}(\beta) = x$, and $\chi_{(c,F)}$ is surjective.

We now show naturality in c. Let $f : c \to d$ be a morphism in $\mathcal{C}$. We wish to show that, for any $F : \mathcal{C}^{\mathrm{op}} \to \mathsf{Set}$ the diagram

$$\begin{tikzcd} \mathrm{Nat}(h_d, F) \arrow[r, "- \circ よ(f)"] \arrow[d, "\chi_{(d,F)}"'] & \mathrm{Nat}(h_c, F) \arrow[d, "\chi_{(c,F)}"] \\ F(d) \arrow[r, "F(f)"] & F(c) \end{tikzcd}$$

commutes. We therefore compute, for an arbitrary natural transformation $\alpha : h_d \Rightarrow F$,

$$\chi_{(c,F)}(\alpha \circ よ(f)) = (\alpha \circ よ(f))_c(\mathrm{id}_c) = \alpha_c(よ(f)_c(\mathrm{id}_c)) = \alpha_c(f \circ \mathrm{id}_c) = \alpha_c(f)$$

Similarly, we compute

$$F(f)(\chi_{(d,F)}(\alpha)) = F(f) \circ \alpha_d(\mathrm{id}_d).$$

However, since α is, itself a natural transformation, our work above shows that

$$F(f)(\alpha_d(\mathrm{id}_d)) = \alpha_c(f).$$

so the diagram commutes, and χ is natural in c.

Finally, we show naturality in F. Let $\beta : F \to G$ be a natural transformation, and c an arbitrary object of $\mathcal{C}$. We want to show that the diagram

$$\begin{tikzcd} \mathrm{Nat}(h_c, F) \arrow[r, "\beta \circ -"] \arrow[d, "\chi_{(c,F)}"'] & \mathrm{Nat}(h_c, G) \arrow[d, "\chi_{(c,G)}"] \\ F(c) \arrow[r, "\beta_c"] & G(c) \end{tikzcd}$$

commutes. We again compute for an arbitrary natural transformation $\alpha : h_c \Rightarrow F$,

$$\chi_{(c,G)}(\beta \circ \alpha) = (\beta \circ \alpha)_c(\mathrm{id}_c)$$

and

$$\beta_c(\chi_{(c,F)}(\alpha)) = \beta_c(\alpha_c(\mathrm{id}_c))$$

This shows that the diagram commutes, and so χ is natural in F. $\square$

Exercise 3.5.6. Show that the Yoneda embedding

$$\mathcal{Y} : \mathcal{C} \to \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathsf{Set})$$

is fully faithful.[18] In particular, if there is a natural isomorphism $\alpha : h_c \cong h_d$, then there is a unique isomorphism $c \cong d$ in $\mathcal{C}$ corresponding to α under the Yoneda embedding.

[18] This is, in fact, what we mean by an *embedding* of categories: a fully faithful functor.

The exercise above is a key consequence of the Yoneda lemma. It tells us that the representable functor associated to an object allows us to retrieve that object up to isomorphism. This now allows us to define universal properties in great generality.

Definition 3.5.7. A *universal property* on a category $\mathcal{C}$ is a functor

$$U : \mathcal{C}^{\mathrm{op}} \to \mathsf{Set}.$$

A pair (c, α), where $c \in \mathrm{Ob}(\mathcal{C})$ and $\alpha : h_c \Rightarrow U$ is a natural isomorphism is called a *representation of* U. We also say that (c, α) *satisfies the universal property* U. We often abuse notation by saying that the object c satisfies U, and leaving the natural isomorphism implicit.

Remark 3.5.8. We will also consider the dual case, that of a functor $U : \mathcal{C} \to \mathsf{Set}$, as a universal property on $\mathcal{C}$. Here an pair satisfying U is an object c and a natural isomorphism $h^c \cong U$.

Example 3.5.9. We now can rephrase Example 3.5.1. Define a functor

$$U_{V,W} : \mathsf{Vect}_k \to \mathsf{Set}$$

by mapping $Z \in \mathsf{Vect}_k$ to the set $\mathrm{Bilin}(V, W; Z)$ of bilinear maps $V \times W \to Z$ and mapping a linear map $f : X \to Z$ to the map

$$\begin{aligned} \mathrm{Bilin}(V, W; X) &\longrightarrow \mathrm{Bilin}(V, W; Z) \\ g &\longmapsto f \circ g \end{aligned}$$

A tensor product of V and W over k is then a representation of $U_{V,W}$.

3.6 Limits and colimits

We now turn to a key application of universal properties: limits and colimits. These generalize notions like: cartesian products, disjoint unions, quotients, subobjects, etc.

Definition 3.6.1. Let $\mathcal{C}$ be a category, and define a functor

$$* : \mathcal{C}^{\mathrm{op}} \to \mathsf{Set}$$

which sends every object of $\mathcal{C}$ to the singleton set $\{*\}$. A *terminal object* of $\mathcal{C}$ is a representation (c, α) for $*$.

Let's unpack what the definition means for a terminal object. Suppose $c \in \mathcal{C}$ is terminal. This means that α provides a natural isomorphism

$$\mathcal{C}(-, c) = h_c(-) \cong \{*\}$$

That is, $\mathcal{C}(d, c)$ is a singleton set for every $d \in c$. Since there is a unique bijection between any two singleton sets, naturality is automatic. The definition therefore amounts to saying that an object c is terminal if and only if there is a unique morphism $d \to c$ for every $d \in \mathcal{C}$.

An object c is then initial if and only if there is a unique morphism $f : c \to d$ for every object d in $\mathcal{C}$.

An *initial object* is a terminal object in $\mathcal{C}^{\mathrm{op}}$, that is, a representation of the functor

$$* : \mathcal{C} \to \mathsf{Set}$$

which sends every object to the singleton set.[19]

Examples 3.6.2.

1. A terminal object in Set is a singleton set. The initial object in Set is the empty set.

2. A terminal object in Top is the unique topological space with underlying set the singleton set. An initial object in Top is the empty topological space.

3. The trivial group $\{e\}$ is both initial and terminal in Grp. When an object is both initial and terminal, we call it a *zero object*.

Warning 3.6.3. Not every category has an initial or terminal object. For instance, viewing a poset P as a category, an object $p \in P$ is terminal if and only if, for all $q \in P$, $p \geq q$.

We now come to a vitally important example of universal constructions in categories: limits.

Definition 3.6.4. Let $\mathcal{C}$ and I be categories, and let $F : I \to \mathcal{C}$ be a functor. We define the *overcategory* of F, $\mathcal{C}_{/F}$ as follows. The objects of $\mathcal{C}_{/F}$ are *cones over* F, and consist of the following data:

1. An object $c \in \mathcal{C}$, called the *cone point*

2. For every $i \in \mathrm{Ob}(I)$, a morphism $\alpha_i : c \to F(i)$ such that, for every morphism $f : i \to j$ in I, the diagram

$$\begin{tikzcd}
& c \arrow[dl, "\alpha_i"'] \arrow[dr, "\alpha_j"] & \\
F(i) \arrow[rr, "F(f)"] & & F(j)
\end{tikzcd}$$

commutes.

A morphism in $\mathcal{C}_{/F}$ is a *morphism of cones* $(c, \{\alpha_i\}_{i \in \mathrm{Ob}(I)}) \to (d, \{\beta_i\}_{i \in \mathrm{Ob}(I)})$. Such a morphism consists of a morphism $g : c \to d$ in $\mathcal{C}$ such that, for every $i \in \mathrm{Ob}(I)$, the diagram

$$\begin{tikzcd}
c \arrow[rr, "g"] \arrow[dr, "\alpha_i"'] & & d \arrow[dl, "\beta_i"] \\
& F(i) &
\end{tikzcd}$$

commutes.

Definition 3.6.5. Let $F : I \to \mathcal{C}$ be a functor. A *limit cone of F in* $\mathcal{C}$ is a terminal object in $\mathcal{C}_{/F}$. We will refer to the cone point c of a limit cone $(c, \{\alpha_i\})$ for F as a *limit* of F. We often write $\lim_I F$ for a limit of F.

[19] An initial object is the *dual* concept to a terminal object — it is obtained by simply reversing the definitions of the arrows.

In category theory, nearly every concept has it's *dual*. The concept that corresponds to swapping the directions of the arrows. As we discuss limits and colimits, I will include dual notions and examples in the margin.

Definition (Dual definition)**.** Let $F : I \to \mathcal{C}$ be a functor. We define the *undercategory* of F, $\mathcal{C}_{F/}$ as follows. The objects of $\mathcal{C}_{F/}$ are *cones under* F (also called *cocones*), and consist of the following data:

1. An object $c \in \mathcal{C}$, called

2. For every $i \in \mathrm{Ob}(I)$, a morphism $\alpha_i : F(i) \to c$ such that, for every morphism $f : i \to j$ in I, the diagram

$$\begin{tikzcd}
& c & \\
F(i) \arrow[ur, "\alpha_i"] \arrow[rr, "F(f)"] & & F(j) \arrow[ul, "\alpha_j"']
\end{tikzcd}$$

commutes.

A morphism in $\mathcal{C}_{/F}$ is a *morphism of cocones* $(c, \{\alpha_i\}_{i \in \mathrm{Ob}(I)}) \to (d, \{\beta_i\}_{i \in \mathrm{Ob}(I)})$. Such a morphism consists of a morphism $g : c \to d$ in $\mathcal{C}$ such that, for every $i \in \mathrm{Ob}(I)$, the diagram

$$\begin{tikzcd}
c \arrow[rr, "g"] & & d \\
& F(i) \arrow[ul, "\alpha_i"] \arrow[ur, "\beta_i"'] &
\end{tikzcd}$$

commutes.

Definition. Let $F : I \to \mathcal{C}$ be a functor. A *colimit cone of F in* $\mathcal{C}$ is an initial object in $\mathcal{C}_{F/}$. We will refer to the cone point c of a colimit cone $(c, \{\alpha_i\})$ for F as a *colimit* of F. We often write $\mathrm{colim}_I F$ for a colimit of F.

Remark. A colimit of $F : I \to \mathcal{C}$ is exactly the same as a limit of $F^{\mathrm{op}} : I^{\mathrm{op}} \to \mathcal{C}^{\mathrm{op}}$. This is what we mean when we say the notions are dual.

Examples.

1. The dual notion of a product in $\mathcal{C}$ is a *coproduct* in $\mathcal{C}$
   (a) In Set or Top, the coproduct is given by the disjoint union, together with the canonical inclusions $X \to X \coprod Y$ and $Y \to X \coprod Y$.
   (b) In Grp, the coproduct of G and H is the free product of groups $G * H$.
   (c) In Ab, the coproduct of G and H is the direct sum $G \oplus H$.

Examples 3.6.6.

1. Let I be the category with two objects and no non identity morphisms. A functor $F : I \to \mathcal{C}$ is uniquely specified by a pair of objects $x, y \in \mathrm{Ob}(\mathcal{C})$. A cone over F consists of an object c and morphisms

$$\begin{tikzcd} & c \arrow[dl, "f"'] \arrow[dr, "g"] & \\ x & & y \end{tikzcd}$$

If the cone $(c, \{f, g\})$ is a limit cone, this means that it is terminal in the overcategory of F, i.e, for any other cone $(d, \{h, k\})$ there is a unique morphism of cones $u : (d, \{h, k\}) \to (c, \{f, g\})$.

Diagramatically, this means that, for any $(d, \{h, k\})$, as in the diagram

$$\begin{tikzcd} & d \arrow[d, dashed] \arrow[ddl, bend right, "h"'] \arrow[ddr, bend left, "k"] & \\ & c \arrow[dl, "f"'] \arrow[dr, "g"] & \\ x & & y \end{tikzcd}$$

There exists a dashed arrow u making the diagram commute. The limit over such a diagram is called a *product* in $\mathcal{C}$. Some examples of products:

   (a) In Set, the product of X and Y is simply the Cartesian product $X \times Y$, together with the projections $X \times Y \to X$ and $X \times Y \to Y$.

   (b) In Top, the product of spaces X and Y is again the product $X \times Y$, equipped with the product topology and the canonical projections.

   (c) In Top_*, the product of (X, x) with (Y, y) is the pointed space $(X \times Y, (x, y))$.

   (d) In Grp the product of G and H is the direct product of groups $G \times H$, equipped with the canonical projections.

   (e) In Ab the product of G and H is the same as in Grp.

Even in an arbitrary category $\mathcal{C}$, a product of x and y is usually denoted $x \times y$.

2. More generally, let I be a category with only identity morphisms (which we can think of as just being a set). Given a diagram $F : I \to \mathcal{C}$, a cone over F consists of an object X and morphisms $p_i : X \to F(i)$ for $i \in \mathrm{Ob}(I)$. A limit of such a diagram is called a *product* of the objects $F(i)$, and denoted by $\prod_{i \in I} F(i)$.

3. Let Pb be the category

$$\begin{tikzcd} & \bullet \arrow[d] \\ \bullet \arrow[r] & \bullet \end{tikzcd}$$

A limit over a functor $F : \mathrm{Pb} \to \mathcal{C}$ is called a *pullback*. In Top, the pullback of a diagram

$$\begin{tikzcd} & X \arrow[d, "f"] \\ Y \arrow[r, "g"'] & A \end{tikzcd}$$

is the subspace $X \times_A Y$ of $X \times Y$ consisting of those pairs $(x, y) \in X \times Y$ such that $f(x) = g(y)$. In Set the pullback is the subset $X \times_A Y \subset X \times Y$ defined in the same way.

4. Consider the poset $\mathbb{N}$ as a category. A limit over a diagram $F : \mathbb{N}^{\mathrm{op}} \to \mathsf{Ab}$ is an *inverse limit* in the classical algebraic terminology. In Set, an inverse limit of a functor F is the subset of the product

$$\prod_{i \in \mathbb{N}} F(i)$$

which consists of elements $(a_i)_{i \in \mathbb{N}}$ such that $f_i(a_{i+1}) = a_i$ for all $i \in \mathbb{N}$.

5. We can consider the category Eq with two objects 0 and 1, and two parallel, non-equal, non-identity morphisms f and g from 0 to 1. A functor $F : \mathrm{Eq} \to \mathcal{C}$ consists of a diagram

$$\begin{tikzcd} F(0) \arrow[r, bend left, "F(f)"] \arrow[r, bend right, "F(g)"'] & F(1) \end{tikzcd}$$

in $\mathcal{C}$. A limit of such a diagram is called an *equalizer*. In Set, an equalizer is given by the subset $A \subset F(0)$ consisting of those $a \in F(0)$ such that $F(f)(a) = F(g)(a)$. In Top, an equalizer is constructed in the same way, and equipped with the subspace topology.

2. Let $\mathrm{Po} := \mathrm{Pb}^{\mathrm{op}}$. A colimit over Po is called an *pushout*. In Top, let

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "g"'] & X \\ Y & \end{tikzcd}$$

be a diagram over Po. The pushout in Top is the topological space $(X \coprod Y)_{/\sim}$, where we define the relation $x \sim y$ if and only if $f(x) = g(y)$. In other words, the pushout is obtained by 'gluing X and Y together along A'

3. A colimit of a diagram $F : \mathbb{N} \to \mathsf{Ab}$ is a *direct limit* in the classical algebraic terminology.

4. Note that $\mathrm{Eq}^{\mathrm{op}} \cong \mathrm{Eq}$. The colimit of a diagram

$$F : \mathrm{Eq} \longrightarrow \mathcal{C}$$

is called a *coequalizer*. In Set, a coequalizer is constructed as the quotient of $F(1)$ by the relation $F(f)(a) \sim F(g)(a)$ for all $a \in F(0)$. In Top, the coequalizer is constructed in the same fashion, and is equipped with the quotient topology.

Lemma 3.6.7. *Let $\mathcal{C}$ be a category, and*

$$\begin{tikzcd} & X \arrow[d, "f"] \\ Y \arrow[r, "g"'] & A \end{tikzcd}$$

a diagram in $\mathcal{C}$ such that f is an isomorphism in $\mathcal{C}$. A commutative square

$$\begin{tikzcd} Z \arrow[r, "q"] \arrow[d, "p"'] & X \arrow[d, "f"] \\ Y \arrow[r, "g"'] & A \end{tikzcd}$$

is a pullback square if and only if p is an isomorphism.

Proof. It will suffice to show that the diagram

$$\begin{tikzcd} Y \arrow[r, "f^{-1}\circ g"] \arrow[d, equal] & X \arrow[d, "f"] \\ Y \arrow[r, "g"'] & A \end{tikzcd}$$

is pullback.[20] Suppose we are given an object $Z \in \mathcal{C}$, and a commutative diagram

[20] The fact that this suffices may be taken as an exercise.

$$\begin{tikzcd} Z \arrow[drr, bend left, "k"] \arrow[ddr, bend right, "\ell"'] & & \\ & & X \arrow[d, "f"] \\ & Y \arrow[r, "g"'] & A \end{tikzcd}$$

We want to show that there is a unique completion to a commutative diagram

$$\begin{tikzcd} Z \arrow[drr, bend left, "k"] \arrow[ddr, bend right, "\ell"'] \arrow[dr, dashed, "q"] & & \\ & Y \arrow[r, "f^{-1}\circ g"] \arrow[d, equal] & X \arrow[d, "f"] \\ & Y \arrow[r, "g"'] & A \end{tikzcd}$$

However, the requirement that the left-hand triangle commute forces $q = \ell$ (and hence if there is a completion, it is unique.), and since $f \circ f^{-1} \circ g \circ \ell = g \circ \ell = f \circ k$, this choice defines a commutative diagram. $\square$

Exercise 3.6.8. Let $\mathcal{C}$ and $\mathcal{D}$ be categories such that every diagram in $\mathcal{D}$ has a limit. Consider a diagram

$$F : I \to \mathrm{Fun}(\mathcal{C}, \mathcal{D}).$$

Show that F has a limit, and that there is a canonical isomorphism $(\lim_I F)(c) \cong \lim_I(F(c))$ in $\mathcal{D}$.

Proposition 3.6.9. *Let $F : I \to \mathcal{D}$ be a functor, and suppose that $\mathcal{D}$ has all coproducts and coequalizers. Then there is a colimit of F in $\mathcal{D}$, and it is equivalent to the coequalizer*

$$\textstyle\coprod_{f\in \mathrm{Mor}(I)} F(s(f)) \overset{A}{\underset{B}{\rightrightarrows}} \coprod_{i\in \mathrm{Ob}(I)} F(i)$$

where $\mathrm{Mor}(I)$ denotes the set of all morphisms of I, and $s(f)$ denotes the source of f. The morphisms A and B in the coequalizer diagram are induced by the canonical inclusion map

$$F(s(f)) \longrightarrow \textstyle\coprod_{i\in \mathrm{Ob}(I)} F(i)$$

and the composite

$$F(s(f)) \xrightarrow{F(f)} F(t(f)) \longrightarrow \textstyle\coprod_{i\in \mathrm{Ob}(I)} F(i)$$

respectively.

Proof. We will build a correspondence between cones over F and cones over the coequalizer diagram above. Denote by

$$G : \mathrm{Eq} \longrightarrow \mathcal{D}$$

the coequalizer diagram from the lemma.

We define a functor

$$\phi : \mathcal{D}_{F/} \longrightarrow \mathcal{D}_{G/}$$

as follows. Given a cone (d, η) under F in $\mathcal{D}$, we assign the cone $\phi(d, \eta)$ with tip d over G, by defining two morphisms:

$$\textstyle\coprod_{i \in \mathrm{Ob}(I)} \eta_i : \coprod_{i \in \mathrm{Ob}(I)} F(i) \longrightarrow d$$

and

$$\textstyle\coprod_{f \in \mathrm{Mor}(I)} \eta_{s(f)} : \coprod_{f \in \mathrm{Mor}(I)} F(s(f)) \longrightarrow d$$

To see that the appropriate triangles commute, we note that (1) the triangles

$$\begin{tikzcd} F(s(f)) \arrow[rr, "\mathrm{id}"] \arrow[dr, "\eta_{s(f)}"'] & & F(s(f)) \arrow[dl, "\eta_{s(f)}"] \\ & d & \end{tikzcd}$$

always commute, and (2) the naturality of η shows that the triangles

$$\begin{tikzcd} F(s(f)) \arrow[rr, "F(f)"] \arrow[dr, "\eta_{s(f)}"'] & & F(t(f)) \arrow[dl, "\eta_{t(f)}"] \\ & d & \end{tikzcd}$$

commute. Thus, $\phi(d, \eta)$ is, in fact, a cone over G. Given a morphism $g : (c, \mu) \to (d, \eta)$, the morphism $\phi(g) : \phi(c, \mu) \to \phi(d, \eta)$ is still given by $g : c \to d$. It is an easy check to show that this still defines a morphism of cones in $\mathcal{D}_{G/}$.

On the other hand, we define a functor

$$\psi : \mathcal{D}_{G/} \longrightarrow \mathcal{D}_{F/}$$

as follows. Given a cone (d, ρ) under G in $\mathcal{D}$, we define the i^{th} component $\psi(d, \rho)_i$ of $\psi(d, \rho)$ to be the composite

$$F(i) \longrightarrow \textstyle\coprod_{i \in \mathrm{Ob}(I)} \xrightarrow{\rho} d$$

with the cone defining the coproduct. Naturality follows from the commutative diagrams

$$\begin{tikzcd} F(s(f)) \arrow[r] \arrow[d, "F(f)"'] & \coprod_{f \in \mathrm{Mor}(I)} F(s(f)) \arrow[d, "B"] \arrow[dr, "\rho"] & \\ & & d \\ F(t(f)) \arrow[r] & \coprod_{i \in \mathrm{Ob}(I)} F(i) \arrow[ur, "\rho"'] & \end{tikzcd}$$

and

$$\begin{tikzcd}
F(s(f)) \arrow[r] \arrow[d, "F(f)"'] & \coprod_{f\in \mathrm{Mor}(I)} F(s(f)) \arrow[d, "A"] \arrow[dr, "\rho"] & \\
F(s(f)) \arrow[r] & \coprod_{i\in \mathrm{Ob}(I)} F(i) \arrow[r, "\rho"'] & d
\end{tikzcd}$$

On morphisms, ψ does not change the morphism of cone tips.

We leave it to the reader to check that ϕ and ψ are weakly inverse functors, which completes the proof. □

Corollary 3.6.10. *The categories* Set*,* Top*,* Ab*,* Grp*, and* Vect_k *admit all (small) limits and colimits.*

Proof. We can apply Proposition 3.6.9 and its dual, and note that the categories listed have all products, coproducts, equalizers, and coequalizers. □

Remark 3.6.11. Proposition 3.6.9 makes more explicit a general description of limits and colimits. Colimits are something like 'quotients of coproducts' and limits are something like 'subobjects of products'. What precisely this means, of course, varies based on the category in question, and can only be a heuristic.[21]

[21] One extreme example of just how heuristic this is, is that limits in $\mathcal{C}$ are colimits in $\mathcal{C}^{\mathrm{op}}$. This means that a description like 'limits are subobjects of Cartesian products' cannot apply in all cases.

Definition 3.6.12. A *retract* in a category $\mathcal{C}$ is a commutative diagram

$$\begin{tikzcd}
 & y \arrow[dr, "f"] & \\
x \arrow[ur, "s"] \arrow[rr, "\mathrm{id}_x"'] & & x
\end{tikzcd}$$

in $\mathcal{C}$. We then say, abusively, that x is a retract of y.

Exercise 3.6.13. Consider a retract diagram $F \to G \to F$ in $\mathrm{Fun}(\mathbb{P}(\{0,1\}), \mathcal{C})$. Show that if G is a pushout diagram, then F is a pushout diagram. Does this statement generalize to colimits of arbitrary shape? Why or why not?

3.7 Groupoids and groups

In the next section, we will need to make heavy use of groupoids. As we will see, groupoids function as an analog of groups, but their formal properties are often better behaved, making them a useful tool to simplify the proofs of powerful theorems. Similarly, the theory of groupoids mirrors the theory of topological spaces in many useful ways, making them an ideal bridge between topology and algebra. Balanced against this, however, is the fact that groups are often computationally simpler than groupoids, and so we will need a dictionary which translates easily between them.

Definition 3.7.1. A *groupoid* is a category $\mathcal{G}$ in which every morphism is an isomorphism.[22] We denote by Grpd the category whose objects are groupoids and whose morphisms are functors.

[22] In category theory, many standard algebraic structures — groups, algebras, etc. — have a "many-object" generalization. Often, one calls the many-object generalization of a *thing* and *thingoid*. In this sense, a groupoid is "like a group, but with many objects" as the next exercise helps to make precise. However, this terminological convention is not uniformly observed. For instance, a category is a multi-object version of a monoid, but no-one calls categories *monoidoids*.

Remark 3.7.2. Notice that for a groupoid $\mathcal{G}$, $\mathrm{End}_{\mathcal{G}}(x) = \mathrm{Aut}_{\mathcal{G}}(x)$. By Lemma 3.1.10 this is a group under the composition of automorphisms.

We have already encountered the key example of groupoids in Example 3.1.2 (6). Namely, for every group G, there is a groupoid BG with one object, and automorphisms given by the group G. To explore how important this construction is, we borrow a definition from topology.

Definition 3.7.3. A groupoid $\mathcal{G}$ is called *path-connected* if, for every two objects $x, y \in \mathrm{Ob}(\mathcal{G})$, there is a morphism from x to y in $\mathcal{G}$.[23] A *path component* of G is a maximal path-connected full subgroupoid. We write $\pi_0(\mathcal{G})$ for the set of path components of a groupoid $\mathcal{G}$.

[23] The topological analogy here will become clearer in the next chapter, where we will see that every topological space X gives rise to a groupoid in which the morphisms are paths in X. The space X is then path-connected if and only if this groupoid is.

Lemma 3.7.4. *Let $\mathcal{G}$ and $\mathcal{H}$ be groupoids. Then the canonical maps $\pi_0(\mathcal{G} \times \mathcal{H}) \to \pi_0(\mathcal{G})$ and $\pi_0(\mathcal{G} \times \mathcal{H}) \to \pi_0(\mathcal{H})$ display $\pi_0(\mathcal{G} \times \mathcal{H})$ as a product. Explicitly, the map*

$$\begin{aligned} \pi_0(\mathcal{G} \times \mathcal{H}) &\longrightarrow \pi_0(\mathcal{G}) \times \pi_0(\mathcal{H}) \\ [(x, y)] &\longmapsto ([x], [y]) \end{aligned}$$

is a bijection.

Proof. This is immediate from the fact that (x, y) is isomorphic to (z, w) in $\mathcal{G} \times \mathcal{H}$ if and only if x is isomorphic to z and y is isomorphic to w. □

Construction 3.7.5. Given a collection $\{\mathcal{G}_i\}_{i \in I}$ of groupoids, we can explicitly construct their coproduct $\coprod_{i \in I} \mathcal{G}_i$ as follows.

- The set of objects is simply the disjoint union of the object sets of the individual groupoids:
$$\mathrm{Ob}\left(\coprod_{i \in I} \mathcal{G}_i\right) = \coprod_{i \in I} \mathrm{Ob}(\mathcal{G}_i).$$
- For any two objects $x, y \in \mathrm{Ob}\left(\coprod_{i \in I} \mathcal{G}_i\right)$ the hom-sets are given by
$$\mathrm{Ob}\left(\coprod_{i \in I} \mathcal{G}_i\right)(x, y) := \begin{cases} \mathcal{G}_i(x, y) & \exists i \text{ s.t. } x, y \in \mathrm{Ob}(\mathcal{G}_i) \\ \varnothing & \text{else.} \end{cases}$$
- The composition and unit morphisms are given by the composition and unit maps of the individual groupoids $\mathcal{G}_i$ when they are not trivial.

It is straightforward, if tedious, to check that this is, indeed, a coproduct in the category Grpd of groupoids.

Lemma 3.7.6. *Let $\mathcal{G}$ be a path-connected groupoid. Then there is a group G and an equivalence of categories $\mathcal{G} \simeq BG$. More generally, if $\mathcal{G}$ is any groupoid, then there are groups $\{G_i\}_{i \in \pi_0(X)}$ and an equivalence $\mathcal{G} \simeq \coprod_{i \in \pi_0(X)} BG_i$.*

Proof. We prove the second statement, as the first is a special case. Let $\mathcal{G}$ be a groupoid. For each path component $i \in \pi_0(\mathcal{G})$ choose a object x_i lying in that path component. Then by Lemma 3.1.10, $\mathrm{End}_{\mathcal{G}}(x_i) = \mathrm{Aut}_{\mathcal{G}}(x_i)$ is a group under composition. We define a functor

$$J : \coprod_{i\in\pi_0(\mathcal{G})} B\mathrm{Aut}_{\mathcal{G}}(x_i) \longrightarrow \mathcal{G}$$

which sends the unique object of $B\mathrm{Aut}_{\mathcal{G}}(x_i)$ to x_i, and sends an element $f \in \mathrm{Aut}_{\mathcal{G}}(x_i)$ to itself (viewed as a morphism from x_i to x_i in $\mathcal{G}$). This preserves composition and identities by construction. Moreover, since every object in $\mathcal{G}$ is isomorphic to one of the x_i (since it lies in the same path component as one of them), the functor J is essentially surjective. Since the induced map on hom-sets is simply the identity

$$\mathrm{End}_{\mathcal{G}}(x) \longrightarrow \mathrm{End}_{\mathcal{G}}(x)$$

the functor J is fully faithful, and thus, by Proposition 3.4.10, is an equivalence of categories. □

Remark 3.7.7. We can, in fact, strengthen the statement of Lemma 3.7.6. If for every object $y \in \mathrm{Ob}(\mathcal{G})$, we choose an isomorphism $f_y : y \to x_{[y]}$ to the chosen representative $x_{[y]}$ of the path component of y, we can define a functor

$$P : \mathcal{G} \longrightarrow \coprod_{i\in\pi_0(\mathcal{G})} B\mathrm{Aut}_{\mathcal{G}}(x_i)$$

which sends every object y to $x_{[y]}$, and sends a morphism $g : y \to z$ to the composite $f_z \circ g \circ f_y^{-1}$. If we let f_{x_i} be the identity, then $P \circ J$ is the identity functor, and so, in fact, we have a *retract* of groupoids which is also an equivalence.

What this proposition tells us is that, in effect, we can think of groupoids as being something like "collections of groups". This is not fully rigorous, but is a quite useful intuition to keep in mind.

In the final section of this course, we will make heavy use of pushouts of groupoids. These are however, computationally challenging to work with, and so we briefly digress to show that in special cases, these agree with ordinary groups.

Construction 3.7.8. Let G and H be groups. We want to define a group $G * H$ which is sort of like a disjoint union of G and H, but such that elements of G do not commute with elements of H. The idea is to let the elements be *formal* products

$$a_1 \cdot a_2 \cdot a_3 \cdots a_k$$

of elements in G and H. This, however, forgets the original group structures of G and H. To add them back in, we identify

$$a_1 \cdot a_2 \cdots a_i \cdot a_{i+1} \cdots a_k = a_1 \cdot a_2 \cdots (a_i a_{i+1}) \cdots a_k$$

whenever a_i and a_{i+1} are both in G, or both in H. We similarly identify

$$a_1 \cdots a_{i-1} \cdot a_i \cdot a_{i+1} \cdots a_k = a_1 \cdots a_{i-1} \cdot a_{i+1} \cdots a_k$$

whenever a_i is the identity element in either G or H.[24]

We call the resulting group the **free product** of G and H, and denote it by $G * H$.

[24] Note that this formally identifies e_G and e_H with the empty product, which is the identity of the free group.

Exercise 3.7.9. Show that the free product is coproduct in the category Grp.

To this definition, we add a little bit of extra structure.

Definition 3.7.10. Suppose we are given homomorphisms of groups

$$G \xleftarrow{\phi} K \xrightarrow{\psi} H$$

We define the **pushout** of these groups to be

$$G *_K H := (G * H)_{/\sim}$$

where we declare $\phi(k) \sim \psi(k)$ for any $k \in K$.[25]

[25] Technically, we are taking the smallest normal subgroup of $G * H$ containing the elements $\phi(k) * \phi(k^{-1})$, and quotienting by that.

Remark 3.7.11. It is worth noting that, while the notation $G *_K H$ does not include ϕ and ψ, the nature of the maps ϕ and ψ is *very* important. For example, consider two diagrams

$$\mathbb{Z} \xleftarrow{0} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$$

and

$$\mathbb{Z} \xleftarrow{\mathrm{id}} \mathbb{Z} \xrightarrow{\mathrm{id}} \mathbb{Z}$$

The pushout of the first diagram is simply the free product $\mathbb{Z} * \mathbb{Z}$, since ψ and ϕ do not give us any new relations.

In the second case, however, we get that the pushout is $(\mathbb{Z} * \mathbb{Z})_{/\sim}$, where $\sim$ is a relation which identifies elements from the second copy of $\mathbb{Z}$ with elements from the first copy of $\mathbb{Z}$. Consequently, we see that the pushout is simply $\mathbb{Z}$ again.

Exercise 3.7.12. Show that the pushout of groups described in Definition 3.7.10 is the pushout in the category Grp.

Proposition 3.7.13. *The functor*

$$B : \mathsf{Grp} \longrightarrow \mathsf{Grpd}$$

which sends a group G to the groupoid BG preserves pushouts.

Proof. Suppose given a pushout square

$$\begin{tikzcd} G \arrow[r, "\phi"] \arrow[d, "\psi"] & H \arrow[d, "\delta"] \\ K \arrow[r, "\gamma"'] & P \end{tikzcd}$$

of groups. Consider a groupoid $\mathcal{A}$ and a cone

$$\begin{tikzcd} BG \arrow[r, "B\phi"] \arrow[d, "B\psi"] & BH \arrow[d, "\eta"] \\ BK \arrow[r, "\mu"'] & \mathcal{A} \end{tikzcd}$$

in Grpd. The commutativity of this square implies that there is an object $x = \eta(*_H) = \mu(*_K)$ in $\mathcal{A}$ such that the diagram factors through the inclusion

$$B\mathrm{Aut}_{\mathcal{A}}(x) \longrightarrow \mathcal{A}$$

as

$$\begin{tikzcd}
BG \arrow[r, "B\phi"] \arrow[d, "B\psi"] & BH \arrow[d] \arrow[ddr, bend left, "\eta"] & \\
BK \arrow[r] \arrow[drr, bend right, "\mu"'] & B\mathrm{Aut}_{\mathcal{A}}(x) \arrow[dr] & \\
 & & \mathcal{A}
\end{tikzcd}$$

Similar considerations show that any morphism $BP \to \mathcal{A}$ which could fulfill the universal property of the pushout would also have to factor through $B\mathrm{Aut}_{\mathcal{A}}(x)$. However, since $\mathrm{Aut}_{\mathcal{A}}(x)$ is a group, and P is a pushout in the category of groups, there is a unique such homomorphism, and thus a unique functor $BP \to \mathcal{A}$ making the diagram commute. As such, the square

$$\begin{tikzcd}
BG \arrow[r, "B\phi"] \arrow[d, "B\psi"] & BH \arrow[d, "B\delta"] \\
BK \arrow[r, "B\gamma"'] & BP
\end{tikzcd}$$

is a pushout square, as desired. $\square$

4

Homotopy and the fundamental group

As we saw with $\mathbb{R}P^1$ and S^1, it is often not too hard to explicitly write down a homeomorphism between two spaces. However, to be able to make meaningful statements about topological spaces, it is necessary for us to be able to say when two spaces are *not* the same (i.e. homeomorphic). At first blush this may seem easy. After all, it should be obvious that two spaces are different. Once one starts looking at an example, however, it is not at all clear how one should go about distinguishing two spaces.

As an example, consider $\mathbb{R}^2$, equipped with the topology induced by the Euclidean metric. and $\mathbb{R}^2 \setminus \{0\}$, equipped with the subspace topology. By inspection, it should be fairly intuitive that these are not homeomorphic spaces, but how do we prove it? The two underlying sets have the same cardinality, and it's not possible to write down every possible map between $\mathbb{R}^2$ and $\mathbb{R}^2 \setminus \{0\}$ and check continuity. So we seem to be stuck.

Our approach to showing that two spaces cannot be homeomorphic will be to study *invariants*, properties, numbers, or more complicated mathematical objects assigned to topological spaces in such a way that two homeomorphic spaces will have the same invariant.

Example 4.0.1. The circle S^1 is not homeomorphic to the interval $[0,1]$. To see this, we notice that, if we had a homeomorphism $f : [0,1] \to S^1$, it would induce a homeomorphism between $[0,1] \setminus \{\frac{1}{2}\}$ and $S^1 \setminus \{f(\frac{1}{2})\}$. However, the former has two path components, and the latter has one path component. Since the number of path components is a homeomorphism invariant, this is a contradiction.

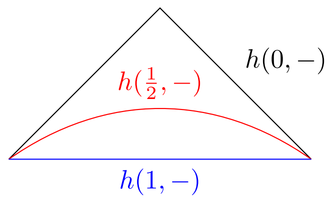
Example 4.0.2. Real Cartesian space $\mathbb{R}^n$ is not homeomorphic to the sphere S^n. The latter is compact, and the former is not, and compactness is a homeomorphism-invariant property.

4.1 Homotopies and homotopy equivalences

Paradoxically, our answer to the question of how to show two spaces are not homeomorphic comes from considering an even weaker notion of equivalence: *homotopy equivalence*, which we can define as an analogue to the equivalences of categories from the previous section. In loose, intuitive terms, two spaces are homotopy equivalent if one can be ‘stretched’ or ‘shrunk’ into another.

Definition 4.1.1. Let X and Y be topological spaces, and $f, g : X \to Y$ continuous maps. A *homotopy* from f to g is a continuous map $h : [0,1] \times X \to Y$ (where $[0,1]$ is equipped with the subspace topology inherited from the Euclidean topology on $\mathbb{R}$) such that $h(0,-) : X \to Y$ is the map f and $h(1,-) : X \to Y$ is the map g. If there is a homotopy from f to g, we say that f and g are *homotopic*.

Example 4.1.2. The image below shows a homotopy $h : [0,1] \times [0,1] \to \mathbb{R}^2$ between paths $[0,1] \to \mathbb{R}^2$.



The idea is that we continuously morph one path into another.

Example 4.1.3. Let $C \subset \mathbb{R}^n$ be a *convex subset*, i.e., a subset such that, for any $x, y \in C$ and any $t \in [0,1]$, the point

$$tx + (1-t)y$$

lies in C. Then given two continuous maps $f, g : X \to C$, we can define the *straight line homotopy* from f to g by

$$H(t,x) = tf(x) + (1-t)g(x).$$

as such, any two such maps are homotopic. Note that if $f(x) = g(x)$ for some $x \in X$, then the straight-line homotopy is constant at x.

Lemma 4.1.4. *Let $[a,b] \subset \mathbb{R}$ be an interval equipped with the subspace topology, and let $f, g : X \to Y$ be continuous maps of topological spaces. The following are equivalent:*

1. *There is a homotopy h from f to g.*
2. *There is a map $H : [a,b] \times X \to Y$ such that $H(a,-) = f$ and $H(b,-) = g$.*

Proof. It is immediate that 1. ⇒ 2. Suppose that 2. holds, and we have such a map H. Define a map

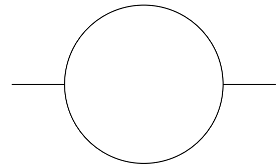
$$q : [0,1] \times X \to [a,b] \times X; \quad (t,x) \mapsto (\rho(t), x)$$

where

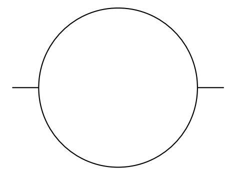
$$\rho(t) := a + t(b-a)$$

It is easy to verify that q is continuous, and therefore $H \circ q : [0,1] \times X \to Y$ is a continous map. However, by definition $(H \circ q)(0,-) = H(a,-) = f$ and $(H \circ q)(1,-) = H(b,-) = g$. Thus $H \circ q$ is a homotopy from f to g. □

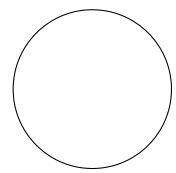
A heuristic depiction of homotopy equivalence might be the following. We consider the space:



This sort of looks like the circle, but they are not homeomorphic. If we remove one of the intersections of the line and the circle, then we get a space with three different path components, but if we remove any single point from the circle, we get a space with only one path component. However, If we are allowed to shrink without tearing our space, we can shrink it to



and then to



yielding the circle. The aim of this section will be to prove rigorous results about this kind of a process.

Lemma 4.1.5. *Let X and Y be topological spaces. The relation*

$$f \sim g \Leftrightarrow f \text{ is homotopic to } g$$

is an equivalence relation on $\mathsf{Top}(X, Y)$*.*

Proof. First, we show reflexivity. Let $f : X \to Y$ be a continuous map. Since the projection $p_2 : [0,1] \times X \to X$ is continuous, the composite $f \circ p_2$ is as well, and provides a homotopy from f to f.[1]

[1] This is sometimes referred to as the *constant homotopy*.

Second, we show symmetry. Let $h : [0,1] \times X \to Y$ be a homotopy. Define $p : [0,1] \to [0,1]$ to send $t \mapsto 1 - t$. This is a homeomorphism (as one can easily verify), and exchanges 0 and 1. Therefore the map $\tilde{h} : [0,1] \times X \to Y$ given by $\tilde{h}(t, x) = h(p(t), x)$ is a homotopy from g to f.

Finally, supose that h is a homotopy from f to g, and k is a homotopy from g to ℓ. We define a map

$$k * h : [0,2] \times X \to Y$$

via

$$(k * h)(t, x) = \begin{cases} h(t, x) & 0 \le t \le 1 \\ k(t-1, x) & 1 \le t \le 2 \end{cases}$$

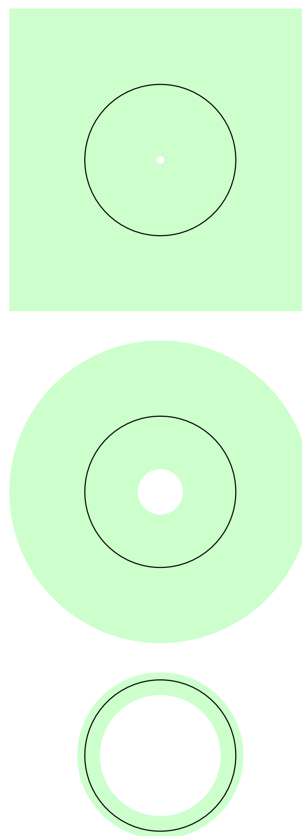
It is straightforward to verify that this is well-defined and continuous, and therefore, by Lemma 4.1.4, f is homotopic to ℓ. □

This now allows us to define our notion of homotopy equivalence:

Definition 4.1.6. Two continous maps $f : X \to Y$ and $g : Y \to X$ are said to be *homotopy inverses* if $g \circ f \sim \mathrm{id}_X$ and $f \circ g \sim \mathrm{id}_Y$. In this situation, we call f (or g) a *homotopy equivalence*, and say that X and Y are *homotopy equivalent*.[2]

[2] This definition should look strangely familiar. If we replace "categories" with "topological spaces", "functors" with "continuous maps", and "natural isomorphisms" with "homotopies", we obtain the definition of an equivalence of categories. The similarity between these two notions can be explained more formally with an appeal to *abstract homotopy theory*, using *model categories*. We will not delve into this perspective in these notes, but it is worth keeping an eye out for such similarities in definitions.

Remark 4.1.7. It is immediate from the definitions that every homeomorphism is a homotopy equivalence.



Example 4.1.8. Consider $X := \mathbb{R}^2 \setminus \{0\}$ and S^1, both with the subspace topology inherited from $\mathbb{R}^2$. There is a canonical inclusion $\iota : S^1 \to X$. We now claim that ι is a homotopy equivalence. Define a map

$$r : X \to S^1; \quad x \mapsto \frac{x}{|x|}$$

Since $x \neq 0$, this is well-defined, and it is easy to check that it is continuous. We note that $r \circ \iota : S^1 \to S^1$ is *equal to* the identity on S^1.

In the other direction, we wish to define a homotopy between $\iota \circ r$ and id_X. Define a continuous map

$$H : [0,1] \times X \to X; \quad (t, x) \mapsto \frac{x}{|x|^t}$$

Note $H(0, x) = \frac{x}{|x|^0} = x$, so $H(0, -) = \mathrm{id}_X$, and that $H(1, x) = \frac{x}{|x|^1} = \frac{x}{|x|} = (\iota \circ r)(x)$. Thus, H is a homotopy from id_X to $\iota \circ R$, and ι is a homotopy equivalence.

Examples 4.1.9. The following examples are quite straightforward, and you should attempt to verify for yourself that they hold:

1. $\mathbb{R}^n$ is homotopy equivalent to the one-point topological space $*$.

2. The Möbius band is homotopy equivalent to S^1.

Definition 4.1.10. If, as in Example 4.1.9 (1), a space X is homotopy equivalent to the one-point topological space $*$, we will call X *contractible*.

Example 4.1.11. Every convex subset of $\mathbb{R}^n$ is contractible.

4.1.1 Homotopy invariants

We want to establish a precise, categorical relation between the notions of homeomorphism and homotopy equivalence. To do this, we will need a category whose isomorphisms are homotopy equivalences.

We call **HoTop** the *naive* homotopy category here because there are more sophisticated notions called 'the homotopy category' in modern mathematics. For our purposes, it will be entirely sufficient to work with the naive homotopy category, and we will often suppress the word 'naive' in our discussions.

Definition 4.1.12. The *(naive) homotopy category* **HoTop** is the category with $\mathrm{Ob}(\mathbf{HoTop}) = \mathrm{Ob}(\mathbf{Top})$. We then define

$$\mathbf{HoTop}(X, Y) := \mathsf{Top}(X, Y)/\sim$$

where $f \sim g$ if and only if f is homotopic to g. We identities $\mathrm{id}_X = [\mathrm{id}_X]$ and a composition $([f], [g]) \mapsto [f \circ g]$.

Lemma 4.1.13. **HoTop** *is a category.*

Proof. We need only show that composition is well-defined. Associativity and unitality then follow from the corresponding statements for Top.

Let $f, g : X \to Y$ be continuous maps, and let $h : [0,1] \times X \to Y$ be a homotopy from x to y. Let $\ell, m : Y \to Z$ be continuous maps, and let $k : [0,1] \times Y \to Z$ be a homotopy from ℓ to m. Define a map $f : [0,1] \times X \to Z$ as the composite

$$\begin{array}{ccccccc}
[0,1] \times X & \longrightarrow & [0,1] \times [0,1] \times X & \longrightarrow & [0,1] \times Y & \longrightarrow & Z \\
(t, x) & \longmapsto & (t, t, x) & & (t, y) & \longmapsto & k(t, y) \\
& & (s, t, x) & \longmapsto & (s, h(t, x)) & &
\end{array}$$

Then note that $f(0, x) = k(0, h(0, x)) = (\ell \circ f)(x)$ and $f(1, x) = k(1, h(1, x)) = (m \circ g)(x)$. It is straightforward to verify that f is continuous, and thus provides a homotopy from $\ell \circ f$ to $m \circ g$. $\square$

Remark 4.1.14. There is a functor $\mathrm{Ho} : \mathsf{Top} \to \mathbf{HoTop}$ which acts as the identity on objects, and sends each morphism f to $[f]$.

Remark 4.1.15. The isomorphisms in **HoTop** are precisely the homotopy equivalences.[3]

[3] This follows by noticing that if $[f]$ and $[g]$ are morphisms in HoTop with $[f \circ g] = [\mathrm{id}]$, then $f \circ g$ is homotopic to id.

Lemma 4.1.16. *Let $\mathcal{C}$ be a category, and $F : \mathsf{Top} \to \mathcal{C}$ be a functor. The following are equivalent:*

1. *If $f, g : X \to Y$ are homotopic, then $F(f) = F(g)$.*
2. *F descends uniquely to a functor $\tilde{F}$ such that the diagram*

$$\begin{tikzcd} \mathsf{Top} \arrow[r, "F"] \arrow[d, "\mathrm{Ho}"'] & \mathcal{C} \\ \mathbf{HoTop} \arrow[ur, "\tilde{F}"'] & \end{tikzcd}$$

commutes.

3. *F sends homotopy inverse maps to inverse maps.*

Proof. To see that 1. $\Rightarrow$ 2., define $\tilde{F}$ to agree with F on objects, and, for each equivalence $[f] : X \to Y$ in **HoTop**, set

$$\tilde{F}([f]) := F(f).$$

By assumption 1., $\tilde{F}$ is well-defined, and this is clearly the only choice of $\tilde{F}$ which makes the diagram commute. Associativity and unitality of $\tilde{F}$ follow directly from the corresponding properties of F.

It is immediate that 2. $\Rightarrow$ 3. We thus need only show 3. $\Rightarrow$ 1. Suppose that $f, g : X \to Y$ are homotopic via a homotopy $h : [0,1] \times X \to Y$. We note that the maps

$$\iota_t : X \to [0,1] \times X; \quad x \mapsto (t, x)$$

are homotopy equivalences for all $t \in [0,1]$. Moreover, a homotopy inverse to both ι_1 *and* ι_0 is the map $p : [0,1] \times X \to X$ which sends $(t, x) \mapsto x$. We can write down a commutative diagram

$$\begin{tikzcd} X \arrow[r, "\iota_0"] \arrow[dr, "f"'] & {[0,1] \times X} \arrow[d, "h"] & X \arrow[l, "\iota_1"'] \arrow[dl, "g"] \\ & Y & \end{tikzcd}$$

in Top. Applying F to the this diagram tells us that

$$F(f) = F(h) \circ F(\iota_0)$$

and

$$F(g) = F(h) \circ F(\iota_1).$$

By 3., the maps $F(\iota_1)$ and $F(\iota_0)$ both have inverse $F(p)$. However, since inverses are unique in any category, this implies $F(\iota_1) = F(\iota_2)$, and thus $F(f) = F(g)$, completing the proof. $\square$

Definition 4.1.17. A *homotopy invariant* is a functor $F : \mathbf{HoTop} \to \mathcal{C}$ for some category $\mathcal{C}$. A *homeomorphism invariant* is a functor $F : \mathsf{Top} \to \mathcal{C}$ for some category $\mathcal{C}$.[4]

The rest of this text will be devoted to the study of homotopy invariants, which provide a flexible and potent way to study topological spaces. Before continuing, however, we introduce a variant of the construction:

[4] Why do we call functors $F : \mathsf{HoTop} \to \mathcal{C}$ homotopy invariants? Any functor sends isomorphisms to isomorphisms, and the isomorphisms in HoTop are the homotopy equivalences. Thus, we can think about a functor $F : \mathsf{HoTop} \to \mathcal{C}$ as a way of associating an object $F(X)$ to every space X in such a way that if $X \simeq Y$, $F(X) \cong F(Y)$. The object $F(X)$ is an invariant of X in the sense that, if two spaces have different (i.e., non-isomorphic) invariants $F(X)$ and $F(Y)$, they must be non-homotopy-equivalent.

Construction 4.1.18. Let Top_* denote the overcategory $\mathsf{Top}_{*/}$. The objects of Top_* can be represented as pairs (X, x) where X is a topological space, and $x \in X$. Morphisms $f : (X, x) \to (Y, y)$ are continuous maps $X \to Y$ such that $f(x) = y$. We refer to Top_* as the *category of pointed topological spaces.*[5]

We define a *pointed homotopy* between morphisms $f, g : (X, x) \to (Y, y)$ in Top_* to be a homotopy

$$h : [0,1] \times X \to Y$$

from f to g such that, for every $t \in [0,1]$, the map $h(t, -) : X \to Y$ is a morphism in Top_*, i.e. preserves basepoints.

We define HoTop_* to be the category whose objects are the same as Top_*, and whose morphisms are pointed homotopy classes of maps in Top_*.

[5] While we briefly discussed overcategories in our exploration of colimits, a brief reminder is in order. The overcategory $\mathsf{Top}_{*/}$ has as its objects continuous maps

$$* \longrightarrow X$$

where $*$ is a fixed 1-point space, and $X \in \mathrm{Ob}(\mathsf{Top})$. Such a map is uniquely determined by the space X, together with the point $x \in X$ which is the image of the map in question.

The morphisms In the overcategory are commutative diagrams

$$\begin{tikzcd} & * \arrow[dl, "\{x\}"'] \arrow[dr, "\{y\}"] & \\ X \arrow[rr, "f"'] & & Y \end{tikzcd}$$

That is, a morphism $f : (X, x) \to (Y, y)$ is a continuous map $f : X \to Y$ such that $f(x) = y$, as described in the main text.

Exercise 4.1.19. Formulate and prove analogues of Lemmas 4.1.13 and 4.1.16 for HoTop_*.

Exercise 4.1.20. Show that there are functors $\mathsf{Top}_* \to \mathsf{Top}$ and $\mathsf{HoTop}_* \to \mathsf{HoTop}$ given by forgetting the basepoints. Conclude that if $f : (X, x) \to (Y, y)$ is a pointed homotopy equivalence of pointed spaces, then $f : X \to Y$ is a homotopy equivalence of spaces.

Remark 4.1.21. We will abuse terminology by refering to functors whose domain is *either* HoTop or HoTop_* as *homotopy invariants.*

4.2 The fundamental group(oid)

We now come to our first really homotopy invariants: the fundamental group and fundamental groupoid. The former is a homotopy invariant of pointed spaces, and the latter is a homotopy invariant spaces.

The underlying idea of the fundamental group is quite simple, though computing fundamental groups can be tricky in practice. The main idea is to consider homotopy classes of paths in X as either elements of a group, or morphisms in a groupoid, with composition given by *concatenation*: forming a new path from two others by laying them end-to-end.

4.2.1 Paths and concatenation

Our first goal is to show that concatenating paths is an associative and unital operation up to homotopy of paths. With this in hand, the definition of the fundamental group and the fundamental groupoid will be almost immediate.

Definition 4.2.1. Recall from Definition 1.3.16 that a *path* in a topological space X going from a point $x \in X$ to a point $y \in X$ is a continuous map

$$\alpha : [0,1] \longrightarrow X$$

such that $\alpha(0) = x$ and $\alpha(1) = y$. As an important notational convention: we will use the symbol $s \in [0,1]$ to represent the parameter of the path. When we consider homotopies of paths, we will use the parameter $t \in [0,1]$ for the homotopy coordinate.[6]

A *homotopy of paths* between two paths α, β from x to y is a homotopy of continuous maps

$$\begin{aligned} H : [0,1] \times [0,1] &\longrightarrow X \\ (t,s) &\longmapsto H(t,s) \end{aligned}$$

from α to β such that, for every $t \in [0,1]$, the path $H(t,-) : [0,1] \to X$ is a path from x to y. That is, it is a homotopy which keeps the endpoints of the paths fixed.

[6] One intuitive way to remember this is that we think of the parameter s of the path as moving through "space". Viewing a homotopy as a "movie", the parameter t then represents "time".

Lemma 4.2.2. *Homotopy of paths is an equivalence relation on the set of paths in X from x to y.*

Proof. Effectively follows from keeping track of the endpoints in the proof that homotopy of continuous maps is an equivalence relation. See the proof of Lemma 4.1.5. □

Definition 4.2.3. Let $\alpha : [0,1] \to X$ be a path from x to y and $\beta : [0,1] \to X$ a path from y to z. The *concatenation* $\beta * \alpha$[7] of α and β is the path from x to z defined by[8]

$$\beta * \alpha(s) = \begin{cases} \alpha(2s) & 0 \le s \le \frac{1}{2} \\ \beta(2s-1) & \frac{1}{2} \le s \le 1. \end{cases}$$

[7] Notice that we order α and β in the same way we would if we were composing functions. This will be of importance when we come to the fundamental groupoid.

[8] This is equivalent to the following, more category-theoretic, definition. Since $\alpha(1) = \beta(0)$, α and β form a cone over the pushout diagram

$$\begin{tikzcd} * \arrow[r, "\{1\}"] \arrow[d, "\{0\}"'] & {[0,1]} \arrow[d, "\frac{1}{2}s"] \\ {[0,1]} \arrow[r, "\frac{1}{2}(s+1)"'] & {[0,1]} \end{tikzcd}$$

in Top. The concatenation $\beta * \alpha$ is the unique morphism to X guaranteed by the universal property of the pushout.

Notation 4.2.4. Let $X \in \mathsf{Top}$ and $x, y \in X$. We denote by $X(x,y)$ the set of paths in X from x to y and by $\Pi_1(X)(x,y)$ the set of path-homotopy classes of paths in X from X to Y. We denote by e_x the constant path $[0,1] \to X$ with value x. For a path $\alpha \in X(x,y)$, we abusively write $\alpha \in \Pi_1(X)(x,y)$ to denote the path-homotopy class of α.

Proposition 4.2.5. *Let X be a topological space.*

1. *For any $x, y, z \in X$, The concatenation of paths defines a well-defined operation*
$$* : \Pi_1(X)(y,z) \times \Pi_1(X)(x,y) \longrightarrow \Pi_1(X)(x,z).$$
2. *For any $x, y, z, w \in X$ and any paths $\gamma \in \Pi_1(X)(x,y)$, $\beta \in \Pi_1(X)(y,z)$, and $\alpha \in \Pi_1(X)(z,w)$, the concatenations*
$$\alpha * (\beta * \gamma) \quad \textit{and} \quad (\alpha * \beta) * \gamma$$
are path-homotopic.
3. *For any $x, y \in X$ and any path α from x to y in X,*
$$\alpha * e_x \quad \textit{and} \quad e_y * \alpha$$
are both path-homotopic to α.
4. *For every path α in X from x to y, there is a path α^{-1} in X from y to x such that $\alpha * \alpha^{-1}$ is homotopic to e_y and $\alpha^{-1} * \alpha$ is homotopic to e_x.*

Proof. (1) is left to the reader as an exercise.

For part (2), notice that, if we define $f:[0,1]\to[0,1]$ by

$$f(s)=\begin{cases}2s & 0\le s\le \frac{1}{4}\\ s+\frac{1}{4} & \frac{1}{4}\le s\le\frac{1}{2}\\ \frac{1}{2}s+\frac{1}{4} & \frac{1}{2}\le s\le 1\end{cases}$$

then $(\alpha*(\beta*\gamma))\circ f = (\alpha*\beta)*\gamma$. As such, it suffices to define a homotopy of paths from f to $\mathrm{id}_{[0,1]}$. But since $[0,1]$ is convex, the straight-line homotopy $H(s,t)=tf(s)+(1-t)s$ suffices.

For part (3) we again note that, defining

$$f(s)=\begin{cases}2s & 0\le s\le\frac{1}{2}\\ 1 & \frac{1}{2}\le s\le 1\end{cases}$$

and

$$g(s)=\begin{cases}0 & 0\le s\le\frac{1}{2}\\ 2s-1 & \frac{1}{2}\le s\le 1\end{cases}$$

we have $\alpha\circ f=e_y*\alpha$ and $\alpha\circ g=\alpha*e_x$. As in part (2), straight-line homotopies then suffice to show part (3).

For part (4), the inverse path is given by

$$\alpha^{-1}(s)=\alpha(1-s).$$

Thus, for example, $\alpha^{-1}*\alpha=\alpha\circ f$, where $f:[0,1]\to[0,1]$ is given by

$$f(s)=\begin{cases}2s & 0\le s\le\frac{1}{2}\\ 2-2s & \frac{1}{2}\le s\le 1.\end{cases}$$

There is a similar formula for $\alpha*\alpha^{-1}$, and so again, straight-line homotopies give the desired solution. ☐

Definition 4.2.6. By parts (1), (2), and (3) of Proposition 4.2.5, there is a category $\Pi_1(X)$ whose objects are the points of x, whose morphisms are the homotopy classes of paths in X, and whose composition operation is the concatenation of paths. Furthermore, by part (4), this category is a groupoid, i.e., all of its morphisms are invertible. We refer to this category as the *fundamental groupoid* of X.

Since $\Pi_1(X)$ is a groupoid, for any object x, the hom-set $\Pi_1(X)(x,x)$ is a group under the concatenation. We call this group the *fundamental group* of the pointed space (X,x) and denote it by $\pi_1(X,x)$.

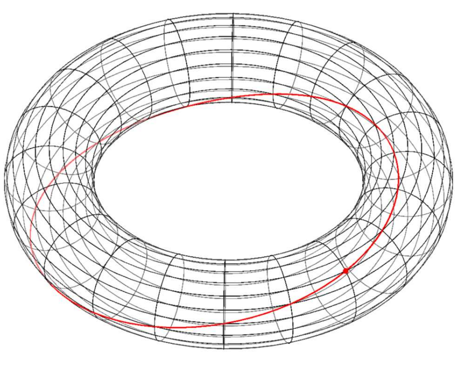
The fundamental group $\pi_1(X,x)$ can equivalently be viewed as the set of homotopy classes of *loops based at* x in X, i.e., continuous maps

$$\alpha:[0,1]\longrightarrow X$$

A loop in X is pretty easy to visualize, since it matches our intuition precisely. Let's consider the example of the torus $S^1\times S^1$. We can define a loop in $S^1\times S^1$ by

$$\begin{aligned}[0,1] &\longrightarrow S^1\times S^1\\ t &\longmapsto (e^{2\pi i t}, e^{2\pi i t})\end{aligned}$$

If we draw this, we get something like:



Where the path is drawn in red, and the basepoint is represented by a red dot.

for which $\alpha(0) = \alpha(1) = x$. If we let $S^1 \subset \mathbb{C}$ denote the unit circle and choose $1 \in S^1$ as a basepoint, the homeomorphism

$$[0,1]_{/0\sim 1} \cong S^1$$

shows that we can view the set of loops in X based at x as the set $\mathsf{HoTop}_*((S^1,1),(X,x))$. That is, there is a bijection of sets

$$\mathsf{HoTop}_*((S^1,1),(X,x)) \cong \pi_1(X,x).$$

One key property of the fundamental group is that, up to isomorphism, it only depends on the path-component of the basepoint.

Corollary 4.2.7. *Let X be a topological space, $x, y \in X$, and $\alpha \in \Pi_1(X)(x,y)$. Then there is an isomorphism*

$$\begin{aligned} \pi_1(X,x) &\xrightarrow{\cong} \pi_1(X,y) \\ \beta &\longmapsto \alpha * \beta * \alpha^{-1} \end{aligned}$$

4.3 Functoriality of the fundamental group(oid)

We will now seek to show that the fundamental groupoid defines a functor

$$\Pi_1 : \mathsf{Top} \longrightarrow \mathsf{Grpd}$$

and that the fundamental group defines a functor

$$\pi_1 : \mathsf{HoTop}_* \longrightarrow \mathsf{Grp}.$$

As before, we will first work with the fundamental groupoid, and then deduce the desired results about the fundamental group.

Something may, at this point, seem strange: the fundamental group is a homotopy invariant, but from what we have written above, it appears that the fundamental groupoid is not. However, this is effectively illusory. While the fundamental groupoid functor does not send homotopy equivalences to isomorphisms, it does send them to *equivalences of categories*. If we were to define a *homotopy category of categories* HoGrpd, whose morphisms are equivalence classes of functors under natural isomorphism, then we would obtain a functor

$$\Pi_1 : \mathsf{HoTop} \longrightarrow \mathsf{HoGrpd}.$$

Construction 4.3.1. Let $f : X \to Y$ be a continuous map of topological spaces. We construct a functor

$$\Pi_1(f) : \Pi_1(X) \longrightarrow \Pi_1(Y)$$

by requiring that, on objects, it sends $x \in X$ to $f(x) \in Y$, and on morphisms, it sends the path-homotopy class of α to the path-homotopy class of $f \circ \alpha$.

Lemma 4.3.2. *For every continuous map of topological spaces $f : X \to Y$, $\Pi_1(f) : \Pi_1(X) \to \Pi_1(Y)$ is a functor. Moreover the assignment of $\Pi_1(f)$ to f defines a functor*

$$\Pi_1 : \mathsf{Top} \longrightarrow \mathsf{Grpd}.$$

Proof. We will show that $\Pi_1(f)$ is a functor. The functoriality of Π_1 is then immediate from the associativity and unitality of the composition of continuous maps.

Suppose that $\alpha \in \Pi_1(X)(y,z)$ and $\beta \in \Pi_1(X)(x,y)$. Unwinding the definitions, we have

$$\Pi_1(f)(\alpha * \beta)(s) = \begin{cases} f(\alpha(2s)) & 0 \leq s \leq \frac{1}{2} \\ f(\beta(2s-1)) & \frac{1}{2} \leq s \leq 1. \end{cases} = \Pi_1(f)(\alpha) * \Pi_1(f)(\beta).$$

Similarly, for $x \in X$, $\Pi_1(f)(e_x) = f \circ e_x = e_{f(x)}$. $\square$

Proposition 4.3.3. *Let $f, g : X \to Y$ be continuous maps, and let H be a homotopy from f to g. Then H defines a natural isomorphism $\Pi_1(H) : \Pi_1(f) \cong \Pi_1(g)$. Moreover, given a point $x \in X$, if $H(-, x)$ is constant, then the component of $\Pi_1(H)$ at x is the identity.*

Proof. Since $\Pi_1(Y)$ is a groupoid, it suffices to show that there is a natural transformation $\Pi_1(H)$.

For each point $x \in X$, we define a path μ_x from $f(x)$ to $g(x)$ in Y by

$$\mu_x(s) = H(s, x).$$

These will be the components of the natural transformation. To see the desired commutativity, we need to show that, given a path α from x to y in X, there is a path homotopy from $\mu_y * \Pi_1(f)(\alpha)$ to $\Pi_1(g)(\alpha) * \mu_x$. We define this homotopy in steps as follows. Let G denote the composite

$$[0, 1] \times [0, 1] \xrightarrow{\mathrm{id}\times\alpha} [0, 1] \times X \xrightarrow{H} Y$$

Then $\mu_y * \Pi_1(f)(\alpha)$ is the composite of G with the path $\beta : [0, 1] \to [0, 1] \times [0, 1]$ defined by

$$\beta(s) = \begin{cases} (0, 2s) & 0 \leq s \leq \frac{1}{2} \\ (2s - 1, 1) & \frac{1}{2} \leq s \leq 1 \end{cases}$$

and $\Pi_1(g)(\alpha) \circ \mu_x$ is the composite of G with the path $\gamma : [0, 1] \to [0, 1] \times [0, 1]$ defined by

$$\gamma(s) = \begin{cases} (2s, 0) & 0 \leq s \leq \frac{1}{2} \\ (1, 2s - 1) & \frac{1}{2} \leq s \leq 1. \end{cases}$$

By Lemma 4.1.13, it thus suffices for us to give a homotopy of paths between β and γ. However, since $[0, 1] \times [0, 1] \subset \mathbb{R}^2$ is convex, there is a straight-line homotopy between them, completing the proof.

By construction, if $H(-, x)$ is constant, then $\mu_x = e_x$, and so the component of $\Pi_1(H)$ at x is the identity e_x. □

Corollary 4.3.4. *If $f : X \to Y$ is a homotopy equivalence, then $\Pi_1(f) : \Pi_1(X) \to \Pi_1(Y)$ is an equivalence of categories.*

Proof. If g is a homotopy inverse to f, then $f \circ g \simeq \mathrm{Id}_Y$. Thus the preceding proposition shows that $\Pi_1(f) \circ \Pi_1(g) \cong \mathrm{Id}_{\Pi_1(Y)}$, and similarly for the other composite. □

Having now established the functoriality of Π_1, we turn to π_1.

Corollary 4.3.5. *There is a functor*

$$\pi_1 : \mathsf{HoTop}_* \longrightarrow \mathsf{Grp}.$$

which sends (X, x) to $\pi_1(X, x)$ and sends a continuous map of pointed spaces $f : (X, x) \to (Y, y)$ to the homomorphism

$$\begin{aligned} \pi_1(f) : \pi_1(X, x) &\longrightarrow \pi_1(Y, y) \\ \alpha &\longmapsto f \circ \alpha. \end{aligned}$$

Moreover, if $f : X \to Y$ is a homotopy equivalence (not necessarily of pointed spaces) then for any $x \in X$, $\pi_1(f) : \pi_1(X, x) \to \pi_1(Y, f(x))$ is an isomorphism of groups.

Proof. If $f : (X, x) \to (Y, y)$ is a map of pointed spaces, then $\Pi_1(f) : \Pi_1(X) \to \Pi_1(Y)$ sends x to y. Moreover, by definition, $\pi_1(f)$ is the map

$$\Pi_1(X)(x, x) \longrightarrow \Pi_1(Y)(y, y)$$

induced by $\Pi_1(f)$. Since $\Pi_1(f)$ preserves composition, this is a group homomorphism. It is immediate from the functoriality of Π_1 that this defines a functor

$$\mathsf{Top}_* \longrightarrow \mathsf{Grp}.$$

To see that this descends to the homotopy category, suppose that $f : X \to Y$ is a homotopy equivalence. Then $\Pi_1(f)$ is an equivalence of categories, and so, in particular, is fully faithful. As such, $\pi_1(f)$ is a bijection, and so an isomorphism of groups. Thus, by Lemma 4.1.16, π_1 is a homotopy invariant of pointed spaces. □

Having now established the most basic properties of fundamental groups, we turn our attention to computing fundamental groups.

Example 4.3.6. Let X be a contractible space, so that the unique map $f : X \to *$ is a homotopy equivalence. By Corollary 4.3.5, this implies that for any $x \in X$,

$$\pi_1(X, x) \cong \pi_1(*, *).$$

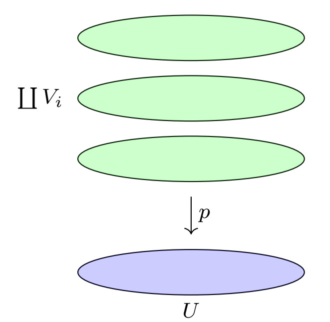
Since there is only one continuous map $[0, 1] \to *$, the latter group is trivial, and so

$$\pi_1(X, x) \cong \{1\}$$

is a trivial group.

However, at this point, this is the only fundamental group we can compute rigorously! To compute more, we will need to think creatively.

Graphically, a covering space should locally look like



4.4 *Coverings and the circle*

We will now aim to compute the fundamental group of the circle S^1. To this end, we will develop the notion of a *covering space*. In a sense, a covering is a surjective map $f : X \to Y$, such that X 'locally looks like Y'.

Definition 4.4.1. Let $p : E \to B$ be a continuous map of topological spaces. We call p a *covering map*, and E a *covering space* of B if, for every $b \in B$, there exists an open subset $b \in U$ such that $p^{-1}(U)$ is a disjoint union $\coprod_{i \in I} V_i$ such that, for each i, $p|_{V_i} : V_i \to U$ is a homeomorphism. We will call each neighborhood U with this property a *covering neighborhood*.

Example 4.4.2. Let X be a topological space, and let K be a set. Define a continuous map

$$p : \coprod_{k\in K} X \longrightarrow X$$

by the universal property of the coproduct to be the identity on each component. Then p is a covering map.

To avoid such examples, we usually focus on covering maps $p : E \to B$ such that E is connected. We will make use of two prototypical examples to clarify covering spaces:

Exercise 4.4.3. Let $S^1 \subset \mathbb{C}$ be the unit circle. Define a continuous map

$$\begin{aligned} p : \mathbb{R} &\longrightarrow S^1 \\ x &\longmapsto \exp(2\pi i x) \end{aligned}$$

Show that p is a covering map.

Exercise 4.4.4. Recall the construction of the real projective space $\mathbb{R}P^2$ from Example 1.2.34. Show that the quotient map

$$S^2 \longrightarrow \mathbb{R}P^2$$

is a covering map.

Exercise 4.4.5. Every covering map is open and surjective, and thus is a quotient map.

The use of covering spaces comes from the associated *lifting properties*:

Definition 4.4.6. Let $p : E \to B$ and $f : Z \to B$ be continuous maps. A *lift* of f (along p) is a continuous map $\widetilde{f} : Z \to E$ such that $p \circ \widetilde{f} = f$, i.e. such that the diagram

$$\begin{tikzcd} & E \arrow[d, "p"] \\ Z \arrow[ur, "\widetilde{f}"] \arrow[r, "f"'] & B \end{tikzcd}$$

commutes.

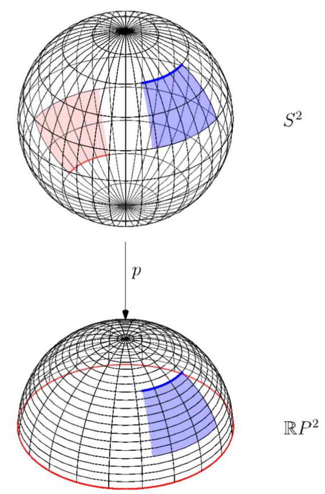
Before proving our main result on liftings, we will need several lemmata.

Lemma 4.4.7. *Let $p : E \to B$ be a covering space, and $f : Z \to B$ any continuous map. Form the pullback in* Top*:*

$$\begin{tikzcd} E \times_B Z \arrow[r] \arrow[d, "\pi"] & E \arrow[d, "p"] \\ Z \arrow[r, "f"] & B \end{tikzcd}$$

Then $\pi : E \times_B Z \to Z$ is a covering space.

In the homotopy lifting property — Theorem 4.4.16 — the initial lift $\tilde{f}$ of f is *essential* to guaranteeing the uniqueness of the lifted homotopy $\tilde{h}$. For example, consider the following two lifts of the same homotopy $[0,1] \times [0,1] \to \mathbb{R}P^2$ to the sphere S^2.



Both the red and blue rectangles above represent lifts of the same homotopy, but each starts from a different lift of same path.

Proof. For each $z \in Z$, we choose a covering neighborhood U of $f(z)$ in B, so that $p^{-1}(U) = \coprod_{i \in I} U_i$, with each U_i mapped homeomorphically to U by p. The set $V := f^{-1}(U)$ is then an open neighborhood of $z \in Z$.

We now note that[9]

$$\coprod_{i \in I} (U_i \times_U V) \cong \left(\coprod_{i \in I} U_i \right) \times_U V = \pi^{-1}(V) \subset X \times_Y Z.$$

[9] The reader can easily check this using the characterization in Example 3.6.6.3.

Moreover, by Lemma 3.6.7, the fact that each

$$p|_{U_i} : U_i \to U$$

is a homeomorphism implies that each map

$$\pi|_{U_i \times_U V} : U_i \times_U V \to V$$

is a homeomorphism. Thus we have found a covering neighborhood V of z. □

Definition 4.4.8. We call a covering map $p : E \to B$ *trivial* (or *a trivial cover*) if it is a homeomorphism.

Definition 4.4.9. Let $p : E \to B$ be a covering map, and let $S \subset B$ be a subspace. We call the pullback covering map

$$\begin{tikzcd} E \times_B S \arrow[r] \arrow[d, "p_S"'] & E \arrow[d, "p"] \\ S \arrow[r, hook] & B \end{tikzcd}$$

the *restriction of* p *over* S and denote it by $p_S : E|_S \to S$.

Lemma 4.4.10. *Let $p : E \to B$ be a covering map such that E is connected. Then p admits a section*[10] *s if and only if p is a homeomorphism.*

[10] Recall that this means that $s : B \to E$ is a continuous map such that $p \circ s = \mathrm{id}_B$.

Proof. Let $b \in B$, and choose an open covering neighborhood U_b of b. Then

$$E|_{U_b} = p^{-1}(U_b) = \coprod_{k \in K_b} V_{b,k}$$

such that p restricted to each $V_{b,k}$ is a homeomorphism to U_b. Let $k(b) \in K$ be the index such that $s(b) \in V_{k(b),b}$, and set $W_b = s^{-1}(V_{k(b),b})$. Since $W_b \subset U_b$, W_b is a covering neighborhood, and since p is a homeomorphism when restriced to $V_{k(b),b}$, we see that $s(W_b)$ is open in $V_{k(b),b}$, and thus in E. By construction

$$p^{-1}(W_b) \cong s(W_b) \amalg \left(\coprod_{k \in K \setminus k(b)} V_{k,b} \cap p^{-1}(W_b) \right).$$

We set $X_b := s(W_b)$ and $Y_b := \coprod_{k \in K \setminus k(b)} V_{k,b} \cap p^{-1}(W_b)$ and note that these are disjoint open sets in E. Moreover, Y_b is disjoint from the image $s(B)$ of s by construction.

Now we can note that the image $s(B)$ of B is the union of the X_b, and thus is open. However, we also note that $E \setminus s(B)$ is the union of the Y_b, and so is open. Since E is connected by assumption, it follows that $E \setminus s(B)$ is empty, and so s is bijective. It follows immediately that $s \circ p = \mathrm{id}_E$, and so p is a homeomorphism. □

Proposition 4.4.11. *Let Z be a topological space, and $p : S \to [0,1] \times Z$ a covering space. Let $I : [0,1] \times Z \to [0,1] \times Z$ denote the identity map, and suppose given a lift $\tilde{I}_0 : \{0\} \times Z \to S$ of $I|_{\{0\}\times Z}$. Given $z \in Z$, there is an open neighborhood $z \in U \subset Z$, and a map $f : [0,1] \times U \to S$ such that*

1. $f|_{\{0\}\times U} = \tilde{I}|_{\{0\}\times U}$
2. *The diagram*

$$\begin{tikzcd} & S \arrow[d, "p"] \\ {[0,1] \times U} \arrow[ur, "f"] \arrow[r, hook] & {[0,1] \times Z} \end{tikzcd}$$

commutes

Proof. We fix $z \in Z$. For each $t \in [0,1]$, we choose a covering neighborhood $U_{(t,z)}$ of (t,z). The images of the $U_{(t,z)}$ under the map $[0,1] \times Z \to [0,1]$ cover $[0,1]$, so by compactness, we can take a finite set $\{U_{t_i,z}\}_{i=1}^n$ which cover $[0,1]\times Z$. By restricting to subsets, we may assume that all of the $U_{(t,z)}$ are of the form $A_{t_i,z} \times M_{t_i,z}$, where $A_{t_i,z}$ is an open[11] interval in $[0,1]$ containing t, and $z \in M_{t_i,z} \subset Z$ is an open subset. We write

$$U = \bigcap_{i=1}^{n} M_{t_i,z}$$

And obtain a finite set $\{[0,a_2^1) \times U, (a_1^2, a_2^2) \times U, \dots (a_1^n, 0]\} \times U$ of covering neighborhoods in $[0,1] \times Z$ which together cover $[0,1] \times U$. We assume, without loss of generality, that $a_1^i < a_1^{i+1} < a_2^i$ for every i.

We now construct a lift inductively. Firstly, $p^{-1}\left([0,a_2^1) \times U\right)$ is a disjoint union of isomorphic copies of $[0,a_2^1) \times U$, each mapped homeomorphically to $[0,a_2^1) \times U$ by p. The lift $\tilde{I}_0|_{\{0\}\times U}$ must take values in precisely one of these copies, which we denote by V. We then want a lift

$$\begin{tikzcd} & V \arrow[d, "p|_V"] \\ {[0,a_2^1) \times U} \arrow[ur, "f_1"] \arrow[r, "\mathrm{id}"] & {[0,a_2^1) \times U} \end{tikzcd}$$

which then must be $f_1 := (p|_V)^{-1} \circ \mathrm{id}_{[0,a_2^1)\times U}$.

Now presume that we have the desired lift f_k on $[0,a_2^k)\times U$. We then consider the preimage $p^{-1}((a_1^{k+1}, a_2^{k+1}) \times U)$, which is a disjoint union of copies of $(a_1^{k+1}, a_2^{k+1}) \times U$. Denote by V the unique sheet of $p^{-1}((a_1^{k+1}, a_2^{k+1}) \times U)$ which has non-empty intersection with the image of f_k. We then need a lift

$$\begin{tikzcd} & V \arrow[d, "p|_V"] \\ {(a_1^{k+1}, a_2^{k+1}) \times U} \arrow[ur, "\ell"] \arrow[r, "\mathrm{id}"] & {(a_1^{k+1}, a_2^{k+1}) \times U} \end{tikzcd}$$

which agrees with f_k on $(a_1^{k+1}, a_2^k) \times U$. As before, this can only be $(p|_V)^{-1} \circ$

[11] Note that an interval is open in $[0,1]$ if it is of the form $[0,a)$, (a,b) or $(b,1]$ for a and b between 0 and 1.

$\mathrm{id}_{(a_1^{k+1},a_2^{k+1})\times U}$, and therefore we have constructed a lift

$$\begin{tikzcd} & p^{-1}([0,a_2^{k+1})\times U) \arrow[d, "p"] \\ {[0,a_2^{k+1})\times U} \arrow[ur, "f_{k+1}"] \arrow[r, "\mathrm{id}"] & {[0,a_2^{k+1})\times U} \end{tikzcd}$$

as desired. □

Corollary 4.4.12. *Let $p : E \to B$ be a covering space.*

1. *If $\gamma : [0,1] \to B$ is a path and $\alpha, \beta : [0,1] \to E$ are two lifts of γ with $\alpha(0) = \beta(0)$, then $\alpha = \beta$.*

2. *If Z is a path-connected space, $f : X \to B$ a continuous map, and $g, h : X \to E$ are two lifts of f which agree at a point $x \in X$, then $g = h$.*

Corollary 4.4.13. *Let Z be a topological space and let $p : S \to [0,1] \times Z$ be a covering space. Suppose given a lift $\tilde{I} : \{0\} \times Z \to S$ of the inclusion $\{0\} \times Z \to [0,1] \times Z$. Then there is a* unique *continuous map $\tilde{h} : [0,1] \times Z \to [0,1] \times Z$ such that*

1. *$\tilde{h}(0,-) = \tilde{I}$*

2. *the diagram*

$$\begin{tikzcd} & S \arrow[d, "p"] \\ {[0,1]\times Z} \arrow[ur, dashed, "\tilde{h}"] \arrow[r, "\mathrm{id}"'] & {[0,1]\times Z} \end{tikzcd}$$

commutes, i.e. $\tilde{h}$ is a lift of the identity on $[0,1] \times Z$.

Proof. For each $z \in Z$, Proposition 4.4.11 shows that we can find a neighborhood $z \in U_z \subset Z$ and a lift $\tilde{h}_z$ on $[0,1] \times U_z$. By Corollary 4.4.12 (1), we find that $\tilde{h}_z$ and $\tilde{h}_y$ agree on $([0,1] \times U_z) \cap ([0,1] \times U_y)$ for any $y, z \in Z$, and thus piece together to form a lift $\tilde{h} : [0,1] \times Z \to [0,1] \times Z$ of the identity starting from $\tilde{I}$. A second application of Corollary 4.4.12 (1) shows that these two properties uniquely determine $\tilde{h}$. □

Lemma 4.4.14. *Let $p : E \to [0,1]$ be a covering map such that $p^{-1}(0)$ is a singleton. Then E is connected.*

Proof. Suppose to the contrary that there is a disconnection $U \amalg V$ of E. By Corollary 4.4.13, it follows that p admits a section s, which without loss of generality we may take to have image equal to U, and note that this implies that $V \cap p^{-1}(0) = \varnothing$. Let $m = \inf\{t \in p(V)\}$. Since $p(V)$ is open and does not contain 0, we see that $m \notin p(V)$. As a result, any open set in $[0,1]$ containing m will contain some points in $p(V)$, whose preimages under p are not singletons (as they contain points of both U and V), but $p^{-1}(m)$ is itself a singleton. Thus, no open neighborhood of m can be a covering neighborhood, a contradiction. □

Corollary 4.4.15. *Let $p : E \to B \times [0,1]$ be a covering map. Then $p : E \to B \times [0,1]$ is trivial if and only if the restriction $p_{B\times\{0\}} : E|_{B\times\{0\}} \to B \times \{0\}$ is trivial.*

Proof. We may, without loss of generality, assume that B is connected, as otherwise we can simply work independently over connected components of B.

On the one hand, suppose that $p : E \to B$ is a trivial covering map. Since a pullback of an isomorphism is an isomorphism, it follows that $p_{B\times\{0\}}$ is a trivial covering map.

On the other hand, suppose that $p_{B\times\{0\}} : E|_{B\times\{0\}} \to B \times \{0\}$ is trivial. By 4.4.13, it follows that E admits a section. It thus remains only for us to see that E is connected. Suppose, to the contrary, that $E \cong U \amalg V$ with U and V non-empty. If $p(U)$ and $p(V)$ were disjoint, then B would be disconnected, a contradiction. Thus we can take $(b, t) \in p(U) \cap p(V)$ and consider the covering space $E|_{b\times[0,1]} \to [0, 1]$. The open sets $U|_{\{b\times[0,1]\}}$ and $V|_{\{b\}\times[0,1]}$ are both non-empty by construction, and so $E|_{b\times[0,1]}$ is not connected. This contradicts Lemma 4.4.14, completing the proof. $\square$

Theorem 4.4.16 (The homotopy lifting property)**.** *Suppose that $p : E \to B$ is a covering space, $h : [0,1] \times Z \to B$ is a homotopy from f to g, and $\tilde{f} : Z \to E$ a lift of f. Then there is a unique lift $\tilde{h}$ of h with $\tilde{h}(0,-) = \tilde{f}$.*

Proof. By pulling back along the map h, we may assume that $B = Z \times [0,1]$ and f is the inclusion $Z \times \{0\} \to Z \times [0,1]$. The theorem then follows immediately from Corollary 4.4.13. $\square$

Using the homotopy lifting property, we can now compute the fundamental group of the circle.

Proposition 4.4.17. *Let $S^1 \subset \mathbb{C}$ be the unit circle, and consider the point $1 \in S^1$ as a basepoint. Then there is an isomorphism of groups*

$$\pi_1(S^1, 1) \cong \mathbb{Z}.$$

Proof. We first define a map

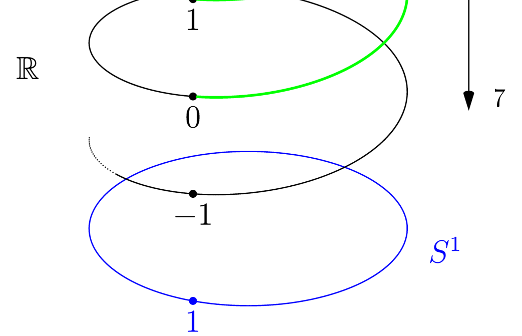
$$f : \pi_1(S^1, 1) \to \mathbb{Z}.$$

To do this, consider the covering space $p : \mathbb{R} \to S^1$ of Exercise 4.4.3, and note that $p^{-1}(1) = \mathbb{Z} \subset \mathbb{R}$. For an element $[\gamma] \in \pi_1(S^1, 1)$, choose a loop γ from 1 to 1 in S^1, And consider this as a homotopy from the constant map $* \overset{\{1\}}{\to} S^1$ to itself. Choose as a lift of the starting point $0 \in \mathbb{R}$. Then by the homotopy lifting property, there is a unique lift $\tilde{\gamma} : [0,1] \to R$ of the loop γ, with the property that $\tilde{\gamma}(0) = 0$. We then define f to be the map which sends

$$[\gamma] \mapsto \tilde{\gamma}(1).$$

We now need to check that this is independent of the choice of representative γ of $[\gamma]$. Suppose that α is another loop, homotopic to γ. Then we have a homotopy

$$h : [0,1] \times [0,1] \to S^1$$



from γ to α, i.e $h(0,-) = \gamma$, $h(1,-) = \alpha$, and $h(-,1) = h(-,0) = 1$. We apply the homotopy lifting property to this map, with $\tilde{\gamma}$ as the lift of $h(0,-)$, to obtain a unique lift $\tilde{h} : [0,1] \times [0,1] \to \mathbb{R}$. Note that $\tilde{h}(0,-) = \tilde{\gamma}$. Moreover, by the uniqueness of lifts, $\tilde{h}(-,0)$ and $\tilde{h}(-,1)$ are the constant paths at 0 and $\tilde{\gamma}(1)$ respectively. Therefore, $\tilde{\alpha}$ is the unique lift of α starting at 0, and $\tilde{\alpha}(1) = \tilde{\gamma}(1)$, so the map f is well-defined.

We now check that f is a homomorphism. Since the constant path at 0 in $\mathbb{R}$ is a lift of the constant path at 1 in S^1, we see that f preserves the group identity. To see that f preserves multiplication, consider loops α and β representing elements in $\pi_1(S^1, 1)$, and let $\tilde{\alpha}$ and $\tilde{\beta}$ be the corresponding lifts starting at 0. Then the map

$$\overline{\beta} : [0,1] \to \mathbb{R}, \quad t \mapsto \alpha\tilde{(1)} + \tilde{\beta}(t)$$

defines the unique lift of β starting at $\tilde{\alpha}(1)$. Consequently, $\overline{\beta} * \tilde{\alpha}$ is the unique lift of $\beta * \alpha$ starting at $0 \in \mathbb{R}$. It is then immediate from the definition that

$$f(\beta * \alpha) = (\overline{\beta} * \tilde{\alpha})(1) = \tilde{\alpha}(1) + \tilde{\beta}(1) = f(\beta) + f(\alpha),$$

so f is a group homomorphism.

Since we can easily define a path $\gamma : [0,1] \to \mathbb{R}$ from 0 to n for any $n \in \mathbb{Z}$, and since $p \circ \gamma$ will be a loop in S^1, the homomorphism f is surjective. It remains only to see that it is injective.

Let $\alpha : [0,1] \to S^1$ be a loop in $\pi_1(S^1, 1)$ such that $\tilde{\alpha}(1) = 0$ in $\mathbb{R}$. Then $\tilde{\alpha} \in \pi_1(\mathbb{R}, 0)$. Since $\mathbb{R}$ is contractible, there is a homotopy h from $\tilde{\alpha}$ to the constant path at 0 fixing the endpoints. Consequently, $p \circ h$ is a homotopy from α to the constant path on 1 in S^1, and so α represents the identity of $\pi_1(S^1, 1)$. □

Remark 4.4.18. While it's great that we've computed $\pi_1(S^1, 1)$, it's worth stepping back for a moment to notice how hard it was. We needed the notion of a covering space, and a lot of propositions about covering spaces, just to formalize our intuition. This is a general feature, not only of the fundamental group, but of homotopy groups[12] more generally.

[12] We won't discuss these now. However, the basic idea is to replace the circle S^1 with the n-sphere S^n in the definition of the fundamental group. The big difference between the fundamental group and the higher homotopy groups is that the higher homotopy groups are *always* Abelian.

We round out this section with a general purpose lemma on covering spaces.

Lemma 4.4.19. *Let $p : E \to B$ be a covering space. Let $b \in B$ and $x \in p^{-1}(b)$. The induced map*

$$p_* : \pi_1(E, x) \longrightarrow \pi_1(B, b)$$

is injective.

Proof. Let α be a loop in E based at x, and let H be a homotopy in B from $p_*(\alpha)$ to the constant loop. We can lift the homotopy H to a homotopy $\tilde{H}$ in E of loops starting from α. However, this implies that $\tilde{H}|_{\{1\}\times I}$ is a lift of the constant loop starting from x, and thus is itself constant. Consequently, $\tilde{H}$ is a homotopy from α to the constant loop. □

4.5 Discrete fibrations, covering spaces, and monodromy

It is entirely possible to study covering spaces only using the fundamental groupoid, tracking basepoints throughout. However, this can be tedious, and often obscures the bigger picture of what is going on. In this section, we will instead work with fundamental groupoids, and derive results about fundamental groups as corollaries.

Before we begin, however, let us try to sketch out a roadmap for the section:

1. We first show that, for a sufficiently nice space B, the topological information of a covering space $E \to B$ over B is equivalent to the *purely algebraic* information of a functor of groupoids $\Pi_1(E) \to \Pi_1(B)$ satisfying certain conditions. We call such functors *discrete fibrations.* This step takes the form of an equivalence of categories
$$\mathsf{Cov}(B) \simeq \mathsf{DFib}(\Pi_1(B))$$
between the category of covering spaces over B and the category of discrete fibrations over $\Pi_1(B)$.

2. Our second step is to show that discrete fibrations can be reformulated as "groupoid actions" — functors into set. For any groupoid $\mathcal{G}$, this means that there is an equivalence of categories
$$\mathsf{DFib}(\mathcal{G}) \simeq \mathrm{Fun}(\mathcal{G}, \mathsf{Set}).$$
This equivalence is a special case of the famed *Grothendieck construction* — a theorem which relates functors to fibrations in great generality.

3. Combining the previous two steps, we see that there is an equivalence of categories
$$\mathsf{Cov}(B) \simeq \mathrm{Fun}(\Pi_1(B), \mathsf{Set}).$$
If B is furthermore path-connected, it follows that $B\pi_1(B) \simeq \Pi_1(B)$, and so a covering space over B is equivalently encoded by a set X equipped with a $\pi_1(B)$-action. This action will be called the *monodromy action.* We can immediately use the monodromy action to classify path-connected covers in terms of subgroups of $\pi_1(B, b)$.

4. By analyzing the automorphism groups of sets with group actions, the above results tell us that, for special covers called *universal covers* — those path connected covers with trivial fundamental groups — the fundamental group of the base space B is simply the automorphism group of the universal cover. Our computation of $\pi_1(S^1, [x])$ was one application of this theorem, and there are many more.

4.5.1 Covering spaces and discrete fibrations

Let us first introduce our dramatis personæ:

Definition 4.5.1. Let B be a topological space, and $p : E \to B$ and $q : F \to B$ covering maps. A *morphism of covering spaces* from p to q over B is a continuous map $f : E \to F$ such that the diagram

$$\begin{tikzcd} E \arrow[rr, "f"] \arrow[dr, "p"'] & & F \arrow[dl, "q"] \\ & B & \end{tikzcd}$$

commutes. The category $\mathsf{Cov}(B)$ has as its objects the covering spaces over B, and as its morphisms the morphisms of covering spaces.

Definition 4.5.2. Let $\mathcal{G}$ and $\mathcal{H}$ be groupoids. A functor $P : \mathcal{H} \to \mathcal{G}$ is called a *discrete fibration (of groupoids)*[13] if, for any morphism $f : x \to y$ in $\mathcal{G}$ and any $\widetilde{x} \in P^{-1}(x)$, there is a unique morphism $\widetilde{f} : \widetilde{x} \to \widetilde{y}$ such that $P(\widetilde{f}) = f$.[14]

A *morphism of discrete fibrations* from $P : \mathcal{H} \to \mathcal{G}$ to $Q : \mathcal{K} \to \mathcal{G}$ is a functor $F : \mathcal{H} \to \mathcal{G}$ such that the diagram

$$\begin{tikzcd} \mathcal{H} \arrow[rr, "F"] \arrow[dr, "P"'] & & \mathcal{K} \arrow[dl, "Q"] \\ & \mathcal{G} & \end{tikzcd}$$

commutes. We denote the category of discrete fibrations over $\mathcal{G}$ by $\mathsf{DFib}(\mathcal{G})$.

Exercise 4.5.3. Verify the following properties for a discrete fibration $P : \mathcal{C} \to \mathcal{D}$.

1. The fibres of P are discrete categories (which we will view as sets).
2. If $\mathcal{D}$ is a groupoid, then $P^{\mathrm{op}} : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}^{\mathrm{op}}$ is also a discrete fibration and $\mathcal{C}$ is a groupoid.

The connection to covering spaces begins with the following observation.

Lemma 4.5.4. *Let $p : X \to Y$ be a covering space. Then $\Pi_1(p) : \Pi_1(X) \to \Pi_1(Y)$ is a discrete fibration.*

Proof. Let $x \in X$ be a point, and let $[\alpha]$ be a morphism from $p(x)$ to y in $\Pi_1(Y)$. Then by the Homotopy Lifting Property (Theorem 4.4.16), there is a unique path $\widetilde{\alpha}$ from x to $\widetilde{y}$ lifting α.

If β is another path in Y, and H is a homotopy from α to β fixing the endpoints, then the Homotopy Lifting Property guarantees the existence of a lift $\widetilde{H}$ starting from $\widetilde{\alpha}$. The homotopy $\widetilde{H}$ restricted to one of the endpoints lifts the constant path, and so by the uniqueness guaranteed by the Homotopy Lifting Property, $\widetilde{H}$ is constant on endpoints. Thus, $\widetilde{H}$ must end in the unique lift $\widetilde{\beta}$ of β starting from x. As such, there is a unique morphism $[\widetilde{\alpha}] : x \to \widetilde{y}$ in $\Pi_1(X)$ lifting α, completing the proof. □

Corollary 4.5.5. *For any topological space B, the fundamental groupoid functor $\Pi_1 : \mathsf{Top} \to \mathsf{Grpd}$ induces a functor*

$$\Pi_1 : \mathsf{Cov}(B) \longrightarrow \mathsf{DFib}(B)$$

[13] There is a much more general notion of fibration of categories — also called a cocartesian fibration — which is central to the *Grothendieck construction* in category theory. Roughly, these do not require unique lifts, but rather, the existence of lifts which are a special kind of morphism, called a *cocartesian morphism.* While we will not discuss the general Grothendieck construction, we will prove a special case of it for discrete fibrations in the coming sections.

[14] It is no accident that this looks exactly like the homotopy lifting property, using morphisms instead of paths. As we will shortly see, it is simply the algebraic incarnation of the path-lifting property of covering spaces. To make the connection clearer, try the following exercise:

Exercise. Let $P : \mathcal{H} \to \mathcal{G}$ be a discrete fibration of groupoids, let $\mathcal{A}$ be a groupoid, and let $H : \mathcal{A} \times [1] \to \mathcal{G}$ be a functor. Denote by $F : \mathcal{A} \times \{0\} \to \mathcal{G}$ the restriction of H to $\mathcal{A} \times \{0\}$. Given a lift $\widetilde{F} : \mathcal{A} \times \{0\} \to \mathcal{H}$ of F, there is a unique lift $\widetilde{H} : \mathcal{A} \times [1] \to \mathcal{H}$ of H such that $\widetilde{H}|_{\mathcal{A} \times \{0\}} = \widetilde{F}$.

Lemma 4.5.6. *Given a commutative diagram of groupoids*

$$\begin{tikzcd} \mathcal{G} \arrow[rr, "F"] \arrow[dr, "P"'] & & \mathcal{J} \arrow[dl, "Q"] \\ & \mathcal{H} & \end{tikzcd}$$

such that P and Q are discrete fibrations and F induces bijections on fibres, then F is an isomorphism of categories

Proof. We need only check that F is fully faithful. Suppose given $f : x \to y$ in $\mathcal{G}$. Then f is the unique morphism over $P(f)$ starting from x, and similarly $F(f)$ is the unique morphism over $P(f)$ starting from $F(x)$. It follows immediately that F is faithful. Now suppose given $g : F(x) \to F(y)$ in $\mathcal{J}$, and let $\overline{g} : x \to \overline{y}$ be the unique lift of $P(g)$ starting from x. Then $F(\overline{g})$ is a lift of $P(g)$ starting from $P(x)$, and so $F(\overline{g}) = g$, showing that F is full. □

It is now quite reasonable to ask when a discrete fibration of groupoids comes from a covering space. To answer this question, we need some mild hypotheses on our base space.

Definition 4.5.7. We will call a topological space X *locally simply connected* if, for every $x \in X$ and every open neighborhood $U \subset X$ of x, there is an open set $V \subset U$ containing x such that, for every basepoint $y \in V$,[15]

$$\pi_1(V, y) \cong 0.$$

[15] Equivalently, we can require that $\Pi_1(V)$ is *indiscrete*, that is, lies in the essential image of the right adjoint to the forgetful functor from Grpd to Set.

The necessary hypotheses on the base Y to construct a covering from a discrete fibration are that Y is locally path-connected and locally simply connected. Before we state and prove that these are sufficient conditions, we need a lemma on the topology of such a Y.

Lemma 4.5.8. *Let Y be a locally path-connected, locally simply connected space. Then the path-connected, simply-connected subsets of Y form a basis of Y.*

Proof. For any $z \in Y$, and any open $z \in U \subset Y$, the fact that Y is locally simply connected implies that there is an open $z \in V \subset U$ such that $\pi_1(V, v)$ vanishes for any $v \in V$. Moreover, since Y is locally path-connected, the path components of an open set are themselves open. Thus, there is a path-connected, simply-connected $\overline{V} \subset U$ which contains z. □

Proposition 4.5.9. *Let Y be a locally path-connected, locally simply connected space. Given a discrete fibration*

$$P : \mathcal{G} \longrightarrow \Pi_1(Y)$$

there is a covering space $\rho_P : X_P \to Y$ and an isomorphism $\psi_P : \Pi_1(X) \to \mathcal{G}$ of discrete fibrations over $\Pi_1(Y)$.

Proof. If such an X exists, its underlying set must be (in bijection with) $\mathrm{Ob}(\mathcal{G})$, and the map ρ_P must agree with P. We thus attempt to topologize $X := \mathrm{Ob}(\mathcal{G})$ so that P becomes a covering space.

We define a basis for the topology on X as follows. Given $x \in X$ and an open $U \subset Y$, denote by $K(U)$ the set of endpoint-preserving homotopy classes of paths which have a representative lying entirely in U. we define

$$V(x, U) := \{z \in X \mid \exists f : x \to z \,\mathrm{in}\, \mathcal{G}\,\mathrm{s.\,t.}\, P(f) \in K(U)\}.$$

We claim that the set of the $V(x, U)$ such U is a path-connected, simply-connected open forms a basis of a topology on X. To see this, let $V(x, U)$ and $V(z, W)$ be two putative basis elements, and suppose $a \in V(x, U) \cap V(z, W)$. Choose a path-connected, simply-connected open $O \subset U \cap W$. We claim that $V(a, O) \subset V(x, U) \cap V(z, W)$. Suppose given $b \in V(a, O)$. By construction, there is a morphism $f : a \to b$ in $\mathcal{G}$ such that $P(f) \in K(O)$. Since $K(O) \subset K(U)$ and $K(O) \subset K(W)$, it immediately follows that $b \in V(x, U) \cap V(z, W)$ showing that we do indeed have the basis of a topology.

We now show that this is a covering map. Given a basic open $V(x, U)$, we claim that $P(V(x, U)) = U$, which will then imply that P is a continuous open map. However, given $y \in U$ we can choose a path α from $P(x)$ to y in U by path connectedness, and since P is a discrete fibration, this lifts to a morphism $\widetilde{\alpha} : x \to \widetilde{y}$ in $\mathcal{G}$. Thus, $\widetilde{u} \in V(x, U)$, and so $P(V(x, U)) = U$.

To complete the proof that $P : X \to Y$ is a covering space, we show that for a basic open $V(x, U)$, the map $P|_{V(x,U)} : V(x, U) \to U$ is a bijection. Suppose that $y, z \in V(x, U)$ such that $P(y) = P(z)$. Choose morphisms $f : x \to y$ and $g : x \to z$ such that $P(f), P(g) \in K(U)$. Then $P(f) \star P(g)^{-1}$ is a loop in U, and since U is simply connected, it is homotopic to the constant loop. Thus, the corresponding morphism $f \circ g^{-1}$ is the unique lift of the identity, and so is an identity morphism in $\mathcal{G}$, so that $y = z$.

Finally, we show that the identity map on objects defines a functor of groupoids $\psi_P : \mathcal{G} \to \Pi_1(X)$ over Y, which will immediately show that it is an isomorphism of discrete fibrations by Lemma 4.5.6. Given a morphism $f : x \to y$ in $\mathcal{G}$, and a representative α of $P(f)$, we will show that the unique lift $\widetilde{\alpha}_x$ of α to a path in X staring from x is a path from x to y. Since the interval is compact, it suffices to show this for $\alpha \in K(U)$ for a basic open U, so that $x, y \in V(x, U)$. However, since $P : V(x, Y) \to U$ is a homeomorphism, y is the only point which can $\widetilde{\alpha}(1)$.

We can then define ψ_P on morphisms by $\psi_P(f) = [\widetilde{\alpha}_x]$. It is immediate that this preserves identities, and functoriality follows from the fact that $\widetilde{\alpha}_{\widetilde{\beta}(1)} \star \widetilde{\beta}_x = \widetilde{\alpha \star \beta}_x$. This completes the proof. $\square$

Lemma 4.5.10. *Let Y by locally path-connected and locally simply-connected. For a covering space $p : X \to Y$, the sets $V(x, U)$ where $U \subset U$ is open, path-connected, and simply-connected form a basis of the topology on X. In particular $p : X \to Y$ can be identified with the covering space constructed from $\Pi_1(p) : \Pi_1(X) \to \Pi_1(Y)$ in Proposition 4.5.9.*

Proof. Let $x \in U \overset{\circ}{\subset} X$. By restricting to open subsets and using the definition of covering space, we may assume, without loss of generality, that $p(U)$ is a covering neighborhood, and U is mapped homeomorphically by p to $p(U)$. Choose a path-connected, simply-connected open subset $p(x) \in W \subset p(U)$. It is immediate by the homotopy lifting property that $p^{-1}(W) \cap U = V(x, W)$, and so we have found a basic open $x \in V(x, W) \subset U$, completing the proof. $\square$

Theorem 4.5.11. *Let Y be a locally path-connected, locally simply connected space. The fundamental groupoid functor induces an equivalence of categories*

$$\Pi_1 : \mathrm{Cov}(Y) \longrightarrow \mathrm{DFib}(\Pi_1(Y)).$$

Proof. By Proposition 4.5.9, Π_1 is essentially surjective. It thus remains only for us to show that Π_1 is fully faithful. Faithfulness is immediate from the definitions, and so it remains for us to see that the functor is full.

To see this, consider a morphism of discrete fibrations

$$\begin{tikzcd} \Pi_1(X) \arrow[rr, "F"] \arrow[dr, "P"'] & & \Pi_1(Z) \arrow[dl, "Q"] \\ & \Pi_1(Y) & \end{tikzcd}$$

It suffices for us to show that F is continuous when considered as morphism of covering spaces.

Let $z \in Z$ and let $V(z, U)$ be a basic open. Given $x \in F^{-1}(V(z, U))$ and $w \in V(x, U)$, there is a morphism $f : x \to w$ in $\Pi_1(X)$ such that $P(f) \in K(U)$ and a morphism $g : Z \to F(x)$ in $\Pi_1(Z)$ such that $Q(g) \in K(U)$. Then $g \circ F(f)$ is a morphism in $\Pi_1(x)$ from z to $F(w)$ such that $Q(g \circ F(f)) = Q(g) \star P(f) \in K(U)$. Thus, $V(x, U) \subset F^{-1}(V(z, U))$, and so F is continuous, as desired. $\square$

4.5.2 Discrete fibrations and the Grothendieck construction

As a final step in our exploration of covering spaces, we will transform the discrete fibrations of the previous section into groupoid actions via the *Grothendieck construction*. Before we continue into abstraction, however, let's try to get a sense of what the monodromy action should be in a more explicit way.

Definition 4.5.12. Let $f : X \to Y$ be a continuous map of topological spaces. For any $y \in Y$, we call the subspace $f^{-1}(y) \subset X$ the *fibre of f over y*.

Exercise 4.5.13. Show that if $p : X \to Y$ is a covering map and $y \in Y$ is a point, then the fibre of p over y is a discrete topological space.

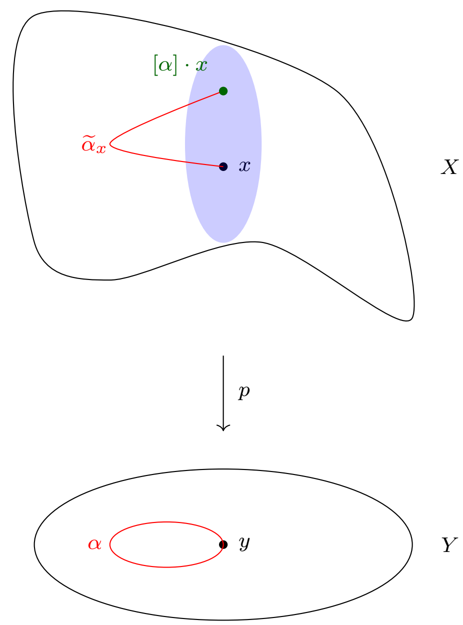
Construction 4.5.14 (The monodromy action)**.** Fix a covering space $p : X \to Y$ such that X is path-connected and locally path-connected, and a basepoint $y \in Y$. We define the *monodromy action* of $\pi_1(Y, y)$ on the fibre $S := p^{-1}(y)$. As follows. Choose a representative $\alpha : I \to Y$ for each element $[\alpha]$ of the fundamental group.

For each $x \in S$, we can then construct the following commutative diagram

$$\begin{tikzcd} * \arrow[r, "\{x\}"] \arrow[d, "\{0\}"'] & Y \arrow[d, "\pi"] \\ I \arrow[r, "\alpha"'] & Y \end{tikzcd}$$

Such a diagram is called a *lifting problem*, and encodes the input necessary input to apply the homotopy lifting property.

When considering the monodromy action, it is useful to keep the following schematic picture in mind.



By the homotopy lifting property, there exists a unique morphism $\tilde{\alpha} : I \to Y$ such that the diagram

$$\begin{tikzcd} * \arrow[r, "\{x\}"] \arrow[d, "\{0\}"'] & Y \arrow[d, "\pi"] \\ I \arrow[r, "\alpha"'] \arrow[ur, "\tilde{\alpha}_x"] & Y \end{tikzcd}$$

commutes. Such a morphism is called a *solution* to the lifting problem. Choose such a solution $\widetilde{\alpha}_x$ for each $[\alpha] \in \pi_1(Y, y)$ and each $x \in S$. We thus obtain a map

$$\begin{aligned} \mu_p : \pi_1(Y, y) \times S &\longrightarrow S \\ ([\alpha], x) &\longmapsto \tilde{\alpha}_x(1). \end{aligned}$$

By the homotopy lifting property, this doesn't depend on the choice of representative $\alpha \in [\alpha]$. To see that this is a group action, we need to check two properties:

- The constant path on $x \in S$ is a lift starting from x of the constant path on y. Thus, the monodromy action satisfies the identity axiom.
- Given $x \in S$, and $[\alpha], [\beta] \in \pi_1(Y, y)$, The concatenation $\widetilde{\alpha}_{\tilde{\beta}_x(1)} \star \widetilde{\beta}_x$ is a lift of $\alpha \star \beta$ starting from x. Thus,

  $$[\alpha \star \beta] \cdot x = \widetilde{\alpha}_{\tilde{\beta}_x(1)}(1) = [\alpha] \cdot ([\beta] \cdot x)$$

  and so the action is compatible with the group multiplication.

Not every action of $\pi_1(Y_y)$ on a set can be obtained as a monodromy action from a path-connected covering space. More precisely

Lemma 4.5.15. *For a path-connected covering space $p : X \to Y$, and any basepoint $y \in Y$, the following are equivalent:*

1. *The monodromy action μ_p of $\pi_1(Y, y)$ on $S := p^{-1}(y)$ is transitive.*
2. *X is path-connected.*

Proof. We first show (2) implies (1). Let $x, z \in S$ be two elements. Note that, since X is path-connected, there is a path $\gamma : I \to Y$ from x to z. Then $p \circ \gamma$ is a loop in Y based at y, and γ is a lift of this loop starting from x and ending at z. Thus

$$[\gamma] \cdot x = z$$

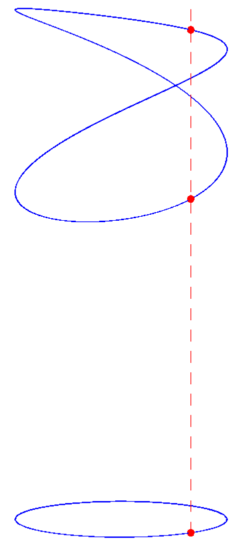
as desired.

To see that (1) implies (2), first note that, by the homotopy lifting property, every element $w \in X$ lies in the same path component as some $x \in S$. It thus suffices to show that given $x, z \in S$, there is a path from x to z. Since the monodromy action is transitive, there is a $[\gamma] \in \pi_1(Y, y)$ such that

$$[\gamma] \cdot x = z.$$

By construction, this means there is a path $\tilde{\gamma}_x : I \to X$ which goes from x to z, completing the proof. □

Below, we draw a 2-fold path-connected covering space of the circle S^1 — this means that each fibre $p^{-1}(x)$ for $x \in S^1$ has cardinality 2.



Because we draw points in the 2-fold cover so that they lie directly above the corresponding point on the circle, the picture of the covering space appears, incorrectly, to have a point of self-intersection. In fact, the space in question is just the circle, and the covering map $S^1 \to S^1$ sends $\theta \mapsto 2\theta$. It is a worthwhile exercise to convince yourself of the transitivity of the monodromy action in this (and similar) cases, by simply drawing lifts of paths.

Now that we have established the monodromy action in more elementary terms, let us reformulate it in terms of discrete fibrations.

Construction 4.5.16 (The discrete inverse Grothendieck construction)**.** Let $P : \mathcal{G} \to \mathcal{H}$ be a discrete fibration. Define a functor

$$U_P : \mathcal{H} \longrightarrow \mathsf{Set}$$

as follows. On objects, send $x \in \mathcal{H}$ to the fibre $P^{-1}(x)$. For a morphism $f : x \to y$, define a map of sets

$$U_P(f) : P^{-1}(x) \longrightarrow P^{-1}(y)$$

by setting $U_P(f)(z)$ to be the target of the unique morphism $\widetilde{f}$ of $\mathcal{G}$ with source z and $P(\widetilde{f}) = f$. We call U_P the *(discrete) inverse Grothendieck construction* of P.

Exercise 4.5.17. Verify that for a discrete fibration $P : \mathcal{G} \to \mathcal{H}$, the inverse Grothendieck construction U_P of P is a functor.

Remark 4.5.18. Let $p : X \to Y$ be a covering space. We will unwind the definitions to relate the monodromy action μ_p to the functor $U_{\Pi_1(p)}$. Notice that, for a basepoint $y \in Y$, $B\pi_1(Y, y) \subset \Pi_1(Y)$ is the full subcategory on the object y. We can thus consider the composite

$$B\pi_1(Y, y) \hookrightarrow \Pi_1(Y) \xrightarrow{U_{\Pi_1(p)}} \mathsf{Set}$$

Given a loop $[\alpha] \in \pi_1(Y, y)$, this composite sends $[\alpha]$ to the map from $\Pi_1(p)^{-1}(y) = p^{-1}(y)$ to itself given sending $x \in p^{-1}(y)$ to the target $\tilde{y}$ of the unique morphism $\widetilde{\alpha}_x : x \to \tilde{y}$ lifting α starting from x. We thus see that this composite is simply the monodromy action.

Definition 4.5.19. For a covering space $p : X \to Y$, we call $U_{\Pi_1(p)} : \Pi_1(Y) \to \mathsf{Set}$ the *monodromy functor*, and denote it by M_p.

Proposition 4.5.20. *The inverse Grothendieck construction defines a functor*

$$U_{(-)} : \mathrm{DFib}(\mathcal{D}) \longrightarrow \mathrm{Fun}(\mathcal{D}, \mathsf{Set}).$$

Proof. Given a morphism

$$\begin{tikzcd} \mathcal{C} \arrow[rr, "F"] \arrow[dr, "P"'] & & \mathcal{E} \arrow[dl, "Q"] \\ & \mathcal{D} & \end{tikzcd}$$

of discrete fibrations over $\mathcal{D}$, we first note that the commutativity of the diagram implies that, for any $d \in \mathrm{Ob}(\mathcal{D})$, F restricts to a map of sets $F_d : P^{-1}(d) \to Q^{-1}(d)$. We then show that the functors F_p form the components of a natural transformation. To see this, we note that the naturality square for $f : b \to d$ in $\mathcal{D}$ can be written as

$$\begin{tikzcd} P^{-1}(b) \arrow[r, "U_P(f)"] \arrow[d, "F_b"'] & P^{-1}(d) \arrow[d, "F_d"] \\ Q^{-1}(b) \arrow[r, "U_Q(f)"'] & Q^{-1}(d) \end{tikzcd}$$

However, for $x \in P^{-1}(b)$ and $\tilde{f} : x \to y$ the unique lift of f in $\mathcal{C}$ starting from x, then $F(\tilde{f}) : F(x) \to F(y)$ is the unique lift of f in $\mathcal{E}$ starting from $F(x)$. Thus, $U_P(f)(x) = y$ and $U_Q(f)(F(x)) = F(y)$, proving naturality. It is then immediate from the definitions that this construction respects composition and identities. □

Theorem 4.5.21. *For any small category $\mathcal{D}$, the inverse Grothedieck construction*

$$U_{(-)} : \mathrm{DFib}(\mathcal{D}) \longrightarrow \mathrm{Fun}(\mathcal{D}, \mathsf{Set})$$

is an equivalence of categories.

Proof (sketch). We will give the construction of a functor

$$\int_{\mathcal{D}} : \mathrm{Fun}(\mathcal{D}, \mathsf{Set}) \longrightarrow \mathrm{DFib}(\mathcal{D})$$

called the *Grothendieck construction*, and leave it to the reader to check that this does, in fact, define a weak inverse to $U_{(-)}$. Given a functor $F : \mathcal{D} \to \mathsf{Set}$, we define $\int_{\mathcal{D}} F$ to be the category whose objects are pairs (d, x) consisting of an object $d \in \mathrm{Ob}(\mathcal{D})$ and an element $x \in F(d)$. A morphism $(d, x) \to (e, y)$ is then a morphism $f : d \to e$ in $\mathcal{D}$ such that $f(x) = y$. There is a forgetful functor $\int_{\mathcal{D}} F \to \mathcal{D}$ given by forgetting the second component of each object.

A natural transformation $\alpha : F \Rightarrow G$ is sent to the functor $Z_\alpha : \int_{\mathcal{D}} F \to \int_{\mathcal{D}} G$ which is given on objects by $Z_\alpha(d, x) = (d, \alpha_d(x))$ and on morphisms by $Z_\alpha(f) = f$. □

Corollary 4.5.22. *Let Y be locally path connected and locally simply-connected.*

1. *The functor*

$$M : \mathrm{Cov}(Y) \longrightarrow \mathrm{Fun}(\Pi_1(Y), \mathsf{Set})$$

which sends a covering to its monodromy functor is an equivalence of categories.

2. *If Y is moreover, path-connected, then restricting along the equivalence $B\pi_1(Y, y) \to \Pi_1(Y)$ yields an equivalence*

$$\mu : \mathrm{Cov}(Y) \longrightarrow \mathrm{Fun}(B\pi_1(Y, y), \mathsf{Set})$$

which sends a cover to its monodromy action.

3. *The equivalence μ descends to an equivalence between the subcategories of path-connected covering spaces of Y and transitive $\pi_1(Y, y)$-actions.*

As a first application of the general framework, we can immediately extract the usual "existence of covers" statements related to fundamental groups.

Corollary 4.5.23. *Let Y be path-connected, locally path-connected, and locally simply-connected. Let $y \in Y$ be a basepoint.*

1. *If $p : X \to Y$ is a path-connected covering space, the monodromy action μ_p is the canonical transitive $\pi_1(Y, y)$ action on the quotient*
$$\pi_1(Y, y)/p_*(\pi_1(X, x))$$
2. *For every subgroup $H \subset \pi_1(Y, y)$, there is a path-connected covering space $p : X \to Y$ such that $p_*(\pi_1(X, x)) = H$. In particular, Y has a simply-connected cover.*

Proof. To see part (1), note that by Lemma 4.5.15, the monodromy action of $\pi_1(Y, y)$ on $p^{-1}(y)$ is transitive. As such, to identify $p^{-1}(y)$ with $\pi_1(Y, y)/p_*(\pi_1(X, x))$, it suffices to show that the stabilizer of x under the action is identified with $p_*(\pi_1(X, x))$. This is immediate from the definition, as a loop based at y is in the stabilizer precisely when it's unique lift starting from x is a loop.

To see part (2) we note that for any subgroup $H \subset \pi_1(Y, y)$, $\pi_1(Y, y)/H$ is a transitive G-set, so we can simply take X to be path-connected cover corresponding to $\pi_1(Y, y)/H$ under the equivalence of categories of 4.5.22. By Lemma 4.4.19, this identifies $\pi_1(X, x)$ with H. □

Definition 4.5.24. Let Y be path-connected, locally path-connected, and locally simply-connected. We call the simply connected cover of Y the *universal cover* and denote it by $\widetilde{Y} \to Y$.

4.5.3 Deck transformations

From the general theory, we can now extract computationally useful criteria. Namely, we can relate the fundamental group of the base of a covering space to the group of "deck transformations" — automorphisms of the covering space. To do this, we fix a path-connected, locally path-connected, locally simply-connected space Y, and a basepoint $y \in Y$. Given a covering space $p : X \to Y$, we consider the monodromy action
$$\mu_p : B\pi_1(Y, y) \longrightarrow \mathsf{Set}.$$
Every such action decomposes uniquely as a coproduct of transitive representations, which can in turn be identified with the action of $\pi_1(Y, y)$ on $\pi_1(Y, y)/H$ for a subgroup $H \subset \pi_1(Y, y)$. As such, it suffices to consider such G-sets. Finally, by the equivalence of categories of Corollary 4.5.22 (1), we can understand automorphisms of covering spaces by understanding automorphisms of $\pi_1(Y, y)$-sets.

Definition 4.5.25. For a covering space $p : X \to Y$, the *group of deck transformations of* p, sometimes denoted Deck_p, is the automorphism group $\mathrm{Aut}_{\mathrm{Cov}(Y)}(p)$.

Definition 4.5.26. For a group G and a subgroup $H \subset G$, we write $N_G(H)$ for the normalizer[16] of H in G. For a G-action on G/H, we denote by $\mathrm{Aut}_G(G/H)$ the group of automorphisms of this G-action in $\mathrm{Fun}(BG, \mathsf{Set})$.

[16] Recall that the normalizer is the largest subgroup of G inside which H is normal.

Lemma 4.5.27. *For any group G and any subgroup $H \subset G$, there is an isomorphism of groups*

$$\mathrm{Aut}_G(G/H) \cong N_G(H)/H.$$

Proof. Let $\eta : G/H \to G/H$ be G-equivariant. Then, in particular, $\eta(H) = uH$ for some $u \in G$, which by equivariance implies that $\eta(gH) = guH$, and so the element u uniquely determines η. Given $u \in G$, we see that the map $\nu : G/H \to G/H$ defined by $\nu(gH) = guH$ is well-defined if and only if, whenever $gH = H$, we have $guH = uH$. Equivalently, this requires that the element $u^{-1}gu \in H$, and so $u \in N_G(H)$. It thus follows that we have a surjective homomorphism

$$N_G(H) \longrightarrow \mathrm{Aut}_G(G/H)$$

whose kernel is those $u \in N_G(H)$ such that $uH = H$, that is, whose kernel is H. This completes the proof. $\square$

As a corollary, we obtain a powerful computational criterion for the computation of fundamental groups.

Corollary 4.5.28. *For a path-connected covering space $p : X \to Y$, there is an isomorphism of groups*

$$\mathrm{Deck}_p \cong N_{\pi_1(Y,y)}(p_*(\pi_1(X,x)))/p_*(\pi_1(X,X)).$$

In particular, if p is a universal cover

$$\mathrm{Deck}_p \cong \pi_1(Y,y).$$

Before we apply this, we note the following

Lemma 4.5.29. *Let $p : X \to Y$ be a path-connected covering space and fix a basepoint $x \in X$ which maps to $y \in Y$. A deck transformation $f : X \to X$ is uniquely determined by $f(x)$.*

Proof. By Corollary 4.5.22 (1), f is uniquely determined by the morphism of transitive $\pi_1(Y,y)$-sets $f|_{\pi_1(Y,y)}$, which by transitivity of the action is uniquely determined by its value on a point. $\square$

Example 4.5.30. As an application, let us fix $n > 1$, and consider the quotient map

$$p : S^n \longrightarrow \mathbb{R}P^n.$$

This is a covering space, as for any point $[x] \in \mathbb{R}P^n$, we can choose an open $x \in U \in S^n$ such that $U \cap (-U) = \varnothing$, and such a U defines a covering neighborhood $p(U)$ of $[x]$. We claim that for any $x \in S^n$

$$\pi_1(S^n, x) \cong 0$$

and so $p : S^n \to \mathbb{R}P^n$ is a universal cover. We will defer the proof of this fact to the next section, but a nice heuristic argument is that any loop in S^n can be homotoped to miss at least one point, and the sphere minus one point is contractible.

We then compute the group of deck transformations Deck_p. Since a deck transformation is uniquely determined by its action on one point in a fibre and p is a 2-fold cover, Deck_p is a subgroup of $\mathbb{Z}_{/2}$. However, the antipode map is clearly a deck transformation, and so $\mathrm{Deck}_p = \mathbb{Z}_{/2}$. By Corollary 4.5.28, it thus follows that

$$\pi_1(\mathbb{R}P^n, [x]) \cong \mathbb{Z}_{/2}.$$

We can, however, do more, and can find representative loops for the elements of $\pi_1(\mathbb{R}P^n, [x])$. By our construction of the isomorphism of Corollary 4.5.28, it follows that the deck transformation ϕ_γ induced an element $[\gamma] \in \pi_1(\mathbb{R}P^2, [x])$ is induced by the lift of γ starting from x. Thus, the unique non-trivial element in $\pi_1(\mathbb{R}P^2, [x])$ is induced by (any) path in S^2 from x to $-x$.

4.6 The Seifert-van-Kampen Theorem

Now that we have ways to compute some simple fundamental groups, we want to extend our reach a bit. We want a statement that allows us to build up the fundamental group of a complicated space X if we can break X apart into pieces whose fundamental groups we know.

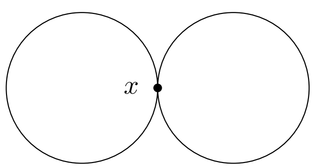
The **wedge sum** of two path-connected topological spaces X and Y is a way of gluing X and Y. Formally, given pointed spaces (X, x) and (Y, y) we define

$$X \vee Y := (X \coprod Y)_\sim$$

where we define $x \sim y$. What this means intuitively is that we glue X to Y at a single point.

This definition depends on the choice of x and y, but when we are just gluing circles together, the choice doesn't matter as much because of the symmetries of the circle.

Example 4.6.1. Consider the *wedge of two circles* $S^1 \vee S^1$, which is the space



Take the point x as a basepoint. We note that we get a non-contractible loop α around the left-hand circle, and a non-contractible loop β around the right-hand circle. Each of these would generate the corresponding fundamental group of a circle, but how do they interact.

It is pretty easy to convince ourselves that the elements of the resulting group will have the form

$$\alpha^{k_1} * \beta^{k_2} * \alpha^{k_3} * \cdots * \alpha^{k_{r-1}} * \alpha^{k_r} \qquad k_i \in \mathbb{Z}$$

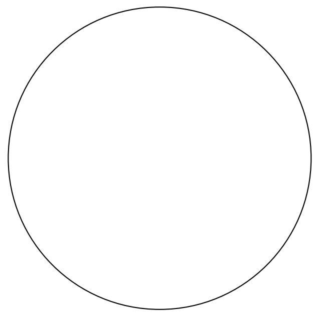
so that $\pi_1(S^1 \vee S^1, x)$ is the free group on $\{\alpha, \beta\}$. More suggestively, this is the free product

$$\pi_1(S^1) * \pi_1(S^1)$$

or the pushout

$$\pi_1(S^1) *_{\pi_1(*)} \pi_1(S^1).$$

Example 4.6.2. Imagine the circle S^1 as a union of two open intervals U and V.



Consider two points, x and y in the intersection of these two intervals. We know that $\Pi_1(S^1)(x,y) \cong \mathbb{Z}$, but we can also break down each path from x to y as a composite

$$\alpha^{k_1} * \beta^{k_2} * \alpha^{k_3} * \cdots * \alpha^{k_{r-1}} * \alpha^{k_r} \qquad k_i \in \mathbb{Z}$$

of paths which lie entirely in either U or V. These must be subjected to the equivalence relation that consecutive paths in e.g., $\Pi_1(U)$, can be composed without changing the equivalence class. While we have not given an explicit formula for the pushout of groupoids, it should not be hard to believe that the fundamental groupoid is given by

$$\Pi_1(S^1) \simeq \Pi_1(U) \amalg_{\Pi_1(U \cap V)} \Pi_1(V).$$

We can generalize these examples in the following theorem.[17]

Theorem 4.6.3 (The Seifert-van-Kampen Theorem)**.** *Let X be a topological space and let $U_1, U_2 \subset X$ such that the interiors of U_1 and U_2 cover X. Then the diagram of groupoids*

$$\begin{tikzcd} \Pi_1(U_1 \cap U_2) \arrow[r] \arrow[d] & \Pi_1(U_1) \arrow[d] \\ \Pi_1(U_2) \arrow[r] & \Pi_1(X) \end{tikzcd}$$

induced by the subspace inclusions is a pushout diagram in Grpd*.*

Proof. We will simply show that the square satisfies the universal property of the pushout. To this end, let $\mathcal{G}$ be a groupoid, and let

$$\begin{tikzcd} \Pi_1(U_1 \cap U_2) \arrow[r] \arrow[d] & \Pi_1(U_1) \arrow[d, "f_{U_1}"] \\ \Pi_1(U_2) \arrow[r, "f_{U_2}"'] & \mathcal{G} \end{tikzcd}$$

be a commutative diagram — a cone under the diagram. We will write $f_{U_1 \cap U_2}$ for the unwritten diagonal functor in the diagram. We first use this to show that a functor $\Pi_1(X) \to \mathcal{G}$ making the appropriate diagrams commute must be of a certain

[17] It is important to point out that this is **not** the classical Seifert-van-Kampen Theorem as it is usually stated, but a generalization thereof. We will deduce the more classical statement about groups from this one.

form — showing uniqueness. We will then verify that this does, indeed, define a functor.

Uniqueness: Suppose $f : \Pi_1(X) \to \mathcal{G}$ commutated with the given cones. Then, given an object $x \in \mathrm{Ob}(\Pi_1(X))$, we know that x must lie in either U_1 or U_2, without loss of generality say U_i. Then commutativity tells us that $f(x)$ must equal $f_{U_i}(x)$. Note that if x lies in both U_1 and U_2, then

$$f(x) = f_{U_1}(x) = f_{U_2}(x) = f_{U_1 \cap U_2}(x).$$

Now suppose we have a morphism $[\alpha] : x \to y$ in $\Pi_1(X)$ represented by a path α. Divide $[0,1]$ into intervals $[0,t_1],[t_1,t_2],\ldots,[t_{k-1},1]$ so that either $\alpha([t_{i-1},t_i]) \subset U_1$ or $\alpha([t_{i-1},t_i]) \subset U_2$ for every $0 < i \leq k$. This can be done for the interiors of U_1 and U_2 by the compactness of the interval, since these form an open cover of X by assumption. Then each path $\alpha_i := \alpha|_{[t_{i-1},t_i]}$ represents a morphism in either $\Pi_1(U_1)$ or $\Pi_1(U_2)$. We can, in fact, assume without loss of generality that the α_i lie in alternating U_j's. As such the fact that f should be a morphism of cones means that we must have

$$f(\alpha) = f([\alpha_k]) * \cdots * f([\alpha_1]) = f_{U_1}([\alpha_k]) * f_{U_2}([\alpha_{k-1}]) * \cdots f_{U_1}([\alpha_1]). \tag{4.1}$$

This then uniquely determines f on morphisms, and so such an f must be unique.

Existence: We now must show that the formulas derived above do, indeed, define a functor. The fact that Equation (4.1) will send identities to identities is immediate. Similarly, if $\alpha : y \to z$ and $\beta : x \to y$ are morphisms in $\Pi_1(X)$ with subdivisions $[0,t_1],\ldots,[t_{k-1},1]$ and $[0,s_1],\ldots,[s_{j-1},1]$, respectively, then $[0,\frac{t_1}{2}],\ldots,[\frac{t_{k-1}}{2},\frac{1}{2}],[\frac{1}{2},\frac{1+s_1}{2}],\ldots,[\frac{1+s_{j-1}}{2},1]$ is a subdivision for $\alpha * \beta$ and using this subdivision it immediately follows that

$$f(\alpha * \beta) = f(\alpha) * f(\beta)$$

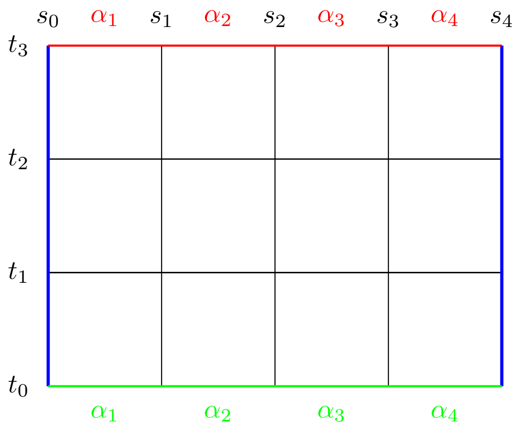
so that such an f will be functorial. There is, however, something subtle and important that must be shown: namely, that Equation (4.1) is well-defined on homotopy classes.

To this end, let $H : [0,1] \times [0,1] \to X$ be a homotopy of paths from α to β and fix subdivisions of the interval corresponding to each of α and β. Choose subdivisions $[0,t_1],\ldots,[t_{r-1},1]$ and $[0,s_1],\ldots[s_{q-1},1]$ of the interval such that each restriction

$$H([t_i,t_{i+1}] \times [s_j,s_{j+1}])$$

is contained entirely in U_1 or U_2, and such that the subdivision $[0,s_1],\ldots,[s_{q-1},1]$ is a refinement of both of the initial subdivisions corresponding to α and β [18]

[18] If you are not convinced we can do this, try to prove it yourself or look up the Lebesgue number lemma.



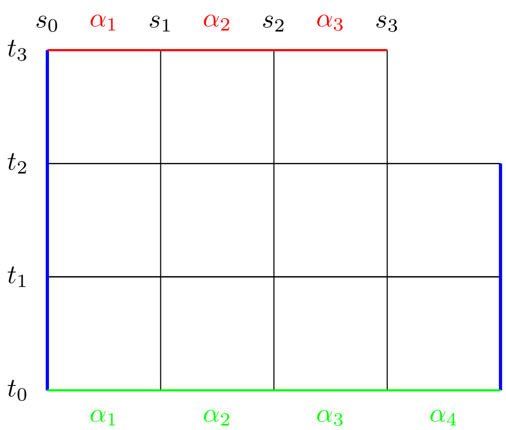
The lines in blue are sent to the basepoint x, and the composite

$$\alpha_4 * \alpha_3 * \alpha_2 * \alpha_1 \simeq \alpha.$$

The composite $\alpha_4 * \cdots * \alpha_1$ can be viewed as an element of G, since each of the α_i lies in either U_1 or U_2. Each little square then represents a homotopy in either U_1 or U_2. Thus, for instance, the top right square in the diagram represents a homotopy in, e.g. U_2 from $e_x * \alpha_4$ to the bottom left path γ of that top right square. Thus,

$$f_{U_2}(e_x * \alpha_4) = f_{U_2}(\gamma)$$

We can thus reduce the expression for the image of α under f to the top right composite:



in G. We can iterate this process, showing that $f(\alpha) = f(\beta)$, as desired. $\square$

Corollary 4.6.4 (The classical Seifert-van-Kampen Theorem)**.** *Let X be a topological space and let $U_1, U_2 \subset X$ such that the interiors of U_1 and U_2 cover X. If U_1, U_2, and $U_1 \cap U_2$ are path-connected, then for any $x \in U_1 \cap U_2$,*

$$\pi_1(X, x) \cong \pi_1(U_1, x) *_{\pi_1(U_1 \cap U_2, x)} \pi_1(U_2, x).$$

Proof. By path connectedness, we can display the diagram

$$\begin{tikzcd} B\pi_1(U_1 \cap U_2, x) \arrow[r] \arrow[d] & B\pi_1(U_2, x) \arrow[d] \\ B\pi_1(U_1, x) \arrow[r] & B\pi_1(X, x) \end{tikzcd}$$

as a retract of the diagram

$$\begin{tikzcd} \Pi_1(U_1 \cap U_2) \arrow[r] \arrow[d] & \Pi_1(U_2) \arrow[d] \\ \Pi_1(U_1) \arrow[r] & \Pi_1(x) \end{tikzcd}$$

Since pushouts are stable under retract (see Exercise 3.6.13), it follows that the former diagram is a pushout diagram. The result then follows by Proposition 3.7.13. □

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